Who Gains from Maternal Employment? Self-Employment, Childcare, and Child Development*

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Abstract

Whether a mother's employment helps or harms her child depends on how well formal childcare substitutes for the time she withdraws, and that substitutability differs sharply by education. I estimate a household life-cycle model of work arrangements, time allocation, and child skill formation on UK panel data, using the paid-self-employment margin to decompose a work arrangement into three prices: flexibility (hours menu and earnings curvature), an earnings level, and earnings volatility. Formal childcare substitutes well for a non-college mother's time but poorly for a college mother's, whose time is her child's most productive input. A quantity-quality trade-off therefore binds only for college mothers. The same entry into paid full-time work lowers a college child's cognitive and non-cognitive skills by 0.22 and 0.15 SD but raises a non-college child's by 0.12 and 0.16 SD. Offering flexible hours to paid employees without lower and more volatile pay raises maternal employment by up to four percentage points and household welfare, while leaving child skills essentially unchanged, because mothers draw the added hours from leisure rather than from time with the child. Child-skill gains run instead through childcare. Pairing family-friendly flexibility with childcare tagged by maternal education makes both groups better off, raising the projected lifetime earnings of children of non-college mothers by about six percent.

Keywords: Child development, work flexibility, self-employment, childcare, human capital formation

JEL Classification: J13, J16, J22, J24

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1 Introduction

Parental time and money are primary inputs into early child development,[1] and the work arrangement, whether and how a parent works, governs how much of each the child receives and how well one substitutes for the other. Whether parental employment, especially the mother's, helps or harms her child is debated (Hsin and Felfe, 2014): some estimates find gains, especially for disadvantaged children (Vandell and Ramanan, 1992), while others find losses (Bernal, 2008). These patterns are consistent with formal childcare substituting for parental time to a degree that varies by education, with the costs of care falling on more advantaged children (Fort, Ichino, and Zanella, 2020). College-educated mothers, who are more productive in the market and face higher reservation wages, may spend more skill-intensive time with their children, so childcare substitutes imperfectly for their time and extra market hours lower child skills. Non-college mothers work less and use formal care less, even where the returns to care are high.[2] Within the household, fathers reallocate little in response to a young child, leaving the adjustment to mothers (Kleven, Landais, and Søgaard, 2019). Because parents make these time and childcare choices within their work arrangement, the arrangement sets a quantity-quality trade-off between market hours and child skills that binds differently by gender and education. This is gaining policy relevance as governments legislate flexible work and expand childcare to ease work-family balance, yet how they affect child skills remains unknown.

In this paper, I develop a framework in which the parental work arrangement determines both the quantity and the composition of the inputs into child development, which differ in skill productivity. The arrangement shapes these inputs through three prices a household faces: i) flexibility, through the hours menu and the earnings curvature, governs how parents allocate time between market work, home production, and leisure, while ii) the earnings level and iii) the earnings volatility govern labor-force entry and investment in the child through formal childcare. I isolate flexibility, valued especially by mothers with young children (Mas and Pallais, 2020), using the margin between paid and self-employment. While paid employment offers little control over hours, self-employment offers a wide, nearly continuous menu, but at lower and more volatile pay. The decomposition lets me pose a counterfactual the self-employment margin alone cannot, extending flexible hours to paid employees without the pay penalty.

I estimate a dynamic household labor supply model with child skill formation, in which forward-looking households make their work arrangement and investment decisions over the life cycle. The model nests two trade-offs. The first weighs parental time with the child against household income: market hours buy income and formal childcare but come at the expense of the parent's own time, which care substitutes for imperfectly. The second involves parental labor supply choice: employment raises household resources and builds human capital but costs time and leisure, and

[1] See e.g. Cunha and Heckman, 2007; Del Boca, Flinn, and Wiswall, 2014; Gayle, Golan, and Soytas, 2015; Caucutt et al., 2026; Attanasio, 2026.

[2] See, e.g., Havnes and Mogstad, 2015; Kottelenberg and Lehrer, 2017; Cornelissen et al., 2018; Carneiro, Cattan, and Ridpath, 2024.

the arrangement supplying the hours differs in flexibility, pay, and volatility. Morever, human capital depreciates during employment gaps and early child skills are self-productive (Cunha, Heckman, and Schennach, 2010), so early investment choices carry lasting returns for the child and the mother's career, and households smooth earnings volatility through savings.

The model captures each trade-off through distinct features. The child skill technology governs the first. Cognitive and non-cognitive skills are persistent and accumulate from parental time and formal childcare through a separate technology for each domain, and a supervision constraint requires the two main inputs to jointly cover a young child's needs, tying formal childcare to the hours the arrangement supplies. The work arrangement governs the second. Each period, couples choose an arrangement and, conditional on it, market hours, home production, consumption, savings, and, for a young child, part-time or full-time formal childcare. Hours options within each arrangement are derived from weighted k-means clustering, and the price of flexibility varies by gender through the hours menu and the hours-earnings curvature, while the variance of earnings prices each arrangement's volatility and the tax-and-transfer system shifts the net return across households.

I draw on UK panel data, primarily the British Household Panel Survey (BHPS) and the Millennium Cohort Study (MCS). The BHPS provides annual waves on a nationally representative sample of UK households from 1991 to 2008, covering labor supply, earnings, and household wealth and I restrict the sample to married and cohabiting couples with biological children and working husbands. The MCS follows about 19,000 children born in 2000-2002 with age-appropriate cognitive and non-cognitive assessments, and I focus on birth to age 7, spanning the pre-school investment period and the early school years in which formal childcare remains a parental choice. I supplement the analysis with the UK Time Use Survey and Labour Force Survey to document time-use patterns and trends by work arrangement. The UK setting is well-suited to this study for two main reasons. First, because health insurance is universal and not tied to employment such that insurance access does not distort the choice between paid and self-employment, unlike in the United States, where job-based coverage can deter it (Fairlie, Kapur, and Gates, 2011). Second, the tax-and-transfer system, simulated with FORTAX (Shephard, 2009), varies across family types and income through working tax credits, child tax credits, and childcare subsidies, shifting work arrangements and childcare use.

I first document how spouses' labor supply and time allocations shift at childbirth. Wives move toward a more flexible arrangement conditional on working despite lower average earnings, while husbands' arrangements remain unchanged. Conditional on the work arrangement, mothers with flexibility reallocate more toward home production than those facing rigid schedules, while fathers use it to work longer market hours, and childcare use differs by education. These patterns translate into distinct trajectories in children's cognitive and non-cognitive skills. Identifying the effect, however, poses several challenges. Parents sort into work arrangements on unobserved preferences that also shape their time and money investments, which, in turn, are jointly determined. They also decide as a household, so one spouse's choice affects the other through shared constraints. These challenges compound over the life cycle, shaping future earnings and the household's capacity to invest.

I estimate the model in two steps. First, I estimate the child production function by joint GMM, combining a control function to address parental input endogeneity with cross-domain instruments for measurement error in skill assessments (Agostinelli and Wiswall, 2025). The tax-and-transfer system shifts the price of childcare and the returns to each arrangement, separately identifying the productivity of parental time and formal childcare by skill domain and parental education. Second, I recover the remaining structural parameters governing household preferences, parental income processes, and their human capital dynamics via simulated method of moments (McFadden, 1989). The estimator matches work arrangement shares, earnings distributions, time allocations, and household transitions over the life cycle.

The structural estimates reveal three mechanisms: how households value flexibility against pay, the productivity of parental inputs, and the constraints that bind them. First, both spouses value the flexibility of self-employment, revealed by its lower disutility of work than paid employment. They use it in opposite directions, the wife toward home production and shorter hours, the husband toward longer market hours. Second, parental inputs differ in productivity. Maternal time is the most productive, more so for cognitive skills and college mothers, while the father's home time is much less productive and raises both skill domains only through its interaction with college. Formal childcare sits between college and non-college maternal time, which generates the education gradient. Formal care is an inferior substitute for a college mother's time but a superior one for a non-college mother's, consistent with quasi-experimental evidence that formal care lowers cognitive skills for advantaged children (Fort, Ichino, and Zanella, 2020) while benefiting disadvantaged ones (Cornelissen et al., 2018). Third, a supervision constraint requires parental time and formal care to jointly cover a young child. Because the father barely adjusts, the mother is the sole margin of adjustment.

The model replicates the child-skill loss when a mother works full-time and uses formal childcare (Bernal, 2008), though the pooled effect masks the heterogeneity. The trade-off binds only for college children, where a mother's entry into full-time work lowers their cognitive and non-cognitive skills by 0.22 and 0.15 SD, while for non-college children, the same entry raises their skills by 0.12 and 0.16 SD. I then ask what flexibility does on its own, offering paid employees flexible hours without self-employment's lower, more volatile pay. Able to choose intermediate hours she lacked under rigid schedules, a mother drawn into work enters below full-time, relies less on formal care, and barely reduces her time with the child. The children of the mothers flexibility draws into work bear a smaller cost than those of full-time entrants, -0.13 and -0.07 SD for college children and $+0.04$ and $+0.03$ SD for non-college. This is a composition effect: these mothers enter below full-time, whereas a mother already working full-time does not cut her hours when offered the menu. In the aggregate the pooled effect is close to null, because mothers protect their time with the child and take the adjustment in market hours out of leisure instead.

This education gradient determines whether flexibility works as a family policy. Family-friendly flexibility, offering reduced and intermediate hours, prevents the college child skill loss while raising employment. Education-targeted free childcare supplies non-college children the formal care they gain from and does little to maternal employment, since the employment gains come from family-friendly flexibility rather than childcare. It spares college children the loss untargeted full-time care would impose. Education makes a better tag than income (Akerlof, 1978) because it is fixed, tracks the productivity gap, and, unlike an income test, invites neither a participation tax nor gaming. Pairing the two policies makes both groups better off. The welfare gains are larger from flexibility than from childcare, because flexibility raises parental welfare through control over hours while childcare works through child skills. For college households, almost all the gain is through the parental welfare channel: family-friendly flexibility is worth about £5,000 a year, with little welfare gains from child skills. For non-college households, flexibility alone raises parental welfare but slightly lowers child skills, so pairing it with targeted childcare delivers both, about £1,000 in parental welfare and £4,100 from child skills. The returns have intergenerational effects. Targeted childcare raises a non-college child's expected lifetime earnings by about six percent (£35,700), almost entirely from childcare gains, while family-friendly flexibility raises the college child's by about half a percent.

This paper contributes to three strands of literature. The first studies parental investment and child development. Early skills predict long-run outcomes (Heckman, Stixrud, and Urzua, 2006; Chetty et al., 2011a), and they form through parental time and money in a dynamic, self-productive technology (Cunha and Heckman, 2007; Cunha, Heckman, and Schennach, 2010; Del Boca, Flinn, and Wiswall, 2014; Attanasio et al., 2020). Building on this, a policy literature evaluates the levers that shape these investments and, through them, child's skills: transfer programs (Mullins, 2022), parenting and early childhood interventions (García et al., 2020), childcare subsidies (Havnes and Mogstad, 2015; Kottelenberg and Lehrer, 2017; Cornelissen et al., 2018; Felfe and Lalive, 2018; Moschini, 2023), childcare markets (Berlinski et al., 2024), and childcare regulations (Garcia-Vazquez, 2023). I contribute a policy margin this literature has not considered, work flexibility, by treating the parental work arrangement as a determinant of both the quantity and the returns of parental inputs.

The second strand studies how maternal employment shapes child development. Early maternal employment, especially in the first year, is often found to lower children's cognitive skills (Waldfogel, Han, and Brooks-Gunn, 2002; Baum, 2003; Ruhm, 2004; James-Burdumy, 2005; Brooks-Gunn, Han, and Waldfogel, 2010), and structural estimates that treat employment and childcare as endogenous turn the effect negative (Bernal, 2008; Bernal and Keane, 2011). Some papers report null or positive effects (Vandell and Ramanan, 1992; Dunifon et al., 2013; Suh, 2026). The sign hinges on what replaces the mother's time: universal childcare has harmed children in some settings (Baker, Gruber, and Milligan, 2008, 2019; Fort, Ichino, and Zanella, 2020) yet helped disadvantaged children in others (Cornelissen et al., 2018; Felfe and Lalive, 2018). Closest work is Verriest (2024), who places household labor supply and childcare policy in a child development model, but I add the paid-self-employment margin and its decomposition into flexibility, an earnings level, and earnings volatility. I reconcile this evidence with an education gradient in how well care replaces the mother's time, and connect it to a literature on schedule flexibility and maternal labor supply that has traced flexibility only to labor-market outcomes. Most work focuses on paid employment, distinguishing *greedy jobs* that reward long hours (Goldin, 2014, 2021) from family-friendly professions (Goldin and Katz, 2016). In the UK, flexible arrangements strengthen mothers' attachment (Chung and Van der Horst, 2018), with gains that are not gender-neutral (Arntz, Yahmed, and Berlingieri, 2022). I carry this flexibility to the child.

The third strand relates to self-employment and household labor supply. The choice of self-employment is often gendered: men value autonomy and being their own boss despite lower earnings (Hamilton, 2000; Taylor, 1996), while women enter for work-family balance and flexible schedules (Boden Jr, 1996; Marlow, 1997; Lombard, 2001; Mattis, 2004; Georgellis and Wall, 2005; Budig, 2006; Wellington, 2006; Joona, 2017; Berniell et al., 2025). That work-family motivation is the flexibility my model prices. At the household level, spouses' self-employment decisions are interdependent as they coordinate work schedules (Devine, 2001; Bryan and Sevilla, 2017), and one spouse's resources and experience shape the other's entry (Bruce, 1999; Parker, 2008). This literature traces self-employment to gendered labor-market outcomes around motherhood, in wages (Harrison, Kray, and Sethi, 2026), firm performance (Bonney, Pistaferri, and Voena, 2025), and entrepreneurship rates (Ferrando et al., 2025; Rutigliano, 2025). Embedding self-employment in a joint model of both spouses, I show how their interdependent work and childcare choices shape the child's skills.

The remainder of the paper is organized as follows. Section 2 describes the data and Section 3 documents the empirical patterns motivating the structural model. Section 4 presents the household life-cycle model and Section 5 outlines the identification and estimation strategy. Section 6 presents the estimation results and Section 7 covers comparative statics and policy experiments. Section 8 concludes.

2 Data

British Household Panel Survey (BHPS) The BHPS is an annual longitudinal household panel survey that began in 1991, sampling approximately 5,500 households in Great Britain. The survey ran until 2008, providing 18 waves of detailed information on family structure, labor market outcomes, and income dynamics. The survey also contains rich information on household finances and assets, including data on home ownership, house values, and mortgage status and values. More detailed asset information, including financial assets, savings, and debts, is available every five years to study longer-term wealth.

For the main analysis, I focus on married and cohabiting couples[3] with no changes in family

[3] I pool married and cohabiting couples because the UK tax-and-transfer system treats them identically over the sample period: income tax is assessed individually, and tax credits and means-tested benefits are assessed on joint couple income under the same rules whether the partners are married or cohabiting.

structure and families with biological children.[4] I condition on husbands always working[5], restrict the sample to women aged 21–51 covering early to mid-career stages, and drop couples missing key demographic and labor market information. The final sample consists of 49,958 observations of 9,619 individuals and 4,871 households. Wages are computed as hourly earnings by dividing weekly earnings by weekly hours in the main job; self-employment income uses reported gross monthly pay and profit converted to a weekly measure.[6] Table A.4 shows the summary statistics, and variable construction is detailed in Appendix A.1.

Millennium Cohort Study (MCS) For child development outcomes, I use the MCS, a nationally representative longitudinal study of approximately 19,000 children born in the UK between September 2000 and January 2002. I focus on early childhood from birth to age 7, the period when skill investments are most productive and their effects most persistent (Cunha and Heckman, 2007; Heckman, Stixrud, and Urzua, 2006). Cognitive skills are measured by the Denver Development Screening Test at 9 months and the British Ability Scales (BAS) at ages 3, 5, and 7 (Frankenburg and Dodds, 1967; Elliott, Smith, and McCulloch, 1997); non-cognitive skills by the Carey Infant Temperament Scale at 9 months and the SDQ Total Difficulties Score at ages 3, 5, and 7. This age window also covers the period in which parents are the primary decision-makers over formal childcare arrangements, before school entry substantially reshapes the childcare environment.[7] I apply the same restrictions as the BHPS (married and cohabiting couples, stable family structure, working fathers) and further restrict to singleton births and mothers aged 21–45 at birth.[8] The final sample consists of 5,219 children and 28,802 parent-wave observations. Further details are in Appendix A.2.

Table A.6 shows that the dominant pattern in both the MCS and the BHPS is a large reallocation of mothers across work arrangements as children age, with no comparable shift for fathers. In the MCS, the non-working share among mothers falls from close to half at 9 months to roughly

[4] Since changes in family composition can affect couples' behavior and child outcomes, I exclude divorced households, which account for around 8% of the sample. This restriction aligns the empirical analysis with the model environment and avoids confounding effects arising from step-parenting, non-resident parents, or complex household arrangements (Case, Lin, and McLanahan, 2000; Ginther and Pollak, 2004; Francesconi, Jenkins, and Siedler, 2010). The descriptive event-study figures in Section 3 restrict to couples with at least one observed child in the panel.

[5] The key margin in the data and model is the work arrangement, particularly the wife's choice across paid employment (part-time and full-time), self-employment, and non-employment. Conditioning on the husband working (7% loss) focuses the analysis on this margin and is consistent with the evidence that paternal labor supply is largely invariant to childbirth (Kleven, Landais, and Søgaard, 2019).

[6] Self-employment income is bottom-coded at zero in the BHPS: self-employed individuals reporting losses are recorded as having zero income. The left tail of self-employment income is therefore truncated, and the estimated self-employment earnings dispersion understates true downside unexplained volatility. I later verify in in Section 7 that the flexibility results are insensitive to this measurement by re-solving the model over a range of self-employment dispersions.

[7] Over 98% of MCS children enter school by age 5. By focusing on ages up to 7, I capture both the pre-school investment period and the early school years in which formal childcare remains a meaningful parental choice, while avoiding confounding from school quality differences.

[8] I keep biological children only and track households with no change in family composition, since separation may affect child development outcomes (Amato and Keith, 1991; Cherlin, Kiernan, and Chase-Lansdale, 1995; Kim, 2011). Twins account for 1.3% of the sample; I exclude them for consistency with the singleton model assumption (Ronalds, De Stavola, and Leon, 2005).

one in four by age 7, with most of the reallocation flowing into part-time paid employment and, to a lesser extent, self-employment. The BHPS displays the same pattern: mothers with young children are substantially more likely to work part-time and less likely to work full-time relative to those without young children, and this gap closes as children age. Fathers, by contrast, remain predominantly in paid employment throughout. This asymmetry between maternal and paternal work arrangement dynamics motivates the model's focus on work arrangement choice as the key margin of maternal adjustment.

Parental time inputs entering the child skill production function are not consistently observed in the MCS.[9] I therefore construct them by statistically matching BHPS time-use data to MCS households based on parent's gender, work arrangement, and education as well as household characteristics including the number of children and child's age.[10] The identifying assumption is conditional exchangeability: within these matched cells, BHPS time allocations are representative of what MCS households would report.[11]

I also use the UK Time Use Survey to validate the imputed time inputs (Appendix C) and to document time-use patterns by work arrangement (Appendix A.4), and supplement the analysis with the Labour Force Survey (LFS), Family Resources Survey (FRS), and O*Net (Appendix A.5).

3 Parental Work Arrangement and Child Investment

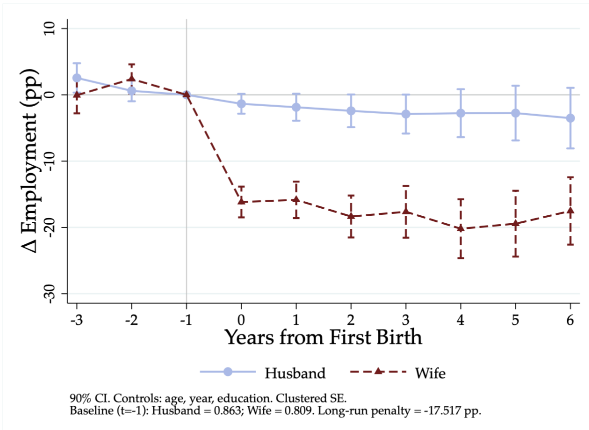
Each work arrangement trades time against money, shaping how parents invest in their children. Paid employment offers a rigid menu of part-time or full-time schedules, self-employment lets hours adjust nearly continuously, and not working frees all time for home production at the cost of earnings. Self-employment's schedule control, though, comes with lower and more volatile pay, so a work arrangement bundles three prices the data cannot separate: flexibility (the control over when and how many hours to work), an earnings level, and earnings volatility. This section documents three empirical patterns, from the mother's shift in work arrangement at childbirth to its consequences for the child, that motivate the structural model.

[9] The MCS records parental investment by frequency (e.g., how many times per week parents read to the child), which does not capture intensity: self-employed mothers report higher frequencies than not working mothers. The MCS includes a time-with-child measure in some waves, but as qualitative bands (nowhere near, not quite, just enough, plenty, more) rather than in real time and I show that they are highly correlated with the BHPS time use (Appendix C).

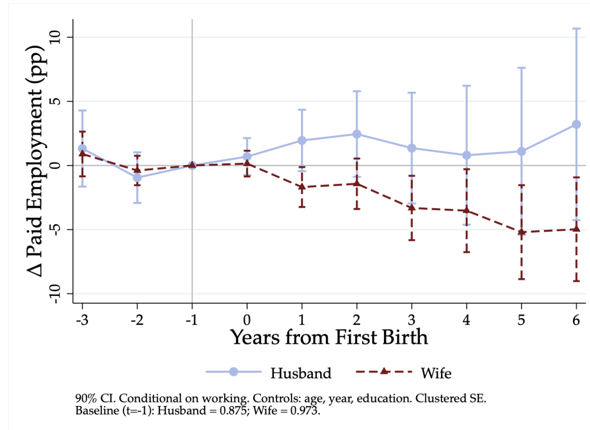
[10] The match uses five cell categories (parent gender, work arrangement, education, number of children, and child age group), which gives imputed maternal and paternal time inputs of 72 and 53 distinct cell values in the estimation sample, so the identifying variation is between-cell by construction.

[11] Differences in the marginal distributions across datasets do not threaten validity because matching is done within each matched cell, so what matters is within-cell comparability, not equal marginal shares. The BHPS confirms that conditional on the five matching variables, time allocation varies little with income. The matched values are also consistent with UK Time Use Survey diary estimates within matched cells (Appendix C). Because the procedure assigns conditional means rather than reconstructing the joint distribution of time inputs and child outcomes, it does not require the conditional independence assumption of classical statistical matching (Ridder and Moffitt, 2007). Because the time input is defined at the arrangement level, the coefficient identifies the effect of moving between arrangements, and the instruments, which shift a household's arrangement, need only be excluded across arrangements rather than within them (Corollary 2, Appendix E).

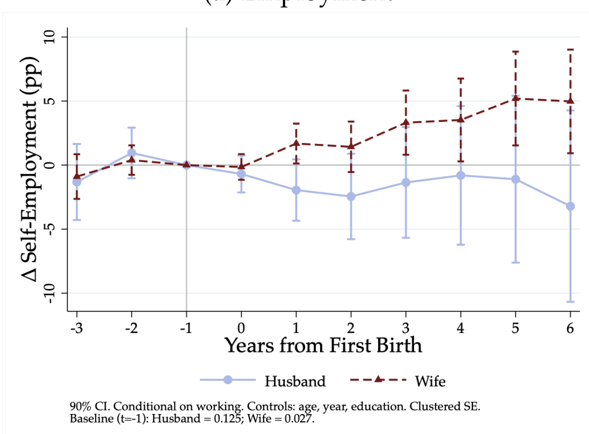
3.1 Work Arrangement at Childbirth



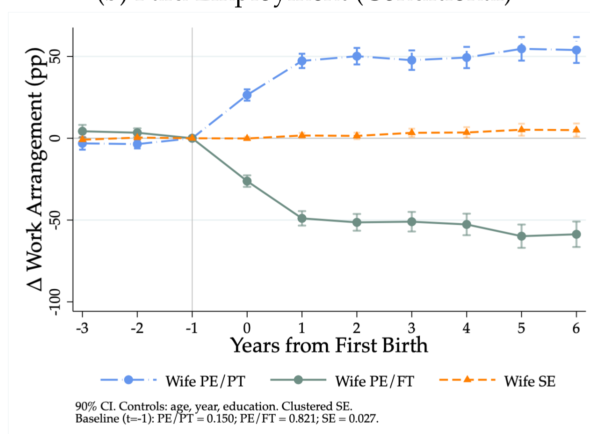
(a) Employment



(b) Paid Employment (Conditional)



(c) Self-Employment (Conditional)



(d) Wife's Work Arrangements (Conditional)

Note: The event is the birth of the first child. The y-axis shows percentage point changes relative to $t = -1$. Panels (b)–(d) are conditional on being employed. Controls include age, year, education, and marital status; standard errors clustered at the individual level. Confidence intervals at 90%.
Source: British Household Panel Study (BHPS).

Figure 1: Event Study: Parental Work Arrangements

The event-study analysis shows that women face a sharp motherhood penalty in employment at first birth (Figure 1 Panel (a)). Women's overall employment falls by about 17 percentage points at birth and remains 15–20 percentage points below the pre-birth trend through the sixth year after birth (Kleven, Landais, and Søgaard, 2019). Work arrangements shift too, and only for mothers. Among those who remain employed, full-time paid employment contracts sharply, part-time paid employment expands, and self-employment rises in the years following birth. This within-employment shift toward self-employment is consistent with mothers seeking work arrangements that offer greater scheduling control (Harkness, Borkowska, and Pelikh, 2019).[12]

[12]Using sequence analysis on UK Understanding Society data, Harkness, Borkowska, and Pelikh (2019) document the same divergence. In the three years after birth, mothers disperse across eight distinct employment clusters, with 7% sorting into self-employment, while fathers vary little, with 62% remaining consistently in full-time employment.

The rise concentrates in unincorporated self-employment, sole traders and freelancers, rather than incorporated firms.[13] Self-employed women also sort into services and personal care rather than the schedule-rigid administrative roles that dominate paid employment (Appendix B.3) (Ferrando et al., 2025). Men, by contrast, show no comparable adjustment in employment or work arrangement across all panels.

These shifts in work arrangement are accompanied by changes in hours worked, income, and home production time (Appendix B.1). Wives reduce market hours and increase home production, which comes with a fall in weekly income in paid employment, while self-employed wives cut hours more sharply but see little change in hourly wages. Husbands' hours, income, and home production again adjust little. Household assets also smooth consumption, which matters given the income-volatility differences across work arrangements that the next subsection documents (Crossley and O'Dea, 2010; Adda, Dustmann, and Stevens, 2017).[14]

3.2 Flexibility, Earnings, and Childcare

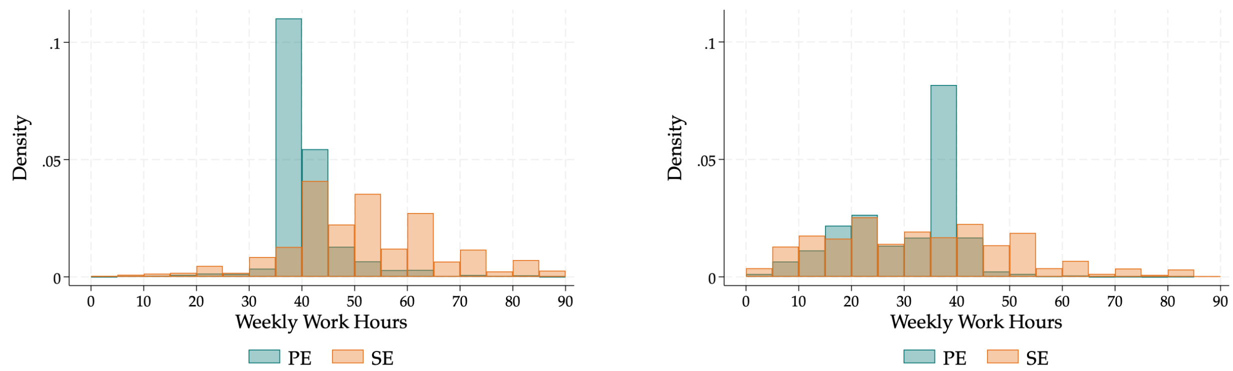
If parents sort into different work arrangements, how do their labor supply choices affect the household's time and money trade-off, especially when children are young? Beginning with time, Figure 2 shows that the hours worked differ across arrangements. Paid-employed workers cluster tightly around the standard full-time hours, with husbands concentrated near 40 hours and wives showing a bimodal distribution at part-time and full-time. Self-employed workers, by contrast, spread broadly across the hours range, with substantial mass at both shorter and longer hours. This dispersion reflects the absence of the rigid wage-hours package that characterizes paid employment (Rosen, 1976; Altonji and Paxson, 1988); self-employed workers set their own schedules and adjust hours in response to changing household needs. They also adjust hours more across years, with higher within-person standard deviation (Appendix B.1).

How parents use this flexibility differs by gender. Figure 3 shows mean hours worked and home production by work arrangement and the presence of a young child (age five or below). Among men, self-employed husbands work about 10 hours more per week than paid-employed husbands, spend similar time in home production, and adjust little with a young child. Among women, self-employed wives with a young child work 8.3 fewer market hours than those without, while for full-time paid-employed wives the difference is only 0.2 hours (Gimenez-Nadal, Molina, and Ortega, 2012). With a young child, self-employed wives average 18.6 hours of home production per week compared to 12.6 hours for full-time paid-employed wives, a 48 percent difference in time available for child investment at home.[15] Self-employed workers also work from home more often,

[13] Rutigliano (2025) and Bonney, Pistaferri, and Voena (2025) use administrative firm records to show large post-birth declines in the performance of female-owned incorporated businesses. Figure B.8 in Appendix B documents the unincorporated concentration in my sample.

[14] The full asset event-study plots are in Appendix B.4.

[15] These home production levels are consistent with diary-based UK estimates. Gimenez-Nadal, Molina, and Zhu (2014) report fathers and mothers devote 0.91 and 1.56 hours per day to housework (about 6 and 11 hours per week), close to the figures here. The UK Time Use Survey confirms the levels (Appendix A.4), though it covers too few self-employed parents to validate the cross-arrangement gap directly. Decompositions by self-employment incorporation status are in

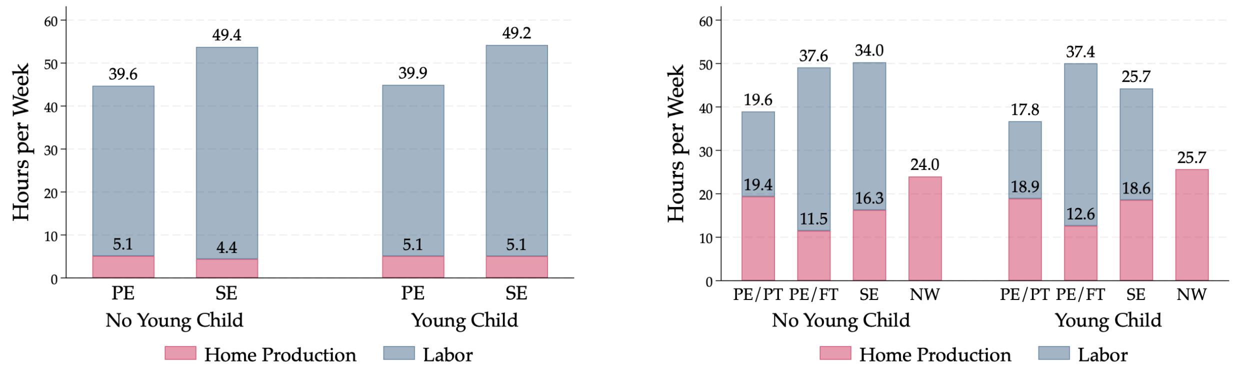


(a) Husband (b) Wife

Note: Bars report the share of person-year observations in 5-hour bins, conditional on working. PE = paid employment; SE = self-employment.
Source: British Household Panel Study (BHPS).

Figure 2: Distribution of Weekly Work Hours by Work Arrangement

especially with young children, take more breaks during the day, and keep more irregular hours (Appendix B.2).



(a) Husband (b) Wife

Note: Bars show mean hours per week in market work and home production by work arrangement and young-child (age 5 or below) status. PE = paid employment; SE = self-employment.
Source: British Household Panel Study (BHPS).

Figure 3: Time Allocation by Parental Work Arrangement

Turning to money, flexibility comes at the cost of lower and more volatile income. Self-employed workers have a wider and flatter weekly earnings distribution than paid employees, with thicker tails on both sides (Appendix B.1 Figure B.2). Table 1 shows the mean and standard deviation of weekly earnings, hours, and hourly wages by work arrangement and presence of a young child. Self-employed parents earn less per week than paid employees, and self-employed husbands also less per hour, though self-employed wives' mean hourly income is higher, reflecting the wide, right-skewed self-employment earnings distribution. Their earnings standard deviations

Appendix B.

are substantially larger, confirming that the distributional spread reflects genuine income volatility rather than compositional differences alone. The within-person standard deviation of log weekly earnings is 1.67 for self-employed husbands versus 0.18 for paid employed husbands, and 1.81 versus 0.25 for wives. Households therefore face a flexibility-risk trade-off: self-employment relaxes hours constraints but exposes them to substantially larger income inequalities.

Table 1: Parental Income by Young Child and Work Arrangement

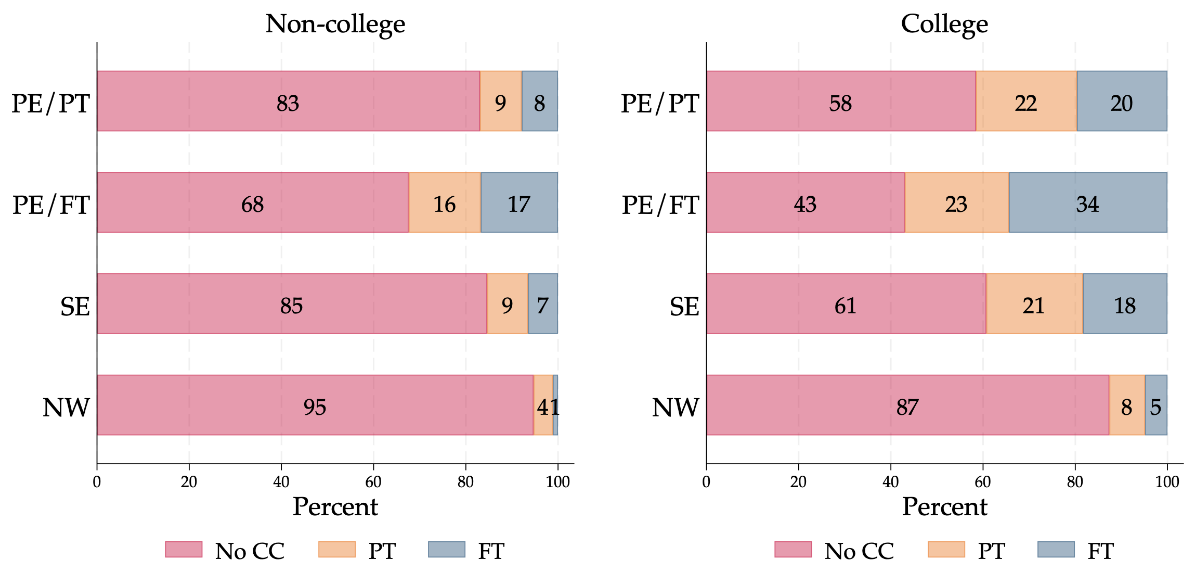
	Weekly Income		Hours Worked		Hourly Income	
Work Arrangement	NChild	YChild	NChild	YChild	NChild	YChild
Husband						
Paid Employed	423.6	479.4	39.5	39.9	10.9	12.2
	(230.0)	(325.1)	(6.6)	(7.0)	(6.1)	(8.5)
Self-Employed	339.6	387.4	48.0	49.2	8.1	9.0
	(391.4)	(390.9)	(15.2)	(14.7)	(13.9)	(14.3)
Wife						
Paid Employed/PT	153.8	148.6	20.7	17.8	7.6	8.2
	(125.6)	(107.1)	(6.6)	(6.6)	(12.5)	(5.8)
Paid Employed/FT	326.7	350.7	37.8	37.4	8.7	9.4
	(166.5)	(159.4)	(3.9)	(3.7)	(4.4)	(4.3)
Self-Employed	236.7	164.7	36.1	25.7	8.0	10.3
	(329.4)	(220.0)	(18.2)	(16.7)	(12.7)	(27.7)

Note: Means and standard deviations (in parentheses) of weekly income (£), weekly hours worked, and hourly wage (£) by parental work arrangement and young-child (age 5 or below) status. Weekly income for self-employed workers is gross monthly pay and profit converted to a weekly measure; self-employment earnings are bottom-coded at zero in the BHPS, so zero-earnings observations are included and pull the self-employed means down relative to positive-earners only. All amounts in 2001 £.
Source: British Household Panel Study (BHPS).

In addition to parental time inputs, sending the child to formal childcare is another lever of investment. Figure 4 shows the share of households using no formal childcare, part-time formal childcare (up to 15 hours/week), and full-time formal childcare, by maternal work arrangement and education.[16] Three patterns stand out. First, self-employed mothers substitute parental time for formal childcare relative to paid employed mothers: among non-college households, 85% of self-employed mothers use no formal childcare, compared with 68% of full-time paid employed mothers. Second, the education gradient is sharp for paid employment but muted for self-employment. College full-time paid employed mothers use full-time formal childcare at a rate of 34%, more than double the 17% rate for non-college full-time paid employed mothers, reflecting both higher earnings and stronger labor market attachment. College self-employed mothers use full-time childcare at only 18%, barely above the non-college rate of 17%, consistent with self-employment

[16]Formal childcare includes childcare centers, childminders, nannies, and breakfast and after-school clubs. I classify breakfast and after-school clubs as formal childcare following Del Bono et al. (2016), as they primarily serve children aged 5. I treat formal childcare as homogeneous in quality within the regulated sector; see Appendix D for supporting evidence.

providing scheduling flexibility that substitutes for formal care regardless of education. Third, non-working mothers overwhelmingly rely on parental care across both education groups.



Note: Bars show the share of mothers by childcare type, work arrangement, and maternal education. No CC = no formal childcare; PT = part-time formal childcare ($\leq$15 hours/week); FT = full-time formal childcare (>15 hours/week).
Source: Millennium Cohort Study (MCS).

Figure 4: Formal Childcare Arrangements by Maternal Work Arrangement and Education

3.3 Implications for Child Development

The time allocation and childcare patterns above translate into measurable differences in early child development outcomes. In Suh (2026), I estimate a child production function via three-stage least squares, treating parental work arrangement and time as jointly determined inputs, using sibling sex composition to instrument for parental time, age-earnings profiles to instrument for income and work arrangement, and lagged work arrangement to identify the state dependence in work arrangement choice. Conditional on parental time and earnings, having a working mother is associated with higher cognitive and non-cognitive skills relative to non-working mothers, with the cognitive advantage concentrated in full-time paid employment and the non-cognitive advantage extending to self-employment. Self-employed fathers also raise non-cognitive skills relative to paid-employed fathers.[17]

[17]The child production function parameters reported here differ from Suh (2026) for two reasons. First, Suh (2026) uses demographic instruments (sibling sex composition and age-earnings profiles). The current estimates use FORTAX-simulated instruments that exploit variation in the UK tax-benefit system, which is plausibly orthogonal to such preferences. Second, Suh (2026) corrects for the joint determination of inputs within the estimating system but does not model selection into work arrangements and childcare use; the life-cycle framework here addresses selection by modeling the work arrangement and hours choices that feed into child skill production.

Translating these patterns into welfare and policy implications requires a structural model, for three reasons. First, the sorting documented in Section 3.1 means that parents select work arrangements based on preferences, human capital, and household resources that independently affect child outcomes, so separating the effect of the arrangement from the preferences that drive the choice requires jointly modeling selection and skill production. This is in a similar vein to Bernal (2008), who models maternal employment and childcare together and finds negative effects on child cognitive skills. Second, the input patterns in Section 3.2 show that parental time and formal childcare respond together to the same work arrangement choice, so their quantities are jointly endogenous. Understanding how parental inputs substitute, and why that substitution differs, especially by maternal education, requires a model that places their productivities within the household's budget and time constraints. Third, the lower and more volatile self-employment earnings and non-linear returns to hours imply that a policy changing incentives shifts the entire household budget set as well as the time constraints, so evaluating childcare subsidies or flexibility mandates requires tracing the household's joint re-optimization.

4 Household Life-Cycle Model

I build a household life-cycle model that connects parental work arrangements to their child's development. Spouses jointly choose a work arrangement, and conditional on it, consumption, time allocations, savings, and formal childcare, and the household budget responds as a whole when incentives change. The household consists of two adults, a wife ($i = w$), a husband ($i = h$), and their biological child(ren).[18] Let t represent the woman's age, from 21 to 51, at which point the household faces a terminal value. Each period covers two years, matching the biennial child development data.[19]

4.1 Household Choices and Preferences

The household enters each period with a state vector which constitutes of each spouse's college degree θ^i, the wife's human capital hc_t^w, household assets a_t, the number of children k_t, and the youngest child's age g_t. The number of children and child's age evolve through a stochastic fertility process: at the end of each period the wife faces a birth probability p_t^w that depends on her education and current number of children.[20] Given this state, the household chooses savings

[18] I include cohabiting couples whose children are biological. Cohabiting couples constitute roughly 22% of UK couples (Office for National Statistics, 2022a) and a larger share of households with young children. The two channels through which cohabitation might differ from marriage in this setting would be change in family structure. Since I do not model divorce or re-partnering, I focus on families with only biological children. During the couple formation, UK tax and transfer treatment is equivalent for cohabitants and spouses, so the budget constraint applies identically to both. Throughout, I refer to the two adults as wife and husband, and as mother and father in periods with a child present.

[19] The choice of model period interval is driven by data availability. For the household life-cycle labor supply, I use the BHPS which is available annually; however, the MCS for child's development skills is available every two years for the age range I focus on.

[20] This assumption does not require that fertility is unconditionally random. It requires only that birth timing is exogenous to unobserved work arrangement preferences conditional on the state vector (age, education, and parity),

a_{t+1}, a work arrangement q_t, and each spouse's hours worked n_t^i and home production time τ_t^i. The work arrangement $q_t \in \mathcal{Q}$ summarizes the joint status of both spouses, indexed by indicators $d_{j,t}^i \in \{0, 1\}$ for individual i in work arrangement j. The husband always works and chooses between paid employment ($j = \text{PE}$) and self-employment ($j = \text{SE}$);[21] the wife chooses among paid employment, self-employment, and not working ($j = \text{NW}$), yielding six mutually exclusive household arrangements ($|\mathcal{Q}| = 6$), from dual paid employment ($q_t = 1$) to a self-employed husband with a non-working wife ($q_t = 6$). When there is a young child aged five or below, the household additionally chooses a formal childcare arrangement $cc_t \in \{0, \text{PT}, \text{FT}\}$ at a cost.[22] The child investment phase ends at age 5, when the child enters school and parents stop making direct time investments.

I assume spouses are paternalistic and value their child's development.[23] Flow utility therefore depends on household consumption C_t, each spouse's leisure l_t^i and home production τ_t^i, the child's latent cognitive and non-cognitive skills $\log \theta_{D,t}$, $D \in \{\text{cog}, \text{ncog}\}$, formal childcare use, and work arrangement-specific non-pecuniary terms.[24] I further allow interaction terms with the presence of a young child, capturing how formal childcare demands and work-arrangement disutilities shift when there is a young child.

which together determine p_t^w. This conditional exogeneity is weaker than full exogeneity and is standard in structural life-cycle models of female labor supply (see e.g. Attanasio, Low, and Sánchez-Marcos, 2008; Blundell et al., 2016). The event-study analysis in Figure 1 supports the assumption: work arrangements show flat pre-trends in the years before first birth, which is inconsistent with parents strategically timing births around planned work arrangement transitions. The forward-looking nature of the model also mitigates the concern because households solve the full life-cycle problem taking the fertility process as given, so optimal work arrangement choices already incorporate the anticipated arrival of children.

[21] In the BHPS sample frame of married women aged 21–51, about 10% of couple-waves have a non-employed husband, and among them, very few have a self-employed wife, so the excluded cells contain almost none of the variation that identifies the work arrangement margin. The husband's employment margin is also the least responsive, as fathers' work arrangements do not respond to childbirth (Section 3; Kleven, Landais, and Søgaard, 2019). Fixing the husband's employment while modeling his arrangement and hours choices is richer than the common practice of treating the husband's earnings as an exogenous income process.

[22] Part-time childcare is 15 hours per week and full-time is 30 hours per week, each multiplied by the hourly childcare price. When the child is older (ages 6–15), childcare is no longer a choice and child-related expenditures are a fixed fraction of household income. See Appendix D for the details.

[23] I assume that the child has no bargaining power. This assumption is common in the literature, especially when the child is young, before school entry shifts the allocation of time and parental decision-making.

[24] I adopt a unitary framework as a benchmark. Although bargaining power might vary over the life-cycle (Lundberg, Pollak, and Wales, 1997; Guo and Xie, 2024), the paper's focus is on joint time-use and child-investment decisions rather than intra-household resource distribution. Appendix M shows the results are robust to relaxing it. Results are robust to re-solving the counterfactuals under education-specific Pareto weights.

$$
\begin{aligned}
\mathcal{U}_t = & \underbrace{\log C_t}_{\text{Consumption}} + \underbrace{\alpha_l^w \log l_t^w + \alpha_l^h \log l_t^h}_{\text{Couple's leisure}} + \underbrace{\alpha_{cog} \log \theta_{cog,t} + \alpha_{ncog} \log \theta_{ncog,t}}_{\text{Child's skills}} \\
& + \underbrace{\left[\alpha_\tau^w + \alpha_{\tau,n}^w \frac{n_t^w}{\bar{n}} + \alpha_{\tau,k}^w \mathbb{1}[ychild_t = 1]\right] \cdot \log \tau_t^w + \left[\alpha_\tau^h + \alpha_{\tau,n}^h \frac{n_t^h}{\bar{n}} + \alpha_{\tau,k}^h \mathbb{1}[ychild_t = 1]\right] \cdot \log \tau_t^h}_{\text{Couple's home production}} \\
& + \underbrace{(\alpha_{cc,PT} + \alpha_{cc,PT}^{ed} \cdot \theta^w)\mathbb{1}[cc_t = PT] + (\alpha_{cc,FT} + \alpha_{cc,FT}^{ed} \cdot \theta^w)\mathbb{1}[cc_t = FT]}_{\text{formal childcare}} \\
& + \underbrace{\sum_{j \in \{PE,SE\}} \left[\alpha_{j,1}^w \log l_t^w + \alpha_{j,2}^w \mathbb{1}[ychild_t = 1] + \alpha_{j,3,PT}^w \mathbb{1}[cc_t = PT] + \alpha_{j,3,FT}^w \mathbb{1}[cc_t = FT]\right] \cdot d_{j,t}^w}_{\text{Wife's (dis)utility by work arrangement}} \\
& + \underbrace{[\alpha_{SE,1}^h \log l_t^h + \alpha_{SE,2}^h \mathbb{1}[ychild_t = 1]] \cdot d_{SE,t}^h}_{\text{Husband's (dis)utility for SE}} + \underbrace{\xi_t}_{\substack{\text{preference} \\ \text{shocks}}}
\end{aligned} \tag{1}
$$

For identification, I normalize the reference categories to a paid-employed father and a non-working, non-college mother; the base parameters $\alpha_{cc,PT}$ and $\alpha_{cc,FT}$ then capture the (dis)utility of part-time and full-time formal childcare for her. The education interactions $\alpha_{cc,PT}^{ed}$ and $\alpha_{cc,FT}^{ed}$ allow college mothers to value formal childcare differently, and the work arrangement interactions $\alpha_{j,3,\mathrm{PT}}^w$ and $\alpha_{j,3,\mathrm{FT}}^w$, $j \in \{\mathrm{PE}, \mathrm{SE}\}$ measure how paid and self-employed mothers differ from the non-working baseline in valuing each intensity.

Two features of the specification require motivation. First, the non-pecuniary terms are important for self-employment. Self-employed workers often earn less and work longer hours than paid employees yet remain self-employed, which income and leisure alone cannot rationalize. The parameters $\alpha_{SE,1}^i$ and $\alpha_{SE,2}^i$ capture the non-pecuniary value of each work arrangement, which may reflect the flexibility and autonomy that draw women and men into self-employment. Together with the hours flexibility in Section 4.3, these amenities explain why parents may accept lower pay to remain self-employed. Second, home production carries a net time cost. If the base weights α_τ^i are negative, parents do not value home production for its own sake but they undertake it because it produces child skills they value, and because a young child must be supervised. The weight also varies with own market hours through $\alpha_{\tau,n}^i$, scaled by hours relative to their maximum $\bar{n}$. A negative value means the effort cost of home production rises with the workday, so a parent working longer supplies less of it. This is a reduced form way to capture effort that is convex in total work, market plus home.

The model features two levels of heterogeneity. Observed heterogeneity enters through parental education, which shifts earnings profiles, formal childcare valuation, and fertility. Unobserved heterogeneity enters through the wife's human capital hc_t^w, which evolves stochastically and is not directly observed by the econometrician. There are four dynamic state variables: household assets a_t, the wife's human capital hc_t^w, and the child's latent cognitive and non-cognitive skills $\log \theta_{D,t}$,

$D \in \{\text{cog,ncog}\}$, indexed by age $g_t \in \{1,3,5\}$.[25] Without loss of generality, I normalize the mean of initial log skills to zero at age 1, since pre-natal investment is unobserved and the level of initial skills is not identified.[26] At the terminal period, when the wife reaches age 51, the household no longer makes choices and faces the terminal value

$$\mathbb{E}V_{T+1} = \psi_{hc}\, hc^w_{T+1} \tag{2}$$

where hc^w_{T+1} denotes the wife's human capital at the end of the life cycle.[27]

4.2 Household Constraints

The household faces three constraints. First, each spouse faces the time constraint across leisure, hours worked, and home production subject to time endowment $\bar{T}$. Home production τ^i_t includes the time a parent spends caring for the child. Second, when there is a young child, the household faces a supervision constraint, which requires that parental home production and formal childcare together meet a minimum care requirement $\bar{T}^{cc}$.[28]

$$\underbrace{l^i_t}_{\text{leisure}} + \underbrace{n^i_t}_{\text{labor}} + \underbrace{\tau^i_t}_{\text{home production}} = \bar{T} \tag{3}$$

$$\tau^w_t + \tau^h_t + \bar{\tau}(cc_t) \geq \bar{T}^{cc}, \qquad cc_t \in \{0, \text{PT}, \text{FT}\}, \quad \bar{\tau} \in \{0, 15, 30\} \tag{4}$$

Meeting this requirement embodies the time-money trade-off: parental care costs hours worked and leisure, while formal childcare costs money. Third, the household faces the intertemporal budget constraint:

$$C_t + M_t + a_{t+1} = (1+r) \cdot a_t + \mathcal{T}(Y^w_{j,t}, Y^h_{j,t}, \mathbf{x}_t) \tag{5}$$

where household disposable income $\mathcal{T}(Y^w_{j,t}, Y^h_{j,t}, \mathbf{x}_t)$ finances consumption C_t, child-related expenditures M_t, and next-period savings a_{t+1} at net interest rate r. For young children, $M_t = p_{cc} \cdot \bar{\tau}(cc_t)$ and for school-age children (ages 6–15), M_t is a fixed share of net income.[29] Assets are subject to a borrowing constraint $a_{t+1} \geq \underline{a}$. I approximate after-tax income using the FORTAX micro-simulation system (Shephard, 2009), which maps gross household earnings and family characteristics to net disposable income. Following Gayle, Hincapié, and Miller (2024), I parametrize the tax-transfer

[25] Ages correspond to 9 months, age 3, and 5 in the MCS. I refer to 9 months as age 1 throughout.

[26] I estimate the initial dispersion from wave-1 MCS measures, as described in Appendix G.6. The test measures are age-standardized, and I normalize the reference measure to load one-for-one on the latent log-skill (Section 5.1); $\log \theta_{D,t}$ is therefore expressed on the scale of the standardized score, so a one-unit change corresponds to one standard deviation of the test-score distribution, the units in which I report all counterfactual skill effects.

[27] The terminal value rewards the wife's human capital but not terminal financial wealth, so the model imposes no bequest motive.

[28] Spousal time endowment is set to $\bar{T} = 112$ hours/week and care requirement is set $\bar{T}^{cc} = 20$ hours/week. Results are robust to the choice of $\bar{T}^{cc}$.

[29] Appendix D describes the calibrated prices of formal childcare per age.

schedule as:[30]

$$\mathcal{T}(Y^w_{j,t}, Y^h_{j,t}, \mathbf{x}_t) = \chi_1(\mathbf{x}_t) + \chi_2(\mathbf{x}_t) \cdot \left(Y^w_{j,t} + Y^h_{j,t}\right)^{\chi_3(\mathbf{x}_t)} \tag{6}$$

where $\chi_1(\mathbf{x}_t)$ are lump-sum taxes or transfers, $\chi_2(\mathbf{x}_t)$ scales net income in gross earnings, and $\chi_3(\mathbf{x}_t)$ captures the degree of tax progressivity.[31] Including $\chi_1(\mathbf{x}_t)$ is essential because self-employed parents may incur losses or report zero income. I allow these coefficients to vary by the child's age to account for benefits that vary by household composition, such as Child Benefit, Working Tax Credit, and Child Tax Credit.[32]

4.3 Couple's Income and Wife's Human Capital

Weekly earnings depend on an earnings rate and hours worked with a curvature $\eta_{i,j}$:

$$Y^i_{j,t} = y^i_{j,t} \cdot \left(n^i_t\right)^{\eta_{i,j}}, \qquad j \in \{PE, SE\} \tag{7}$$

where $\eta_{i,j}$ captures the hours-earnings curvature specific to each work arrangement. Only when $\eta_{i,j} = 1$ does $y^i_{j,t}$ equal hourly earnings; otherwise average hourly earnings, $Y^i_{j,t}/n^i_t = y^i_{j,t}(n^i_t)^{\eta_{i,j}-1}$, vary with hours. The curvature reflects an asymmetry in flexibility. In paid employment, firms offer tied wage-hours packages (Rosen, 1976) and workers face sharp constraints on within-job hours adjustment (Altonji and Paxson, 1988). Self-employment removes this constraint as it allows workers to set the schedule where hours are nearly continuously adjustable. Flexibility enters through two channels: the curvature and the hours menu itself, which is coarser in paid employment and wider in self-employment. Self-employment bundles this flexibility with a lower and more volatile wage process, given below, so the household's arrangement choice trades the value of the flexible hours against the level and risk of pay. Appendix F documents the grid construction, evidence for within-person hours adjustment, and robustness to grid size.

Since the husband always works, his experience accumulates deterministically with age rather than as a separate stochastic human capital state. His log hourly earnings follow a quadratic age profile (Van der Klaauw, 1996; Attanasio, Low, and Sánchez-Marcos, 2008):

$$\log y^h_{j,t} = \phi^h_{j,0} + \phi^h_{j,1}\theta^h + \phi^h_{j,2}t + \phi^h_{j,3}t^2/100 + \varepsilon^h_{j,t} \tag{8}$$

The wife's hourly earnings are a function of her education and human capital:

$$\log y^w_{j,t} = \phi^w_{j,0} + \phi^w_{j,1}\theta^w + \phi^w_{j,2}hc^w_t + \varepsilon^w_{j,t} \tag{9}$$

[30] This functional form is standard in the structural labor supply literature (Heathcote, Storesletten, and Violante, 2014; Blundell, Pistaferri, and Saporta-Eksten, 2016; Agostinelli et al., 2024).

[31] The UK taxes spouses individually, but over the sample period the individual schedule is a personal allowance plus a nearly flat basic rate, so the rate aggregates across spouses and the work arrangement fixes the number of allowances the household uses, which χ_1 and χ_2 absorb. The progressivity χ_3 captures comes from the family-assessed credits and benefits, which depend on joint income by statute. Conditional on the work arrangement, household net income is then approximately a function of household gross income, and I fit the coefficients to FORTAX net incomes computed under the actual individual-tax and family-assessment rules.

[32] More details are described in Appendix A.3.

where the initial distribution $hc_1^w \sim F(hc^w|\theta^w)$ is estimated conditional on education. Income shocks are correlated across spouses to capture assortative matching on unobservable characteristics; conditional on the work arrangement q_t, they are drawn from a bivariate normal distribution:

$$\begin{bmatrix} \varepsilon_{j,t}^w \\ \varepsilon_{j,t}^h \end{bmatrix} \sim \mathcal{N}\Big(\mathbf{0},\, \boldsymbol{\Sigma}_{j_t^w, j_t^h}\Big), \qquad \boldsymbol{\Sigma}_{j^w, j^h} = \begin{pmatrix} (\sigma_{j^w}^w)^2 & \rho\, \sigma_{j^w}^w \sigma_{j^h}^h \\ \rho\, \sigma_{j^w}^w \sigma_{j^h}^h & (\sigma_{j^h}^h)^2 \end{pmatrix}$$

where ρ represents the correlation between unobserved income shocks across spouses.[33]

Wife's human capital evolves as an unobserved discrete state $\mathcal{H} = \{L, M, H\}$ (Gayle, Lott, and Shephard, 2019). Because women enter the panel at different ages and have different levels of education, the entry distribution conditions on age at entry as a proxy for initial condition:

$$P(hc_1^w = hc|\theta^w, age_1) = \frac{\exp\big(\iota_{0,hc} + \iota_{1,hc}\theta^w + \iota_{age,hc} \cdot age_1\big)}{\sum_{hc' \in \mathcal{H}} \exp\big(\iota_{0,hc'} + \iota_{1,hc'}\theta^w + \iota_{age,hc'} \cdot age_1\big)} \tag{10}$$

where I normalize $\iota_{0,M} = \iota_{1,M} = \iota_{age,M} = 0$ relative to the medium human capital state. Conditioning on age_1 prevents the initial distribution from pooling women at different career stages. Current human capital affects hourly income, while hours worked governs its stochastic evolution. When not working, her human capital can only remains or depreciate. This loss of earning capacity during non-employment is the channel behind the persistent motherhood penalty.

$$\Pi^{NW} = \begin{pmatrix} 1 & 0 & 0 \\ \delta_{NW} & 1 - \delta_{NW} & 0 \\ 0 & \delta_{NW} & 1 - \delta_{NW} \end{pmatrix} \tag{11}$$

At the maximum hours of the wife's menu, human capital can depreciate, remain, or appreciate:

$$\Pi^{W} = \begin{pmatrix} 1 - \kappa_L & \kappa_L & 0 \\ \delta_{FT} & 1 - \delta_{FT} - \kappa_M & \kappa_M \\ 0 & \delta_{FT} & 1 - \delta_{FT} \end{pmatrix} \tag{12}$$

For intermediate hours, the transition matrix is a convex combination of the two extremes, weighted by:

$$\omega(n_t^w) = \left(\frac{n_t^w}{\overline{n^w}}\right)^{\nu_{hc}}, \quad \nu_{hc} > 0, \qquad \Pi(n_t^w) = (1 - \omega(n_t^w)) \cdot \Pi^{NW} + \omega(n_t^w) \cdot \Pi^{W} \tag{13}$$

Here $\overline{n^w}$ is the largest hours option across the wife's arrangements, so full accumulation ($\omega = 1$, the matrix Π^W) is reached at that maximum. A value of ν_{hc} close to one implies accumulation scales roughly linearly with hours intensity; large values imply that only hours near the top of the menu produce meaningful accumulation. Following Heckman and Singer (1984) and Keane and Wolpin (1997), the transition probabilities capture the expected evolution of human capital conditional on

[33] I assume a common correlation parameter ρ across work arrangements, so the heterogeneity in dependence between parents' earnings is instead captured through the work arrangement-specific variances of the income shocks.

observables and labor supply choices, integrating over unobserved heterogeneity within each state. The transition parameters are $\{\boldsymbol{\iota}, \boldsymbol{\delta}, \boldsymbol{\kappa}, \nu_{hc}\}$.

4.4 Child Production Function

Child cognitive and non-cognitive skills, $D \in \{\text{cog}, \text{ncog}\}$, evolve dynamically over developmental stages g_t.[34] The evolution of each skill follows a log-linear production function with heterogeneous elasticities by parental education:

$$\begin{aligned}\log\theta_{D,g_t+1} = & \underbrace{\log A_{D,g_t}}_{\text{TFP}} + \gamma_{D,1}\underbrace{\log\theta_{D,g_t}}_{\text{own skill}} + \gamma_{D,2}\underbrace{\log\theta_{D',g_t}}_{\text{cross skill}} + \underbrace{\gamma_{D,3}\log\tau^w_{g_t} + \gamma_{D,4}\log\tau^w_{g_t}\times\theta^w}_{\text{mother's time}} \\ & + \underbrace{\gamma_{D,5}\log\tau^h_{g_t} + \gamma_{D,6}\log\tau^h_{g_t}\times\theta^h}_{\text{father's time}} \\ & + \underbrace{\gamma_{D,7}\,cc^{PT}_{g_t} + \gamma_{D,8}\,cc^{FT}_{g_t} + \gamma_{D,9}\,cc^{PT}_{g_t}\times\theta^w + \gamma_{D,10}\,cc^{FT}_{g_t}\times\theta^w}_{\text{formal childcare}} + \gamma_{D,11}\underbrace{sib_{g_t}}_{\text{siblings}} + \nu_{D,g_t+1}\end{aligned} \tag{14}$$

Each skill depends on its own and the cross-domain lagged skill, both parents' time, formal childcare, and the presence of siblings, plus a mean-zero production shock ν_{D,g_t+1}. Formal childcare enters as separate part-time and full-time, and the college interactions allow the productivity of each childcare intensity to vary with maternal education. Because the elasticities $\gamma_{D,\cdot}$ are indexed by skill domain, maternal time, paternal time, and formal childcare can have different returns to cognitive versus non-cognitive skills; this domain-specific productivity is what makes the allocation of parental time across spouses matter for child outcomes.

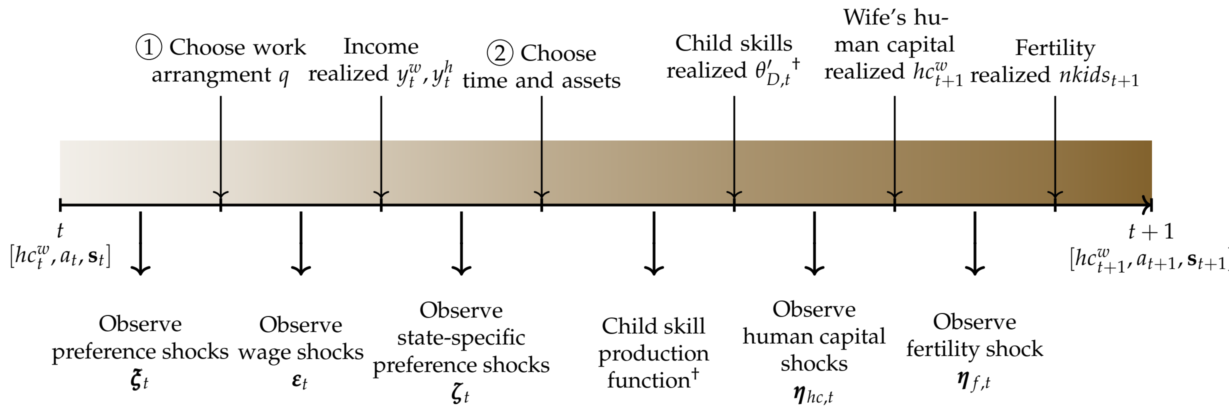
4.5 Model Timeline and Solution

Figure 5 illustrates the intratemporal timeline, including the child skill realization step that occurs after Stage 2 allocations are finalized. The model is initialized with state vector $[\theta^w, \theta^h, hc^w_1, k_1, g_1, a_1]$, where wife's initial human capital $hc^w_1 \sim F(hc^w \mid \theta^w, \text{age}^w_1)$ is drawn conditional on her education and entry age. Note that there are four dynamic states: mother's human capital, child cognitive and non-cognitive skills, and assets. Each period follows a two-stage intratemporal structure.

Stage 1 The household receives realizations of the preference shocks ξ_t and decides the optimal work arrangement choice. The ex-post value function for Stage 1 is:

$$\tilde{\mathcal{V}}^{s1}(\Omega^{s1}, \boldsymbol{\xi}) = \max_{q\in\mathcal{Q}}\left\{\mathcal{V}^{s2}_q(\Omega^{s1}) + \xi_q\right\} \tag{15}$$

[34]The subscript g_t is used for notational consistency, even though the model is indexed by the mother's age t. The child production function becomes relevant only after a fertility shock occurs, such that $g_t \in \{1, 3, 5\}$ denotes the child's age in subsequent developmental periods.



†Applies only when a young child is present; childless couples skip the childcare choice in Stage 2 and this step.

Figure 5: Intratemporal Timeline

Because preference shocks are Type-I Extreme Value, the conditional choice probability has a closed-form logit representation (Rust, 1987):

$$p(d_q = 1|\Omega^{s1}) = \frac{\exp(\mathcal{V}_q^{s2}(\Omega^{s1})/\sigma_\xi)}{\sum_{q\in\mathcal{Q}} \exp(\mathcal{V}_q^{s2}(\Omega^{s1})/\sigma_\xi)} \tag{16}$$

and the ex-ante value function is written as:

$$\mathbb{E}_{\boldsymbol{\xi}}[\tilde{\mathcal{V}}^{s1}(\Omega^{s1}, \boldsymbol{\xi})] = \sigma_\xi \log\Big(\sum_{q\in\mathcal{Q}} \exp\Big(\frac{\mathcal{V}_q^{s2}(\Omega^{s1})}{\sigma_\xi}\Big)\Big) + \sigma_\xi \gamma_E \tag{17}$$

where σ_ξ is the scale of the preference shocks, common across all alternatives, and γ_E is the Euler–Mascheroni constant.

Stage 2 Conditional on the work arrangement, the household draws wage shocks $\boldsymbol{\varepsilon}$ and state-specific preference shocks $\boldsymbol{\zeta}$, then maximizes flow utility over combination of discretized choices $r \in \mathcal{R}$. With a young child, the choice set adds formal childcare, $r = (n^w, n^h, \tau^w, \tau^h, a', cc)$, and the child skill production function enters the problem:

$$
\begin{aligned}
\tilde{V}^{s2}_q(\Omega^{s2},\boldsymbol{\zeta}) = \max_{r\in\mathcal{R}}\Big\{\hat{V}^{s2}_{q,r}(\Omega^{s2}) + \zeta_{q,r}\Big\} &= \max_{r\in\mathcal{R}}\Big\{\mathcal{U}(\cdot) + \zeta_{q,r} + \beta\mathbb{E}[V^{s1}(\Omega^{s1,\prime})|\Omega^{s2},r]\Big\} \\
\text{s.t}\quad (\mathrm{TC}^w)\quad & l^w + \tau^w + n^w = \bar{T} \\
(\mathrm{TC}^h)\quad & l^h + \tau^h + n^h = \bar{T} \\
(\mathrm{Sup})\quad & \tau^w + \tau^h + \bar{\tau}(cc) \geq \bar{T}^{cc}, \quad cc \in \{0, PT, FT\} \\
(\mathrm{BC})\quad & C + M + a' = (1+r)\cdot a + \mathcal{T}(Y^w, Y^h, \mathbf{x}) \\
(\mathrm{I})\quad & Y^i_j = y^i_j \cdot (n^i)^{\eta_{i,j}}, \quad j \in \{PE, SE\} \\
(\mathrm{IC}^h)\quad & \log y^h_j = \phi^h_{j,0} + \phi^h_{j,1}\theta^h + \phi^h_{j,2}\, t + \phi^h_{j,3}\, t^2 + \varepsilon^h_j \\
(\mathrm{IC}^w)\quad & \log y^w_j = \phi^w_{j,0} + \phi^w_{j,1}\theta^w + \phi^w_{j,2} hc^w + \varepsilon^w_j \\
(\mathrm{HC}^w)\quad & hc^w \quad \text{subject to transitions} \\
(\mathrm{CPF}^D)\quad & \log\theta'_D \quad \text{evolves according to Equation 14}
\end{aligned}
\tag{18}
$$

The conditional choice probability takes the same logit form:

$$
p(d^r = 1|\Omega^{s2}, q) = \frac{\exp(\hat{V}^{s2}_{q,r}(\Omega^{s2})/\sigma_\zeta)}{\sum_{r'\in\mathcal{R}}\exp(\hat{V}^{s2}_{q,r'}(\Omega^{s2})/\sigma_\zeta)} \tag{19}
$$

and the ex-ante value function is:

$$
\mathcal{V}^{s2}_q(\Omega^{s2}) \equiv \mathbb{E}_{\boldsymbol{\zeta}}[\tilde{V}^{s2}_q(\Omega^{s2},\boldsymbol{\zeta})] = \sigma_\zeta \log\Big(\sum_{r\in\mathcal{R}}\exp\Big(\frac{\hat{V}^{s2}_{q,r}(\Omega^{s2})}{\sigma_\zeta}\Big)\Big) + \sigma_\zeta\gamma_E \tag{20}
$$

where σ_ζ is the scale of the state-specific preference shocks, common across alternatives.

Human capital updates at the end of each period, after Stage 2. The household's continuation value therefore integrates over two sources of uncertainty, the stochastic transition of the wife's human capital and the fertility shock; child skills evolve in parallel but drop out of the choice problem by log-additivity:

$$
\begin{aligned}
\mathbb{E}\Big[V^{s1}(\Omega^{s1,\prime})\,\Big|\,\Omega^{s2}, r\Big] = \sum_{hc^{w\prime}\in\{L,M,H\}} \Pi\big(n^w(r)\big)_{hc^w,\,hc^{w\prime}}\Big\{ & p^w_t\, V^{s1}\big(a',\, hc^{w\prime},\, \Omega^{s1,\prime}\,\big|\,\text{yes child}\big) \\
& + \big(1-p^w_t\big)\, V^{s1}\big(a',\, hc^{w\prime},\, \Omega^{s1,\prime}\,\big|\,\text{no child}\big)\Big\}
\end{aligned}
$$

To solve the model, I use backward induction from the terminal period T, using the terminal value in Equation 2. For tractability, I restrict backward recursion to reachable states, given each household's initial data observation.[35] At each period, I first solve Stage 2, computing

[35]For example, a wife observed at age 35 with two children cannot have a third birth in the already-completed periods; the reachable state space conditions on observed initial conditions.

conditional choice probabilities and expected value functions over the discrete grids of choices $(n^w, n^h, \tau^w, \tau^h, a')$ given realized income shocks, then integrate over income shocks to obtain the Stage 1 expected value function over work arrangements. The Stage 1 value at period t serves as the continuation value entering period $t-1$. Because of the log-additivity in utility, the value functions are additively separable in the skill states (Appendix E.1), so optimal parental investment decisions are independent of the current skill level in each period. Hence, I solve the model over two dynamic states (hc^w_t, a_t) instead of four, making the model more tractable. I provide more computational details in Appendix G.

5 Identification and Estimation Strategy

This section describes the identification and estimation strategy for the structural parameters, those governing the child skill production functions, household preferences, parental income processes, and wife's human capital accumulation.

5.1 Identifying the Child Production Function

The child production function estimation faces two econometric challenges: measurement error in observed skill measures and endogeneity of parental inputs. First, cognitive and non-cognitive skills are latent and observed only through assessment scores that face measurement error. Following Agostinelli and Wiswall (2025), I specify the measurement system for each skill domain $D \in \{cog, ncog\}$ at child age g_t as:

$$\log \widetilde{\theta}^m_{D,g_t} = \mu^m_{\theta_D,g_t} + \alpha^m_{\theta_D,g_t} \log \theta_{D,g_t} + \epsilon^m_{\theta_D,g_t} \tag{21}$$

where $\log \widetilde{\theta}^m_{D,g_t}$ is observed measure m, $\log \theta_{D,g_t}$ is the latent skill, and ϵ^m is measurement error, uncorrelated across measures. I normalize one reference measure per domain to fix scale and location ($\mu^1_{\theta_{D,g_t}} = 0$, $\alpha^1_{\theta_{D,g_t}} = 1$). The loading on the second measure is identified via an instrument Z_D correlated with $\log \theta_{D,g_t}$ but orthogonal to measurement error:[36]

$$\alpha^2_{\theta_{D,g_t}} = \frac{\mathrm{cov}(\widetilde{\theta}^2_{D,g_t}, Z_D)}{\mathrm{cov}(\widetilde{\theta}^1_{D,g_t}, Z_D)} \tag{22}$$

I use lagged cross-domain skills as Z_D: lagged non-cognitive scores for the cognitive skills and vice versa.

[36] The cognitive skill is measured by the gross-motor scale of the Denver Developmental Screening Test at 9 months, the British Ability Scales (BAS) Naming Vocabulary score at ages 3 and 5, and the BAS Word Reading score at age 7. The non-cognitive skill is measured by the mood scale of the Carey Infant Temperament Scale at 9 months and the Strengths and Difficulties Questionnaire total difficulties score at ages 3, 5, and 7. The second same-domain measure, which is the instrument for identification, is the Bracken School Readiness score at age 3 and the BAS Pattern Construction score at ages 5 and 7 for cognitive skills, and the emotional self-regulation score for non-cognitive skills. All scores are standardized within wave using survey weights, and the 9-month measures enter the production function as the lagged skill for the age-3 outcome.

The second concern is input endogeneity. Parental inputs (time and formal childcare) are subject to the household constraints. Households who invest differently may differ in unobserved determinants of child skills, which the model does not rule out. In particular, the channel of formal childcare goes through the supervision constraint, which ties its take-up to labor supply and it responds to the price of care and in turn couples it to parental time. Therefore, I estimate the child production function instrumenting (τ^w, τ^h, cc) with FORTAX-simulated instruments spanning distinct UK policy channels via joint GMM. I provide more details on the exogeneity of parental inputs and estimation of child skills in Appendix E.3.

5.2 Identifying Preferences, Income, and Human Capital

All parameters are jointly identified by the full set of moments, but distinct sources of variation discipline distinct parameters. Appendix H develops the formal identification and Table 2 provides the moment-parameter mapping. Here, I describe the general identification strategy.

Preferences The preference weights are identified from how households choose hours worked, home production hours, and formal childcare as their budget and child status vary. The degree to which households sort into higher hours as the net-of-tax return to work rises identifies the leisure weights. How households divide time between home production and leisure, and how that division shifts with the arrival of a young child, identifies the home production weights. The workload interactions are identified separately from the levels by the within-person slope of home production in own market hours. With the child production function estimates, the same allocation identifies the child-skill weights α_{cog} and α_{ncog}, since the child production function fixes how much skill that time produces. Shifts in work arrangement shares around childbirth and the variation in part-time versus full-time childcare take-up identify the work arrangement amenities and formal childcare valuations.

Income and Human Capital Log hourly income by work arrangement and education identifies the earnings level, the within-arrangement variance of earnings identifies the earnings volatility, and the gradient of log weekly earnings in log hours within an arrangement identifies the curvature $\eta_{i,j}$ that prices flexibility. The within-couple earnings covariance identifies the shock correlation. Wife's income dynamics identify her human capital process. Re-entry earnings after a non-working spell pin down the depreciation rate δ_{NW}, and the income growth among continuously employed wives identifies the appreciation rates from the low and medium states, κ_L and κ_M, and the working depreciation rate δ_{FT}. The gap between full-time and part-time growth identifies the hours curvature ν_{hc}, since lower hours worked build less human capital. For initial human capital, the mean entry income identifies only the mean, so I add the standard deviation of entry earnings by education as a second moment to separate the low and medium states. Since that dispersion combines the human capital premium with transitory income noise, I use the lagged auto-covariance of the wife's income to separate the two, which links to the persistent component to human capital.

5.3 Estimation Procedure

Estimation relies on a fully specified parametric model and proceeds in two stages. First, I preset a subset of parameters: the discount factor $\beta = 0.98^2$ and the gross return on assets $(1+r)^2$ with annual risk-free interest rate $r = 0.0469$, each compounded over the two-year model period.[37] In the first stage, I estimate directly from the data the probability of child arrival and the parameters of the tax-and-transfer system based on family composition and parental labor market characteristics (Equation 6). The child skill production function is estimated via joint GMM, correcting for skill measurement error and endogeneity of parental inputs.

In the second stage, I recover the remaining structural parameters using Method of Simulated Moments (McFadden, 1989). The estimation targets empirical moments capturing key patterns in work arrangements, earnings, and time allocations across households and the presence of a young child. Let $\mathcal{M}$ be the vector of sample moments. For each of S simulated paths of household choices over the wife's age, I compute the corresponding simulated moments and match them to their data counterparts. The simulated moment vector $\widetilde{\mathcal{M}}_S(\Theta)$ is determined by the vector of structural parameters Θ and the draws used in the data generating process. The SMM estimator minimizes the weighted distance between simulated and actual moments:

$$\widehat{\Theta}_{S,W} = \underset{\Theta}{\operatorname{argmin}} \Big(\mathcal{M} - \widetilde{\mathcal{M}}_S(\Theta)\Big)' W \Big(\mathcal{M} - \widetilde{\mathcal{M}}_S(\Theta)\Big) \tag{23}$$

where W is a symmetric, positive-definite weighting matrix.[38]

6 Estimation Results

6.1 Child Skill Production Function Estimates

The GMM estimates in Table 3 show that child cognitive and non-cognitive skills are persistent, so early gaps persist and transmit across domains. The productivity of maternal time, strongest for cognitive skills and for college-educated mothers, is consistent with maternal time being a quantitatively important early input (Del Bono et al., 2016; Del Boca, Flinn, and Wiswall, 2014), with educational activities with parents being the most productive use of a child's time (Fiorini and

[37]For the UK, Blundell, Pistaferri, and Saporta-Eksten (2016) assume a discount factor of 0.98 and Bolt et al. (2023) and Jordà et al. (2019) set the interest rate to $r = 0.0469$. Weekly earnings and hours are annualized by the fifty-two weeks in a year, so income, consumption, and the childcare cost $p_{cc}\bar{\tau}$ are annual amounts, while the discount factor and asset return compound over the two-year period, an annual-flow, biennial-dynamics timing.

[38]I use the diagonal of the inverse of the variance of the empirical moments. Since the moments are computed using two separate data sources, I calculate the variance-covariance matrix for each dataset independently. In forming the final weighting matrix, I assume that the datasets are statistically independent, which means the off-diagonal blocks representing covariances between the two datasets are zero. Additional computational details are in Appendix G.

Table 2: Moments and Parameter Identification

Moments	Parameters
Work arrangement choice	
Couple's and household work arrangement shares by child bin	$\alpha^w_{j,1}$, $\alpha^w_{j,2}$, $\alpha^h_{SE,1}$, $\alpha^h_{SE,2}$
Labor supply and time allocation	
Couple's hours worked by child bin	α^i_l, α^i_τ, α_{cog}, α_{ncog}
Couple's home production by child bin Within-person slope of home production on hours worked	α^i_τ, $\alpha^i_{\tau,k}$, $\alpha^i_{\tau,n}$
Income processes	
Couple's log hourly income by work arrangement and education (mean, std)	$\phi^w_{j,k}$, $\phi^h_{j,k}$
Husband's log hourly income by work arrangement and age (mean, std)	$\phi^h_{j,k}$
Couple's log hours by work arrangement (mean, std)	$\eta_{w,j}$, $\eta_{h,j}$
Correlation between husband's and wife's income	ρ
Human capital dynamics	
Wife's initial log hourly income by education (mean, std)	$\boldsymbol{\iota}$, $\phi^w_{j,2}$, σ^w_j
Entry-age slope of wife's initial log income by education	ι_{age_1}
Within-person lag auto-covariance of wife's log income	$\phi^w_{j,2}$, σ^w_j
Wife's log income after non-employment vs employment spells	δ_{NW}
Wife's log income growth, continuously employed by age Std. of wife's log income growth	κ_L, κ_M, δ_{FT}
Wife's income growth gap, full-time vs lower hours	ν_{hc}
Childcare choices	
PT/FT childcare usage by wife's work arrangement	$\alpha_{cc,PT}$, $\alpha_{cc,FT}$, $\alpha^{ed}_{cc,PT}$, $\alpha^{ed}_{cc,FT}$, $\alpha^w_{j,3,PT}$, $\alpha^w_{j,3,FT}$

Note: BHPS and MCS are used to compute data moments. The BHPS, with its longitudinal panel structure, provides rich life-cycle information on household work arrangements, income, and time investments. The MCS is used for estimating the child skill production functions and moments related to formal childcare usage. PE denotes paid employment; SE denotes self-employment. Parameters follow the notation of Section 4; $i \in \{w, h\}$ for wife and husband, and $j \in \{PE, SE\}$ indexes the work arrangement, and bold symbols denote vectors.

Keane, 2014), and with educated mothers composing their time to match the child's developmental stage (Kalil, Ryan, and Corey, 2012; Hsin and Felfe, 2014). The mother-father productivity ranking matches Del Boca, Flinn, and Wiswall (2014).

The formal childcare estimates reproduce the central finding of the quasi-experimental literature:[39] returns concentrate among less advantaged children (Cornelissen et al., 2018; Felfe and Lalive, 2018; Havnes and Mogstad, 2015; Kottelenberg and Lehrer, 2017), children of advantaged families gain little or lose (Baker, Gruber, and Milligan, 2008; Fort, Ichino, and Zanella, 2020), and the UK part-time entitlement itself shows small, fading average effects (Blanden et al., 2016). For part-time versus full-time, I find returns concentrating in part-time care for non-college children, consistent with the German expansion of full-day care lowered children's socio-emotional well-being relative to half-day care (Felfe and Zierow, 2018). The sibling coefficients match resource dilution for cognitive skills (Blake, 1981) and the social-skill benefit of siblings for non-cognitive skills (Downey and Condron, 2004).

Table 3: Child Production Function Estimates

	Cognitive$_{t+1}$		Non-Cognitive$_{t+1}$	
Variable	Coef	SE	Coef	SE
TFP intercept	−1.136	0.565	−0.846	0.501
Cognitive	0.748	0.089	0.149	0.082
Non-Cognitive	0.154	0.073	0.721	0.069
log(Mother's time)	0.317	0.127	0.209	0.123
log(Mother's time)× College	0.111	0.021	0.103	0.021
log(Father's time)	0.059	0.296	−0.028	0.262
log(Father's time)× College	0.045	0.020	0.058	0.019
PT Childcare	1.117	0.445	1.624	0.409
FT Childcare	0.675	0.286	0.496	0.268
PT Childcare × Mother's college	−1.120	0.438	−1.657	0.400
FT Childcare × Mother's college	−0.517	0.265	−0.283	0.251
Sibling	−0.168	0.043	0.072	0.040
Observations	6,726			

Note: Joint GMM estimates of the child production function. The system stacks four first-stage moment conditions for the parental inputs with the second-stage CPF moment condition, correcting measurement error in lagged skills via cross-domain skill instruments and endogeneity of τ^w, τ^h, cc via a control function built from FORTAX-simulated instruments from distinct UK policy channels; see Appendix E.2 for instrument construction, moment conditions, and diagnostics. Cognitive skills are measured using the Denver Developmental Screening Test at 9 months and the British Ability Scales (BAS) verbal scores at ages 3 and 5. Non-cognitive skills are measured using the Carey Temperament Assessment Tool at 9 months and the Total Difficulties Score from the Strengths and Difficulties Questionnaire (SDQ) at ages 3–7. All skill measures are standardized by age. Standard errors are analytical two-step sandwich standard errors clustered at the child-household level.

[39] The child production function treats formal childcare as homogeneous in quality. Both WTC and the age-3 free entitlement operate within the regulated sector so if families sorting into higher-quality settings also invest more on unobservables, the childcare coefficients would partly reflect quality-driven selection.

6.2 Preferences, Income, and Human Capital Estimates

The utility estimates in Table 4 show both spouses value leisure, and a young child sharply raises the time cost of work, with large negative young-child interactions (Attanasio, Low, and Sánchez-Marcos, 2008; Blundell et al., 2016). The wife's direct leisure weight is lower than the husband's, consistent with time-use evidence that mothers' non-market hours are predominantly home production and care while fathers' are predominantly leisure. The workload interactions confirm the gender asymmetry: the wife's is negative, so her home time is crowded as her market day lengthens and she cuts home production more than proportionally when moving toward full-time work, while the husband's is small and positive, matching his flat home-production response in the data. Both parents place positive utility weights on the child's cognitive and non-cognitive skills, and these weights are what sustain home production against its costs because the skill return to home time must offset the forgone leisure and the young-child disutility of home production. The value for leisure is lower for self-employment than that of paid employment, which rationalizes self-employment at lower average hourly income. This includes self-employment's non-pecuniary benefits (Hamilton, 2000; Hurst and Pugsley, 2011) and of workers' willingness to pay for schedule control (Mas and Pallais, 2020; Lachowska et al., 2026). Together with the hours-earnings non-linearity discussed below, these preferences produce the observed gender asymmetry in self-employment hours. Part-time childcare carries a large base disutility for non-college mothers that the college interaction offsets, consistent with the child production function estimates, where college-educated mothers face near-zero net skill returns from part-time care and so need less compensating disutility to rationalize their take-up. The wife's human capital enters utility positively, capturing dynamic returns to labor market attachment beyond current earnings.

The income estimates in Table 5 confirm the earnings patterns in Section 3. Paid employment pays a higher baseline and rises with wife's human capital, while self-employment pays a lower baseline but offers larger returns to college for women, and earnings dispersion is substantially higher in self-employment for both spouses (Giupponi and Xu, 2020). The hours-earnings schedules of the two arrangements bend in opposite directions for women. Paid-employed wives face a mildly convex schedule, so hourly earnings rise with hours and part-time carry a pay penalty Goldin (2014). Self-employed wives face a concave schedule, with curvature well below one, so their hourly earnings fall as hours rise. Flexible work therefore offers diminishing returns to long hours while rigid work rewards them, which is why self-employed wives choose shorter, more flexible hours and why paid employment pulls hours upward. Men face the same convex paid-employment schedule and long hours earn the premium, whereas a near-proportional schedule for self-employment as they have access to longer hours with no cost in earnings. Between spouses, the positive correlation in earnings shocks reflects their shared exposure to common shocks.

The wife's human capital estimates in Table 6 indicates that her human capital depreciates sharply when she is not working and barely at all under full-time employment, so labor-market attachment is the key mechanism for preserving earning capacity. This relates to the motherhood penalty because an unemployment spell around childbirth depresses the wife's human-capital

Table 4: Preference Parameter Estimates

Description	Coef	SE
Wife's leisure	0.111	0.035
Husband's leisure	0.499	0.059
Wife's home production	-0.055	0.017
Wife's home production × own hours	-0.173	0.017
Wife's home production × young child	-1.717	0.134
Husband's home production	-0.052	0.008
Husband's home production × own hours	0.054	0.012
Husband's home production × young child	-0.505	0.058
Child's cognitive skills	0.258	0.099
Child's non-cognitive skills	0.386	0.107
Childcare		
PT childcare	-8.044	0.724
FT childcare	-3.062	0.217
PT childcare × college	7.626	0.707
FT childcare × college	2.186	0.192
Work arrangement interactions		
Wife PE × leisure	0.065	0.005
Wife PE × young child	-0.503	0.049
Wife PE × PT childcare	0.031	0.039
Wife PE × FT childcare	-0.114	0.039
Wife SE × leisure	-0.106	0.008
Wife SE × young child	-0.391	0.083
Wife SE × PT childcare	0.035	0.085
Wife SE × FT childcare	-0.082	0.065
Husband SE × leisure	-0.103	0.010
Husband SE × young child	-0.131	0.033
Wife's human capital terminal value	4.788	0.718
Variance of preference shock: Stage 1	0.530	0.022
Variance of preference shock: Stage 2	0.063	0.004

Note: The own-hours rows report the workload interactions $\alpha^{i}_{\tau,n}$, the change in the home production weight as own market hours move from zero to the largest menu option $\bar{n}$. The reference category for childcare base parameters correspond to non-working, non-college mothers. College interactions capture differential childcare valuation by maternal education, reflecting that the child production function implies different skill returns to childcare by education. Work arrangement interactions are deviations from the non-working for the wife and paid employment for the husband. Details for computing standard errors are explained in Appendix G.5. PE denotes paid employment; SE denotes self-employment.

Table 5: Couple's Log Hourly Income Process Parameters

Description	Coef	SE
Wife PE		
Constant	0.968	0.058
College	0.374	0.067
Human capital	0.530	0.030
Standard deviation	0.181	0.027
Curvature	1.073	0.040
Wife SE		
Constant	-0.147	0.115
College	0.698	0.383
Human capital	0.220	0.153
Standard deviation	2.902	0.156
Curvature	0.655	0.116
Husband PE		
Constant	0.423	0.014
College	0.408	0.021
Wife's age	0.090	0.001
Wife's age^2/100	-0.110	0.003
Standard deviation	0.423	0.061
Curvature	1.068	0.020
Husband SE		
Constant	1.377	0.240
College	0.363	0.102
Wife's age	-0.002	0.007
Wife's age^2/100	-0.029	0.014
Standard deviation	2.863	0.115
Curvature	0.988	0.049
Correlation between husband's and wife's income shock	0.187	0.043

Note: Wife wage equations depend on education and a three-level human capital state (low, medium, high); husband wage equations follow a deterministic age profile. Details for computing standard errors are explained in Appendix G.5. PE denotes paid employment; SE denotes self-employment.

state, and the loss has persistent effects.

Table 6: Wife's Human Capital Parameters

Description	Coef	SE
Depreciation when not working	0.524	0.021
Depreciation when working at maximum hours	0.029	0.020
Curvature	0.360	0.044
Appreciation from low	0.335	0.024
Appreciation from medium	0.132	0.039
Initial Human Capital Probabilities		
Non-college		
P(low HC)	0.409	0.128
P(medium HC)	0.330	0.118
P(high HC)	0.261	0.126
College		
P(low HC)	0.088	0.132
P(medium HC)	0.436	0.344
P(high HC)	0.476	0.288
Entry-age gradient (multinomial logit)		
Age at entry × low HC	2.245	1.525
Age at entry × medium HC	1.672	1.023

Note: The initial human capital probabilities are the entry-state shares implied by the estimated multinomial logit, computed by education for a woman entering the labor market at the youngest age, 21; their standard errors follow from the delta method. The entry-age gradient rows report the raw logit coefficients on age at entry which shift these shares as entry age rises. Details for computing standard errors are in Appendix G.5.

6.3 Model Fit

The model fits the work-arrangement, hours, home-production, and income-distribution moments well across households and child-presence states. It also reproduces untargeted moments, including education-specific work arrangement shares, hours and wages of households with older children, as well as labor supply elasticities. The first-stage tax-and-transfer and fertility processes fit the data well (Appendix I) and Appendix J documents the comparisons.

6.4 Model Mechanisms

Three mechanisms govern how the work arrangement reaches child development: households' preferences, the productivity of their inputs, and the constraints they face. The arrangement's three prices enter through the first and third. *Mechanism 1, preferences:* spouses value leisure, home production, and their child's cognitive and non-cognitive skills, and a young child raises the cost of market work. They also value flexibility, the schedule control an arrangement provides, so the wife sorts into self-employment despite its lower pay, where flexibility is priced. *Mechanism*

2, productivity: parental inputs differ in child-skill productivity. Mothers have a comparative advantage in both domains, more so in cognitive skills, and formal childcare sits between college and non-college maternal time in productivity, so replacing maternal time with formal care trades down to an inferior input for college children and up to a superior one for non-college children. The quantity-quality trade-off therefore binds asymmetrically, and the same entry into work carries opposite child consequences. *Mechanism 3, constraints:* the supervision constraint couples spouses' time, mostly the mother's, requiring parental time and formal care to jointly cover a young child; the budget constraint is where the arrangement's pay binds, self-employment's lower earnings and higher volatility. Table K.20 shows the mapping of parameters.

I probe each mechanism with a comparative static, altering the component that governs it and re-solving while holding the rest at their estimated values. *Mechanism 1:* the valuation of child skills drives parental time. Removing child skills from utility leads parents to cut home production and skills fall, while a household that values only consumption and child skills raises them substantially, moving both parents, especially the mother, into self-employment for its flexibility. The value of leisure and the disutility of home production are what hold child investment below its potential. *Mechanism 2:* the father's time is the least productive input. Reallocating his home production to the mother raises cognitive skills, so the household loses nothing when his time is replaced, provided the mother or formal care covers the supervision it satisfied. *Mechanism 3:* the supervision constraint determines who supplies that time. Removing it lets mothers cut home production and formal childcare and move into paid employment, again differing by education. The arrangement's flexibility and pay then pull maternal labor supply in opposite directions: granting paid employment an expanded hours menu (flexibility holding earnings curvature) draws mothers into work at intermediate hours and lowers college children's cognitive skills, while raising paid-employment dispersion to the self-employment level pushes maternal non-employment up. Eliminating self-employment of its income dispersion instead draws husbands into it.

The statics show why home production concentrates on the mother: her time is productive, the father's is not and he does not adjust, and formal childcare fills the remaining gap for non-college educated mothers. The same forces make her home production insensitive to her own wage, as the child-skill motive and the supervision constraint pin it, which are determined by the elasticities.

7 Work Flexibility as a Family Policy

7.1 What Is the Value of Flexibility?

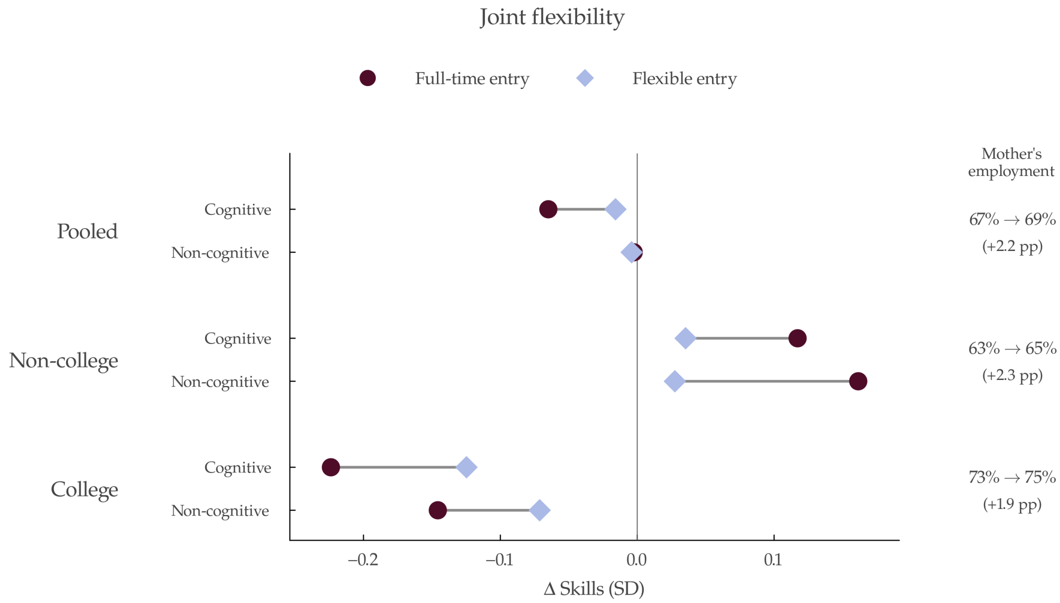
I first validate the model against Bernal (2008)'s finding that maternal full-time work with childcare lowers child cognitive skills, then evaluate the value of flexibility and its trade-off. Flexibility is a price the model can grant without self-employment's lower, riskier pay, which is the margin I isolate here. I evaluate flexibility by granting paid employed workers an expanded hours menu while holding their pay, hours-earnings curvature, and risk at their paid-employment values, so the counterfactual evaluates the hours-menu component of flexibility, control over the quantity of hours. It anchors to UK's *Right to Request Flexible Working*, which is a statutory right for workers to request changes to when and how long they work that aims to help parents combine employment with family life. The policy was introduced by the Employment Act 2002 (in force April 2003) for parents of young children, extended to all employees in 2014, and made a day-one right under the Employment Relations (Flexible Working) Act 2023. However, employers may refuse only on statutory business grounds (House of Commons Library, SN01086).[40] Flexibility reaches the child through changes in hours and I proxy the policy that extends paid-employment hours menu to shorter, intermediate, and longer hours.[41]

Figure 6 confirms that child cognitive skills fall on average when a mother enters full-time employment and uses childcare, with a larger loss for college children. Full-time entry lowers a college child's cognitive skill by 0.22 SD and non-cognitive skill by 0.15 SD but raises a non-college child's by 0.12 and 0.16 SD, respectively, so the pooled effect masks the heterogeneity across maternal education. Table K.21 decomposes in detail by work arrangement and self-employment is the gentlest entry margin for college children, since it preserves the most maternal time, which supports treating self-employment as the flexibility margin (Suh, 2026). The mothers flexibility draws into work enter below full-time, a different group from the baseline full-time entrants, so their children bear a smaller cost than those of full-time entrants, and that cost shrinks as entry hours fall. The gap is compositional, reflecting who enters and at what hours rather than a lower cost for any given full-time mother, who does not cut her hours when offered the menu. Because flexibility raises maternal employment by about two percentage points, its margin is who works, not the inputs of any given child. Averaged over all households, where most mothers already work and take the freed hours as leisure, the effect on children is close to zero, consistent with the welfare and lifetime-earnings results below.

Behind the split is a quantity-quality trade-off, between the mother's market hours and her child's skills, that binds for college mothers. Because a college mother's time is her child's most productive input and formal care is an inferior substitute, raising the quantity of her market work draws down input quality the market cannot replace. A non-college mother does not face this bind, since formal care substitutes well for the time she withdraws, so her entry raises household income at no quality cost. The flexible menu relaxes this trade-off for the mothers it draws into

[40] RRFW overlaps my sample period. About a fifth of paid-employed mothers of young children report flexitime after 2003, a limited form of flexibility that shifts when they work around fixed core hours but leaves the number of hours fixed. The model prices control over the quantity of hours, which governs time with the child and which self-employment's near-continuous menu provides but flexitime does not. A difference-in-differences around the 2003 reform finds no effect on flexitime take-up or maternal labor supply, so it does not confound the baseline. Appendix F.5 presents the estimates and external evidence on the 2014 expansion finds the response emerges as reduced hours among women, not men (Xue et al., 2025).

[41] The choice for expanded hours is explained in Appendix F.4. The counterfactual abstracts from demand-side frictions by assuming workers can freely choose among work arrangements conditional on the available hours menu. This free-choice assumption biases the estimated response in two offsetting ways. On one side, if some workers are self-employed because they are shut out of paid employment rather than because they prefer it, the model reads their choice as attachment to self-employment; it then overstates that attachment and understates how many would move into paid employment once flexibility is available, biasing the response down. On the other, the frictionless setup ignores the employer costs, refusals, and demand-side frictions that would limit access to flexible hours in practice, so it overstates take-up and biases the response up, so the net sign is ambiguous.



Note: This figure plots the effect on a child's cognitive and non-cognitive skill at age 7 of the mother entering paid work during the child's ages 1–5, relative to remaining out of work, by maternal education. Circles are full-time entry, the treatment-on-treated effect for the mothers who enter full-time in the baseline; diamonds are flexible entry, the treatment-on-treated effect for the mothers the wider menu draws into work, at the intermediate hours they choose. The right column reports the rise in the maternal employment rate.

Figure 6: Effects of Flexibility on Child Skills

work: instead of the part-time-or-full-time choice, they add market hours while her supply of home production is relatively inelastic.

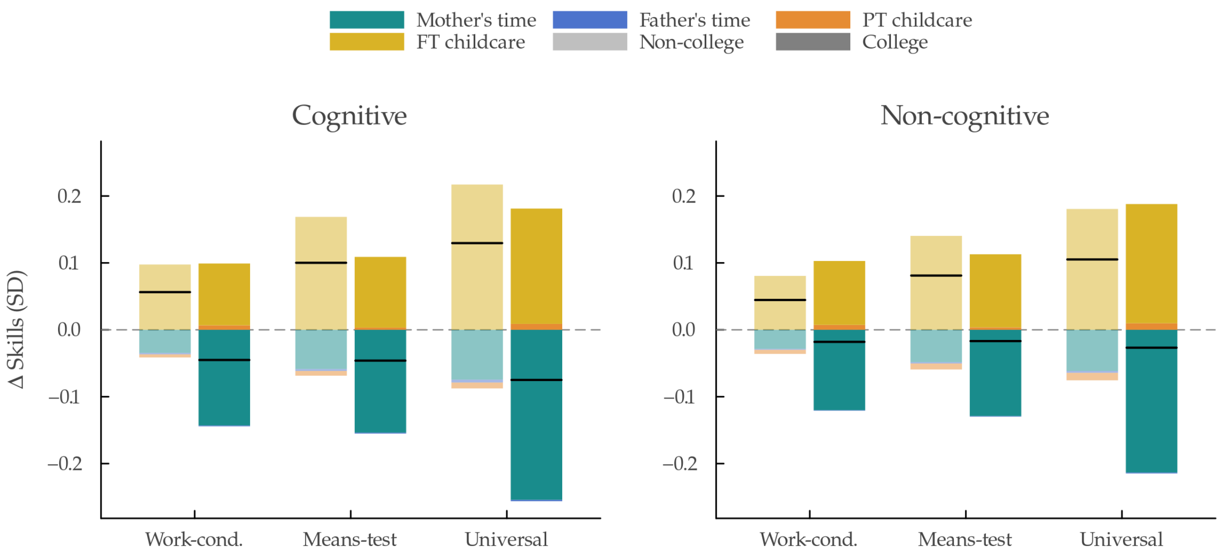
Child skills barely move because mothers protect their time with children, absorbing the added work out of leisure. Under joint flexibility, her work arrangement shifts 2.5 percentage points into paid employment and 2.2 points out of non-work, and among those working she raises her market hours by about 1.2 while cutting home production by only 0.2, with the father's time unchanged. The small residual loss concentrates on the mothers who instead use flexibility to lengthen their hours, the lower-wage earners within each education group who work longer to raise income, and among them on college children (Figure K.20).[42]

Flexibility draws parents into paid employment, and employment responds through whichever spouse has flexibility. Directing it to the wife raises her participation the most, by 3.9 percentage points; directing it to the husband moves him from self-employment into paid employment, trading self-employment's flexible hours for comparable hours at lower income volatility while her hours barely change; granting it to both draws both in (Figure K.19). Given a wider menu, parents work somewhat more and vary their hours more across periods, consistent with workers valuing longer hours, more so for men (Lachowska et al., 2026).[43]

[42]A cluster bootstrap of the production function confirms this: the pooled cognitive loss of 0.018 SD has a 95% confidence interval of $[-0.037, -0.002]$, the college loss of 0.031 SD is significant ($[-0.065, -0.006]$), the non-college effect is not distinguishable from zero ($[-0.020, +0.001]$), and college children lose 0.021 SD more than non-college children ($[-0.046, -0.003]$).

[43]It also reflects gendered sorting on hours: women concentrate in lower-hours positions (Gallen, Lesner, and Vejlin,

If flexibility is the instrument for children of college mothers, formal childcare is the instrument for children of non-college mothers. I evaluate three eligibility rules: i) *Work-conditional*, which is free for working families, withdrawn once household income exceeds £100,000, mirroring the UK's tax-free and 30 hours free childcare; ii) *Means-tested* for households in the bottom third of the net income distribution, mirroring the childcare element of the Working Tax Credit and Universal Credit and the free entitlement for disadvantaged children; iii) *Universal* for all families, mirroring the universal free early-education entitlement.[44] Figure 7 decomposes the change in cognitive and non-cognitive skills at age 7, relative to the baseline, into the four child production function input channels by a Shapley decomposition, and Table K.22 reports the underlying input changes and numerical detail.[45]



Note: Stacked-bar Shapley decomposition of the change in cognitive (left) and non-cognitive (right) skill at age 7, relative to the baseline, into the four child production function input channels, split by maternal education. Within each policy the left bar (lighter) is non-college and the right bar (solid) is college; the black tick marks the net change. Free formal childcare is divided by its income rule: work-conditional (free for working families with household income below £100,000), means-tested (free for the bottom third of net income), or universal (free for all). See Table K.22 for numerical detail.

Figure 7: Skill Effect Decomposition of Free Formal Childcare by Maternal Education

Childcare subsidies raise the skills of children of non-college mothers. Free care relaxes the household budget constraint, drawing mothers modestly from not-working into paid work when eligibility is conditional on work and lowering their home production, while part-time childcare is essentially unchanged. The subsidy operates through full-time care. Universal provision raises a

2019) and report more satisfaction than men with fewer hours (Kahn and Lang, 1995).

[44]I focus on free formal childcare as it is in current debate in the UK. Results on childcare subsidies such as lowering the price yields similar results.

[45]For each subset S of the four inputs, I re-simulate the household's skill trajectory with inputs in S at counterfactual values and the rest at baseline, and input k's Shapley contribution is the weighted average of its marginal effect across subsets that exclude k.

non-college child's cognitive skill by 0.13 SD and non-cognitive by 0.11 SD, while lowering a college child's by 0.08 and 0.03 SD, as their mothers substitute productive maternal time for less-productive formal care and cut home production. Paternal home production does not respond, because of his lower productivity in child skills and formal childcare supplies the supervision of the mother's added work.

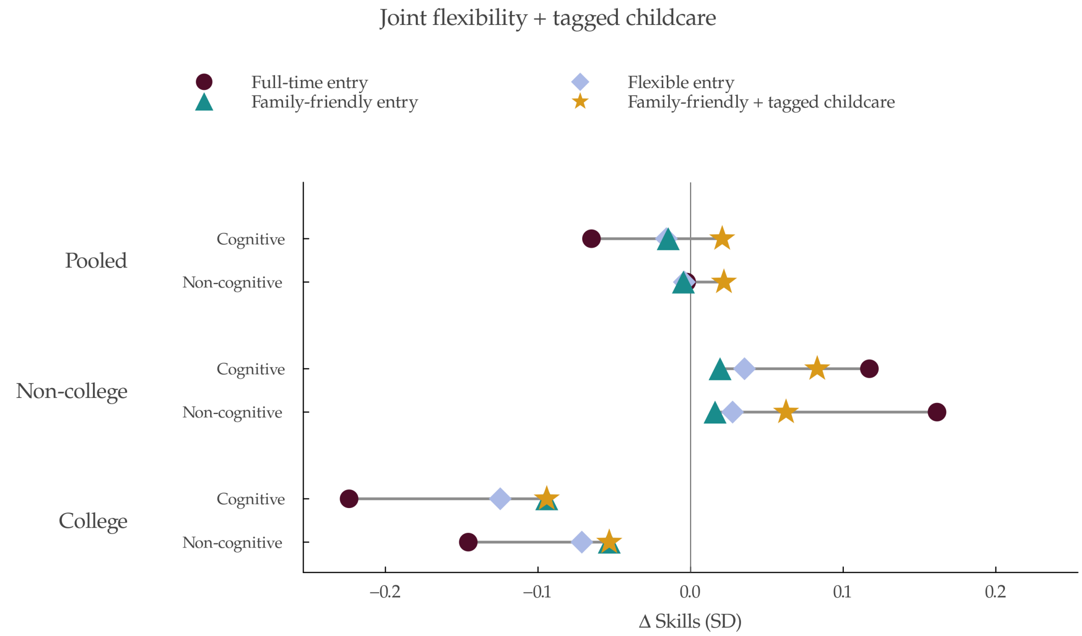
This educational gradient is consistent with the childcare literature. UK evidence confirms the gains for disadvantaged children, most sharply in Sure Start, whose GCSE benefits were six times larger for children eligible for free school meals (Carneiro, Cattan, and Ridpath, 2024), with smaller, fading average gains from the part-time entitlement (Blanden et al., 2016). The losses for advantaged children appear under full-day provision, as in Norway, Quebec, and Italy (Havnes and Mogstad, 2015; Baker, Gruber, and Milligan, 2008; Fort, Ichino, and Zanella, 2020). The same universal provision carries a large maternal payoff, which is where the trade-off is critical. In fact, the Quebec program raised the mothers' employment for two decades after the children left care and their long-run earnings by twice the participation effect (Baker, Gruber, and Milligan, 2026). Universal childcare therefore buys maternal earnings at the price of advantaged children's development. Self-employment is the other side of this trade-off because it is the margin through which mothers reconcile work and childrearing without purchased care, substituting their own flexible time for formal childcare, so extending flexibility raises maternal employment at essentially no cost to children. Childcare and flexibility are thus alternative instruments for maternal work with opposite consequences for the child.

7.2 Family-Friendly Flexibility and Targeted Childcare

Neither childcare nor flexibility as evaluated above is optimal for children: full flexibility carries a small college cost, and universal childcare cannot compensate for her time loss. I now design each instrument that raise child development. These policies have important implications on intergenerational returns and because they differ by maternal education, the design also determines whether a policy narrows or widens the skill gap children would face.

Specifically, I combine two designed instruments: i) *Family-friendly flexibility*, which offers a reduced-hours menu, closer to the shorter-hours policies now debated in the UK, such as the four-day week, than to full flexibility; and ii) *Education-tagged childcare*, which provides free care to non-college households only. Conditioning free childcare on maternal education is a form of tagging (Akerlof, 1978), where a transfer tied to a fixed, observable trait that correlates with its return is more efficient than income testing (Nichols and Zeckhauser, 1982; Mankiw and Weinzierl, 2010). Education is the better tag here because it is fixed, so a mother cannot game it by entering or exiting work as she could an income test, and it captures both household income and which children gain from formal care.[46] Table K.23 provides the input changes and numerical detail.

[46]The estimation sample is limited to two-parent households, so single mothers, whom childcare subsidies most directly target, are out of sample. Because the tag conditions on maternal education, it would still reach them, as single mothers are disproportionately non-college. They also resemble the low-income households in the model, who are more budget-constrained and use little formal care (Blundell and Shephard, 2012), so free provision would raise their take-up.



Note: This figure plots the effect on a child's cognitive and non-cognitive skill at age 7 of the mother entering paid work during the child's ages 1–5, relative to remaining out of work, by maternal education. Circles are full-time entry, diamonds full flexible entry, triangles the family-friendly flexibility, and stars the package of family-friendly flexibility with tagged childcare.

Figure 8: Effects of Family-Friendly Flexibility and Tagged Childcare on Child Skills

Figure 8 builds on the previous Figure 6 and shows that the family-friendly flexibility prevents the college quantity-quality trade-off.[47] Such flexibility raises maternal employment by about 2 percentage points by drawing her into intermediate part-time hours they would have preferred (Booth and van Ours, 2013). On its own, though, family-friendly flexibility does less for non-college children. It draws mothers into working reduced hours and away from full-time formal care, so they gain less than the children of full-time entrants. Bundling it with tagged childcare restores that productive input and delivers the non-college gain. In fact, the bundled policy makes both education groups better off. Family-friendly flexibility protects college children's maternal time, and tagged childcare supplies non-college children's formal care.

Value of Flexibility and Childcare What is the household's willingness to pay for the policies? Table 7 decomposes annual household welfare through consumption equivalent variation (Appendix L) into a *parental* component, covering consumption, leisure, work-arrangement amenities, and childcare costs, and a *child skill* component, the CEV to the household for child skill returns. The policy costs differ by instrument. The childcare subsidy has a direct fiscal cost, the subsidized

The estimated childcare gain is therefore a lower bound on their children's gain, since it is the low-care households whose children gain most from a shift into formal care.

[47]Effects are similar when flexibility is granted either only to the wife or the husband, more so for the former case (Figure K.21).

expenditure on formal childcare.[48] On the other hand, flexibility has no direct fiscal cost in the model, since employer obligations are not modeled, but treating it as zero gives an upper bound on net welfare rather than a realistic estimate.[49] Workers value flexibility, requiring a 12% higher wage to forgo ideal hours (Lachowska et al., 2026) and willing to forgo about 20% of wages to avoid employer-discretionary scheduling (Mas and Pallais, 2017), so in a competitive labor market employers must bear some cost to provide it, otherwise firms will offer full flexibility and cut wages (Flabbi and Moro, 2012). To bound net welfare, I scale wages by the 12% productivity loss that Emanuel and Harrington (2024) estimate for flexible work. I use this employer-side estimate rather than the worker-side valuations above. Because the model is partial equilibrium, the relevant object is the social cost of flexibility, so wages equal to the marginal product of labor and that cost is the employer's productivity loss.

Table 7: Policy Welfare by Maternal Education

	Welfare (£/hh/yr)					
Policy	Total	Parental	Child Skills	Cost (£)	Δ Parental (£)	Δ Child Skills (£)
Non-college mother						
Family-friendly flexibility	+4279	+4345	−66	[−42, 0]	[4303, 4345]	[−108, −66]
Tagged childcare	+831	−3280	+4111	−707	−3987	+3404
Combined	+5158	+1015	+4143	[−731, −702]	[284, 313]	[3412, 3441]
College mother						
Family-friendly flexibility	+5236	+5061	+176	[−60, 0]	[5001, 5061]	[116, 176]

Note: Population-average willingness to pay in consumption-equivalent variation over all young-child households, by the wife's education. Parental and Child Skills sum to Total. Family-friendly flexibility grants the reduced-hours menu to both spouses; tagged childcare provides free formal childcare to non-college households only; combined applies both. College households are excluded from tagged childcare, so for them tagged childcare has no effect and the combined package coincides with family-friendly flexibility, the single college row shown. Cost is the employer productivity loss at 12% of affected wages for flexibility (range [−12%, 0]) following Emanuel and Harrington (2024), or the government childcare expenditure per household-year with a young child for that education group. Δ Parental = Parental + Cost; Δ Child Skills = Child Skills + Cost, with Cost entered as the non-positive welfare adjustment shown.

These welfare values average over all households with young children, in contrast to the entry effects in Figure 8, which condition on the mothers a policy newly draws into work. In college households, most of parents already work, so family-friendly flexibility raises welfare slightly from child skills, because incumbent mothers use the reduced-hours to protect time with their children. The split then shows who captures each policy's welfare and how much they are willing to pay. For non-college households, family-friendly flexibility is a parental gain, raising parental welfare

[48]This direct cost overstates the net fiscal burden, since higher maternal employment returns part of it. Baker, Gruber, and Milligan (2026) estimate that the long-run rise in mothers' tax payments and fall in benefit receipt recovers 75 to 117 percent of the upfront cost of Quebec's universal childcare program.

[49]The model prices flexibility from the self-employment income process, which is imperfectly measured: self-employment income is bottom-coded at zero in the data, its estimated dispersion is large, and the concave wife hours-earnings curvature shapes the hours response. I therefore probe sensitivity to the self-employment process rather than bound bottom-coding directly, re-solving for both spouses under lower dispersion (standard deviations of 2.0 and 1.5) and flatter curvature (0.85 and linear). The self-employment share increases across these cases, but the flexibility results remain robust. Maternal employment gain stays between 1.8 and 2.4 percentage points, mothers still enter at intermediate hours, and the value of flexibility stays positive and within about five percent of its baseline.

by about £4,300 a year with essentially no welfare gains from child skills. Tagged childcare is its mirror image, with parents' welfare falling by about £3,300 as they work more and bear the formal-childcare disutility, offset by child-skill gains of about £4,100, at a fiscal cost near £700. The combined package delivers both. For college households, their welfare runs entirely through flexibility, with a large gain in parental welfare of about £5,000. Flexibility is worth more to college households, while non-college households gain the larger share of their welfare through their child skill gains.

Table 8: The Value of Policy by Maternal Education

Policy	Total	Channel (% of lifetime consumption)			
	(% cons.)	Consumption	Wife's time	Husband's time	Child skills
Non-college mother					
Family-friendly flexibility	+5.84	+4.82	+0.15	+0.95	−0.08
Tagged childcare	+1.07	+0.51	−5.08	+0.35	+5.29
Combined	+6.97	+5.35	−5.03	+1.32	+5.33
College mother					
Family-friendly flexibility	+5.54	+4.10	+0.13	+1.15	+0.16

Note: The value of each policy in consumption-equivalent variation, the percent of lifetime consumption a household would give up to have it, by the wife's education. The channel columns decompose the total by the utility margin it flows through: consumption (a household public good), the wife's and husband's time (leisure and home production), and child skills; the two time channels are spouse-specific, consumption and child are shared. Because the household is unitary the split is by margin, not by spouse. Means over households with at least one young-child period. College households are excluded from tagged childcare, so the combined package coincides with family-friendly flexibility for them, the single college row shown.

Table 8 shows the value of family-friendly flexibility and childcare, in percent of lifetime consumption, decomposed by utility channel and maternal education. Because household utility is unitary, I separate parental from child welfare but cannot divide the parental gain between the two spouses, since consumption is a household public good and only leisure and home production are spouse-specific. Both education groups value flexibility, but for different reasons. Non-college households value family-friendly flexibility almost entirely through consumption, the income from entering work at intermediate hours, with little from the time or child skill channels. College households value it through consumption too, but also through a positive schedule amenity and a positive child-skill channel, since the reduced-hours menu preserves the maternal time their children draw on. When the policies are combined, non-college families receive a child-skill gain from childcare on top of the income from work, and for college families, an income-and-amenity gain from flexibility.

7.3 Returns to Child Skills and Lifetime Earnings

The child skill welfare values measure parents' willingness to pay for their child's skill gain. Parents internalize only part of their child's future earnings, so I also report a gross lifetime-earnings benchmark for the skill gain. This exercise is an earnings projection, distinct from the willingness-to-pay values. I value a standard deviation of skill through its effect on lifetime earnings, following the approach of Chetty et al. (2011a) and the estimates of Mullins (2022), who finds that a one standard deviation increase raises lifetime earnings by 32% for cognitive skills and 18% for non-cognitive skills.[50] I apply this to the UK context, using a UK expected-lifetime-earnings present value of about £590,600.[51] A standard deviation of cognitive skill is then worth roughly £189,000 and of non-cognitive skill about £106,300. UK evidence confirms that both carry into adult earnings (Carneiro, Crawford, and Goodman, 2007; Hanushek et al., 2015), so these figures are an upper benchmark. The cognitive and non-cognitive projections are reported separately and are not summed without accounting for their joint return.

Table 9: Monetary Returns to Policy-Induced Child Skill Changes by Maternal Education

	Δ Skills (SD)		Lifetime earnings (£)			
Policy	Cog	Ncog	Cog	Ncog	Total	% of PV
Non-college mother						
Family-friendly flexibility	−0.00	−0.00	−200	−100	−400	−0.1
Tagged childcare	+0.13	+0.11	+24,500	+11,200	+35,700	+6.0
Combined	+0.13	+0.10	+24,100	+11,000	+35,100	+5.9
College mother						
Family-friendly flexibility	+0.01	+0.01	+2,200	+900	+3,100	+0.5

Note: Present value of the change in a child's expected lifetime earnings by the wife's education, from the population-average skill change over all young-child households, decomposed into cognitive and non-cognitive channels. The cognitive (non-cognitive) channel applies a 32% (18%) earnings return per standard deviation (Chetty et al., 2011a; Mullins, 2022) to a UK expected-lifetime-earnings present value of £590,600, about £189,000 (£106,300) per SD; the final column expresses the total as a share of that present value. Skill changes are mean changes in log skills at age 7 within each education group. College households are excluded from tagged childcare, so the combined package coincides with family-friendly flexibility for them, the single college row shown.

Table 9 translates each policy's population skill change in the present value of a child's lifetime earnings, by maternal education and split into cognitive and non-cognitive channels. For non-college children, tagged childcare is by far the largest lever, raising expected lifetime earnings by about £35,700 (6.0%), while family-friendly flexibility is near zero, so the combined package delivers essentially the childcare gain. For college children, family-friendly flexibility yields a small positive gain of about £3,100 (0.5%). The gains fall on the cognitive channel for flexibility, through the mother's time, and on both channels for childcare.

[50] Chetty et al. (2011a) report that a one-percentile-rank gain in kindergarten test scores is associated with a $132 rise in age-27 earnings in the raw data, and $94 after controlling for parental characteristics; their Project STAR randomization identifies classroom-quality effects, with a class one standard deviation higher in quality raising age-27 earnings by about 3 percent. Scaling the raw gradient by the roughly 34 percentile ranks in a standard deviation gives a cross-sectional cognitive return of about $4,500 per SD annually. Mullins (2022) estimates both the cognitive and the non-cognitive return from PSID-CDS earnings in young adulthood.

[51] Median full-time earnings of £35,000 over 45 working years, discounted at 3.5% and scaled by the 75% UK employment rate (Office for National Statistics, 2024).

8 Conclusion

As flexible work and free childcare expand, how a mother's employment reaches her child has become a central question for family and labor policy. Existing work traces flexibility and childcare to women's labor supply, earnings, and welfare; this paper is the first to quantify their consequences for child development. Using the margin between paid and self-employment, I decompose a work arrangement into three prices: flexibility, an earnings level, and earnings volatility. A structural life-cycle model then traces how that choice reaches the child and evaluates work flexibility and childcare policies.

Granting flexibility to paid employed workers without the pay penalty has small effects on children on average, which mask sharp heterogeneity by maternal education. Full-time employment of a college-educated mother is associated with child skill loss, which flexibility alleviates; for a non-college mother the sign reverses, and flexibility instead dampens the gain. The difference is a quantity-quality trade-off that binds only for college mothers because her time is her child's most productive input, and the market does not offer a good substitute for it, so adding market hours draws down child quality. Flexibility eases the trade-off by letting her enter the labor market while keeping her home-production time nearly intact. A non-college mother faces no such bind, since formal childcare substitutes well for the time she withdraws, so she can enter work at no cost to her child. Which policy reaches a child thus depends on which input is scarce, the mother's time for college children and formal care for non-college children.

In welfare terms, flexibility delivers larger parental gains than childcare, because parents value flexibility. These results speak directly to policy debate. The 2023 Employment Relations (Flexible Working) Act made the right to request flexible working a day-one right for all UK employees. My findings imply that expanding effective access to flexible hours raises maternal employment and household welfare substantially, with little effect on children on average. To reach children, the instruments must be designed and targeted by education: a family-friendly of reduced-hours menu turns flexibility's small college cost into a gain, and childcare tagged by maternal education delivers the non-college gain while sparing college households the loss universal provision imposes.

Two extensions remain for future work. The first endogenizes the supply of flexibility. Treating it as a costly amenity that firms choose to provide, in general equilibrium, would weigh its cost against its returns to children. The second endogenizes fertility. Because flexibility and childcare subsidies lower the time and money cost of children, they would shift not only investment per child but also the number of children (Davis et al., 2026), further affecting the quantity-quality margin.

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Appendix A Data and Variable Construction

Appendix A.1 British Household Panel Survey

Household Identifier: BHPS individuals are identified by an individual specific identifier across all waves. Individuals are also identified by the person number (*pno*) in household (*hidp*). Each wave, individuals have a unique combination of *pno* and *hidp*; however, household identifiers cannot be linked across waves. Since household structures can change over time (e.g. couples get divorced, the child ($age < 16$) at wave t becomes independent and forms his or her own household at wave $t+1$), BHPS does not include a longitudinal household identifier.[52]

Table A.1: BHPS Sample Selection Criteria

Criteria	Obs	Ind	HH
Original Sample	329,993	43,206	
Keep adults (age > 15)	257,502	35,570	
Keep singles or couples	195,043	24,885	
Drop couples after divorce	183,048	24,846	14,075
Keep married couples, stable family structure	174,961	24,846	14,075
Keep wife age 21–51	91,366	14,923	7,586
Drop if missing education	86,666	13,545	7,463
Drop if missing labor market characteristics	75,517	13,069	7,305
Keep working husbands	57,980	10,809	5,717
Keep households with natural children (max 3)	55,101	10,325	5,461
Drop early or late pregnancies	54,048	10,085	5,330
Keep both couples observed	49,958	9,619	4,871

Since BHPS does not provide a longitudinal household identifier, I construct one from the cross-sectional person identifiers. I restrict to reference persons and their partners, extract the woman's individual identifier (*pidp_women*) for each household-wave, and assign a sequential *hh_id* using *pidp_women* as the grouping key. This identifier is unique across couples and stable across waves as long as the woman remains in the sample. I keep only households whose children are all biological, and drop a household entirely if it ever becomes a blended or step-family.

Marital status: Marital status takes value 1 for married and cohabiting couples and 0 otherwise. I retain each household until the first observed dissolution event and drop all subsequent waves, so each retained observation comes from the stable two-parent spell before any separation. Among households that dissolve, the child is on average 6 years old at the time of separation.

Employment status: Individuals are considered working if they did paid work last week and have work. For wives, I distinguish four work arrangements: paid employed part-time (PE/PT, $\leq$ 30 hours per week), paid employed full-time (PE/FT, $>$ 30 hours per week), self-employed, and not working. The 30-hour threshold follows Manning and Petrongolo (2008). For husbands, I distinguish paid employed and self-employed; the sample conditions on husbands being always

[52] More details are described in the Understanding Society: website.

working.

Self-employment incorporation status: BHPS asks self-employed individuals whether they hire employees and the nature of their employment. I define *incorporated* as those who own a business or practice and have employees and *unincorporated* as those who own a business or practice but do not have employees or who work for others, both, or other form of self-employment. This employee-based split identifies employer versus own-account self-employment, which proxies for legal incorporation.

Educational attainment: The highest educational qualification is reported for each individual. I construct a binary indicator for college degree, which takes value 1 if the individual holds a degree or higher. Those with A-levels, GCSEs, other qualifications, or no formal education are assigned a value of 0. I construct a categorical variable *educ_level*: low for GCSE or lower, medium for A-level, and high for degree or other higher degree.

Income: BHPS asks the usual/non-usual payments and the usual/non-usual weeks of payments. I first replace non-usual payments with usual payments and compute weekly payment by dividing the gross-pay by the number of weeks received. For hourly payment, I use the number of hours worked. Self-employed individuals do not report the number of weeks so I use information on their monthly self-employed gross pay and profit. For all working individuals, I compute the weekly gross pay. The timing of occupation and income variables is described in Table A.2. I deflate earnings and wages with the CPI (2001=100).

Assets: BHPS asks households questions on home ownership and home values every year. If adequate, they are asked about whether or not they have a mortgage and the corresponding amount. Asset information is collected every 5 years (wave 5, 10, 15), where individuals are asked in greater detail on financial issues. Assets are categorized into savings, investments, and debt. Savings include both regular and non-regular savings. Investment includes money from national savings certificates, premium bonds, unit trusts, personal equity plans, shares in the UK or foreign countries, NS/BS insurance bonds, and other investments or securities. Debt include owing money from hire purchase, personal loan, credit cards, mail order purchase, DSS social fund loan, individual loans, and other. Further details on the survey instruments are described in Table A.3. I measure savings, investments, and debt as point-in-time stocks at the wealth waves (5, 10, and 15), and construct net financial assets = savings + investment − debt, the model's asset state. Net home value is the home value (for homeowners) minus the outstanding mortgage balance. Home ownership in the sample is 80% for couples with children and 83% for couples without, consistent with Crossley and O'Dea (2010) (Table 3.7). The share owning outright is low, as households with dependent children are less likely to have paid off a mortgage. For missing home ownership between waves, I impute to the closest year.

Table A.2: Timing of Occupation and Income

Variable	**Question**	**Timing**
Work Arrangement		
Did paid work last week	*"Did you do any paid work last week that is in the seven days ending last Sunday either as an employee or self employed?"*	*t*
Employee or self-employed: current job	*"Are you an employee or self-employed?"*	*t*
No. of hours normally worked per week	*"Thinking about your (main) job, how many hours, excluding overtime and meal breaks, are you expected to work in a normal week?"*	*t*
Occupation (SOC): current main job	*"What was your (main) job last week? Please tell me the exact job title and describe fully the sort of work you do."*	*t*
Income		
Gross pay at last payment	*"The last time you were paid, what was your gross pay that is including any overtime, bonuses, commission, tips or tax refund, but before any deductions for tax, national insurance or pension contributions, union dues and so on?"*	
Pay period (weeks): last gross pay	*"How long a period did that (pay) cover?"*	
Was last gross pay the usual amount	*"Your gross pay last time was (amount at E23a). Is this the amount you usually receive (before any statutory sick pay or statutory maternity pay)?"*	
Usual pay	*"How much are you usually paid?"*	
Usual pay: pay period (weeks)	*"What period does that (pay) cover?"*	
Total gross earnings in last 12 months	*"In all, before tax and other deductions, how much have you earned in the past 12 months?"*	
Self-Employment		
S/emp: hours normally worked per week	*"How many hours in total do you usually work a week in your job?"*	*t*
Monthly self employed profit	*"Computes a monthly self-employed profit variable for self-employed respondents who draw up profit and loss accounts. Inapplicable for those who are not self-employed, and the self employed who do not draw up accounts. (cf. AJSPAYG). Includes imputed data."*	
Monthly self employed gross pay	*"Computes a monthly self-employed gross pay variable if a self-employed respondent does not draw up profit and loss accounts. Inapplicable for those who are not self employed, and the self employed who draw up accounts. (cf. AJSPROF). Includes imputed data."*	

Table A.3: Asset Variables

Variable	**Question**
Amount in account(s) – regular saving	"*And about how much do you currently have in your savings account(s)? Please do not include money you have in share purchase schemes or Personal Equity Plans (PEP) schemes. ... In the last 12 months.*"
Amount in account(s) – non-reg saving	"*About how much do you currently hold in this/these account(s)? ... In the last 12 months.*"
Amount held in investments	"*Thinking of all your investments, about how much do you have invested in total? ... In the last 12 months.*" "*Which ones? National Savings Certificate*" "*Which ones? Premium Bonds*" "*Which ones? Unit trusts*" "*Which ones? Personal Equity Plans*" "*Which ones? Shares (UK or foreign*" "*Which ones? National Savings/Building Society/Insurance Bonds*" "*Which ones? Other investments, government or company securities*"
Currently owe money	"*I would like to ask you now about any other financial commitments you may have apart from mortgages and housing related loans. Do you currently owe any money on the things listed on this card (43)? ... In the last 12 months.*"
House owned or rented	"*Does your household own or rent this accommodation or does it come rent-free?*"
Amount borrowed at purchase: mortgagee	"*How much did you borrow originally when you bought the property, that is excluding any later additions to the mortgage?*"
Amount still owed on mortgage	"*About how much do you still owe on the mortgage(s) or loan(s) on this property?*"
Last total monthly mortgage payment	"*How much was your total last monthly instalment on the morgage(s) or loan(s)?*"
Value of property: home owners	"*About how much would you expect to get for your home if you sold it today?*"

Table A.4: Summary Statistics by Young Child: BHPS

		No Young Child	Young Child
Wife	Age	39.13	32.76
		(8.218)	(5.065)
	Annual pay	11044.7	7176.8
		(9199.1)	(8486.2)
	College degree	0.253	0.308
		(0.435)	(0.462)
	Work arrangement		
	Paid Employed/PT	31.81%	42.04%
	Hours worked	19.58	17.81
		(6.750)	(6.573)
	Home production	19.37	18.90
		(9.930)	(10.724)
	Paid Employed/FT	48.30%	19.34%
	Hours worked	37.61	37.42
		(3.850)	(3.698)
	Home production	11.49	12.63
		(7.113)	(7.556)
	Self-Employed	6.13%	4.99%
	Hours worked	34.03	25.69
		(17.797)	(16.676)
	Home production	16.25	18.57
		(10.320)	(13.013)
	Not Working	13.77%	33.64%
	Home production	23.98	25.66
		(13.375)	(13.789)
Husband	Age	41.49	35.01
		(8.858)	(5.601)
	Annual pay	22360.6	23469.8
		(13637.2)	(13722.6)
	College degree	0.259	0.303
		(0.438)	(0.460)
	Work arrangement		
	Paid Employed	81.74%	83.28%
	Hours worked	39.61	39.85
		(6.887)	(6.966)
	Home production	5.11	5.08
		(4.560)	(4.733)
	Self-Employed	18.26%	16.72%
	Hours worked	49.37	49.15
		(14.755)	(14.724)
	Home production	4.39	5.07
		(4.603)	(6.456)
Household	Net assets	11140.1	8094.2
		(43961.0)	(59131.0)
	Net home value	111678.8	110326.6
		(112006.9)	(109169.1)
	Annual household pay	33405.3	30646.6
		(17138.1)	(16528.7)
Observations		17,468	7,511

Note: Sample: married couples, BHPS waves 1–18, wife aged 21–51, husband always working. No young child = no child aged 0–5 present; young child = at least one child aged 0–5. Work arrangement categories sum to 100%. Home production hours from BHPS housework questions. Net assets = savings plus investment holdings minus debt. Net home value = home market value minus mortgage. All monetary values in 2001 £. Standard deviations in parentheses.

Appendix A.2 Millennium Cohort Study

For the main analysis, I use MCS Sweeps 1–4 (9 months to age 7). In Sweep 1, 18,552 children were interviewed across England, Wales, Scotland, and Northern Ireland; the sample is nationally representative. In Sweep 2, an additional 1,389 families were added for whom addresses were reached too late for Sweep 1; I do not include them. I restrict to singleton births with both parents present, and keep children observed in at least two consecutive sweeps so that skill transitions are identified. Table A.5 describes the sample selection steps, yielding 5,219 children and 28,802 parent-wave observations.

Table A.5: MCS Sample Selection Criteria

	All		Wave 1		Wave 2		Wave 3		Wave 4	
	Obs	Child	Obs	Child	Obs	Child	Obs	Child	Obs	Child
Original sample (Sweeps 1–4)	116,179	19,201	34,044	18,388	28,844	15,579	28,014	15,236	25,277	13,847
Singleton birth only	104,816	18,951	31,080	18,147	25,654	15,381	25,363	15,041	22,719	13,681
Drop if added after 2001	101,658	18,147	31,080	18,147	24,445	14,595	24,314	14,369	21,819	13,097
Both natural parents in household	86,972	15,684	27,848	14,964	21,280	11,705	20,270	10,852	17,574	9,499
Mother's age at birth 21–45	81,449	14,377	25,740	13,802	19,904	10,940	19,137	10,236	16,668	8,997
Drop if missing child development outcomes	76,773	14,125	24,550	13,117	17,921	9,770	18,496	9,858	15,806	8,476
Drop if missing child characteristics	75,931	13,936	24,281	12,966	17,749	9,673	18,294	9,746	15,607	8,367
Drop if missing parent labor characteristics	61,481	13,207	19,238	10,491	11,869	6,623	16,338	8,799	14,036	7,603
Drop if missing parent demographics	60,817	13,081	19,221	10,481	11,682	6,528	15,933	8,592	13,981	7,574
Drop if missing childcare	60,199	13,071	19,220	10,480	11,587	6,474	15,427	8,357	13,965	7,558
Drop if father not working	54,382	12,299	17,145	9,441	10,414	5,887	14,054	7,660	12,769	6,954
Drop households with half sibling	45,455	10,287	14,701	8,039	8,836	4,930	11,592	6,279	10,326	5,576
Keep if 2 consecutive waves observed	32,599	6,373	6,654	3,814	8,004	4,526	9,735	5,347	8,206	4,511
Keep both parents	28,802	5,219	5,680	2,840	6,956	3,478	8,776	4,388	7,390	3,695

Incorporated vs Unincorporated Status: The MCS does not include an explicit variable for incorporation status. I use whether the self-employed individual has employees as a proxy: *incorporated* for those with employees, *unincorporated* (sole trader) for those working alone or with partners but without employees. As in the BHPS, this identifies employer versus own-account status rather than legal incorporation directly.

Income: MCS includes gross and net income, the derived OECD equivalised income, and predicted family income. Hawkes, Plewis, and Verropoulou (2008) and Anyaegbu (2010) document missing income in the MCS and show within-household and within-individual correlation in non-response. To address this issue, MCS imputed missing and continuous income using interval regression with observable household characteristics such as housing tenure, parents combined labor market status and parents' combined highest educational attainment. They also derived both McClements scale (McClements, 1977) and OECD equivalence scale Hagenaars, De Vos, and Zaidi (1994) for family income (Centre for Longitudinal Studies, 2020).[53] Instead of using derived

[53]The OECD equivalised income scale is widely used in benefit analysis. According to the OECD scale, the first adult

variables, I calculate weekly gross pay based on the total gross pay amount and the number of periods during which it was received in the last financial year. For self-employed individuals, I only observe the net (take-home) income in the last financial year. Hourly wage is calculated by dividing weekly income by the average number of hours worked. Observations with hourly wages below the 2001 minimum wage of £3 for paid employment (Department for Business, Energy & Industrial Strategy, 2019), below £1 for self-employment, and above £150 were excluded from the analysis. All incomes are deflated to 2001 pounds.

Child Development Outcomes: Cognitive skills are measured using age-appropriate tests (Frankenburg and Dodds, 1967; Elliott, Smith, and McCulloch, 1997). At 9 months, I use the Denver Development Screening Test. At ages 3 and 5, children take the BAS Naming Vocabulary Test, in which they name objects depicted in 36 colored pictures shown by a trained interviewer. At age 7, they take the BAS Word Reading Test, in which they read aloud a series of 90 words arranged in increasing difficulty. All responses are recorded via Computer Assisted Personal Interviewing (CAPI), and I use the age-adjusted ability scores.

For non-cognitive skills, I use the Carey Infant Temperament Scale (Carey and McDevitt, 1978) at 9 months and the Total Difficulties Score from the Strengths and Difficulties Questionnaire (SDQ) at ages 3 to 7. The Carey scale measures infant temperament (the child's characteristic behavioral style and patterns of response to stimuli) and is designed for infants aged 4 to 8 months. The SDQ is a parent-completed behavioral screening questionnaire widely used in research and clinical settings (Goodman, 1997, 2001). It consists of 25 items across five components: (i) emotional symptoms, (ii) conduct problems, (iii) hyperactivity/inattention, (iv) peer relationship problems, and (v) prosocial behavior. Each item is rated on a three-point scale (not true, somewhat true, certainly true). The first four components sum to the Total Difficulties Score, where a higher score indicates more behavioral problems. Following the literature (Del Bono and Ermisch, 2009; Nicoletti and Tonei, 2020), I reverse-code the score so that a higher value reflects better socio-emotional skills.

Appendix A.3 Tax and Transfer System: Fortax

For the tax and transfer system, the FORTAX library must be first installed, which is accessible from the online library written by Shephard (2009): Github Page. From the BHPS, I first get the data on both mother and father's gross weekly income, hours worked, self-employment status, and child's age. I compute the household disposable income given the parent labor market and family characteristics.

During the sample period (2001–2006), the UK government introduced several tax reforms to support working families and reduce income inequality. In 2003, the government replaced the Working Families' Tax Credit (WFTC) with two targeted programs: the Working Tax Credit (WTC) and Child Tax Credit (CTC). While WTC provided supplemental income for low-earning

accounts for 0.67 weight, the spouse 0.33, dependent child between 14 and 18 years old 0.33, and the child aged under 14 years old 0.20. The baseline is set to 1 for couples without children. This measure is relatively complete in the data; however, it captures annual income at the household level rather than the individual level which is critical in my analysis, I use this scale to cross-validate the individually computed annual income.

Table A.6: Summary Statistics by Child Age: MCS

	9 Months	Age 3	Age 5	Age 7
Child Outcomes				
Cognitive‡	9.14	78.31	112.75	115.03
	(1.56)	(15.36)	(15.07)	(26.33)
Non-Cognitive‡	57.36	31.52	33.88	33.75
	(5.92)	(4.55)	(4.16)	(4.58)
Child Characteristics				
Female	0.49	0.49	0.49	0.49
	(0.50)	(0.50)	(0.50)	(0.50)
Parity	0.48	0.23	0.13	0.09
	(0.50)	(0.42)	(0.34)	(0.29)
Birth weight (g)	3465.80	3455.03	3442.58	3439.11
	(545.38)	(551.98)	(559.04)	(551.45)
Low birth weight (<2.5kg)	0.04	0.04	0.05	0.05
	(0.20)	(0.20)	(0.21)	(0.21)
Number of siblings	0.63	0.93	1.10	1.19
	(0.67)	(0.63)	(0.60)	(0.58)
Mother's Characteristics				
Work arrangement				
Paid Employed/PT	34.7%	34.0%	46.0%	49.1%
Paid Employed/FT	12.8%	10.1%	12.6%	15.8%
Self-Employed	4.2%	6.3%	6.7%	7.5%
Not Working	48.4%	49.6%	34.6%	27.6%
Age	31.39	33.53	35.62	37.53
	(4.36)	(4.38)	(4.45)	(4.48)
Annual gross income	7590.59	7276.52	9918.98	12132.53
	(11467.02)	(12529.34)	(15083.64)	(17789.45)
College degree	0.30	0.28	0.28	0.27
	(0.46)	(0.45)	(0.45)	(0.45)
Father's Characteristics				
Work arrangement				
Paid Employed	83.7%	82.0%	81.7%	81.8%
Self-Employed	16.3%	18.0%	18.3%	18.2%
Age	33.78	35.96	37.97	39.82
	(5.17)	(5.22)	(5.16)	(5.04)
Annual gross income	32919.07	36941.51	39098.49	46853.22
	(27178.95)	(35806.84)	(40069.74)	(79865.50)
College degree	0.30	0.28	0.29	0.29
	(0.46)	(0.45)	(0.45)	(0.45)
Household Characteristics				
Equivalized weekly family income†	430.36	441.04	441.33	452.25
	(203.25)	(216.42)	(198.80)	(193.00)
N	2,840	3,478	4,388	3,695

Note: Sample: MCS singleton births, both parents present, no change in family structure, mother aged 21–45. ‡ Raw test scores: Denver Development Screening Test (9 months), British Ability Scales (ages 3–7) for cognitive; Carey Infant Temperament Scale (9 months), SDQ Total Difficulties (ages 3–7) for non-cognitive. † OECD equivalized income scale. All monetary values in 2001 £. Standard deviations in parentheses.

working adults, CTC offered financial support to families with children regardless of employment status. These credits aimed to alleviate child poverty and encourage workforce participation among low-income households. Given these reforms, I calculate net household income for the relevant period and adjust it to £2001 for consistency.

Appendix A.4 UK Time Use Survey

I use the UK Time Use Survey (2000–2001) (Office for National Statistics, 2003) to validate the parental time measures. The BHPS remains the primary source for home production hours because the TUS covers self-employed parents in only a single wave, providing insufficient variation by work arrangement. I use time-diary measures rather than the MCS frequency-based proxies (e.g., times per week reading to the child) because frequency conflates intensity with duration and carries higher social desirability bias (Hofferth, 2009).

The TUS covers 11,664 individuals aged 8 or over. I restrict to married households with the wife aged 21–45 and the youngest child under 9, using the youngest child's age as a proxy for the child of interest (Bolt et al., 2023). I construct two composite measures: (i) time with the child, covering physical care and supervision, teaching, reading/playing/talking, and travel escorting to or from education; and (ii) housework time, covering food preparation, housekeeping, and gardening. The resulting sample is described in Table A.7.

Table A.7: UKTUS Sample Selection Criteria

	Obs
Original sample	11,664
Households with women aged 21–45	6,123
Married couples	2,608
Youngest child under 9 years old	1,378
Completed both weekday and weekend diaries, husband working	2,080

Appendix A.5 Additional Datasets

I supplement empirical evidence from the Labour Force Survey (LFS), the Family Resources Survey (FRS), and O*Net on labor market characteristics by work arrangement. LFS is the largest household survey with measures related to the labor market and income and I use the five-quarter longitudinal version to explore the individual-level variation in working hours. Unfortunately, the Labour Force Survey does not have information on self-employment income, which is crucial in my analysis; hence, I use the annual Family Resources Survey (FRS) for more details on earnings. The sample is from 1997 to 2021/2, consisting of household heads and their spouses of ages 16-64, after full-time education and not missing labor market characteristics (e.g. employment status and hours worked). To cover all UK, I look from 1997 onwards.

To get more details by occupation codes, I construct a work flexibility measure following Goldin (2014) and Bang (2022) and classify into Time pressure, Contact with others, Establishing and maintaining interpersonal relationships, Structured vs. unstructured work, and Freedom to make decisions. Using the O*Net dataset, I first normalize the values to mean 0 and variance 1 and get the average per 3 digit, 2 digit, and 1 digit groups. Since the UK dataset is based on UK SOC Codes which are different than the O*Net occupation codes and there were several changes in the SOC codes after the 2000s, 2010 and 2020, I describe in detail how the O*Net is mapped to the UK datasets. First, I convert UK SOC2000 to ISCO-08 following Office for National Statistics (2022b).[54] Then, I crosswalk between the International Standard Classification of Occupations (ISCO-08) and the 2010 Standard Occupational Classification (SOC).[55] Lastly, I match ONET SOC codes from 2010 to 2019 (O*NET Resource Center, 2019). The sequential mapping is intended to reduce errors in converting the codes.[56] Using these measures, I find that self-employed workers have more flexibility than paid employed workers for both gender.

Appendix B Additional Motivational Facts

This appendix documents additional empirical patterns that support the findings in Section 3. Figure B.1 shows how labor market outcomes evolve around first birth separately by gender and work arrangement. Figure B.8 documents self-employment by incorporation and sectoral status. Figure B.9 documents household asset dynamics around first birth.

Appendix B.1 Labor Market Outcomes

Figure B.1 reports event-study estimates for hours worked, home production, log weekly income, and log hourly wage around first birth, separately for husbands and wives within each work arrangement. Wives reduce market hours and increase home production at birth regardless of work arrangement, but the adjustment is sharpest for self-employed mothers. For paid-employed wives, the decline in weekly income tracks the hours reduction, on the other hand, for self-employed wives, weekly income falls less sharply because hourly wages do not decline and, in some specifications, rise modestly, consistent with positive selection remaining in the labor market after birth. Husbands show limited adjustment across all outcomes and work arrangements, confirming the gender asymmetry documented in Section 3.

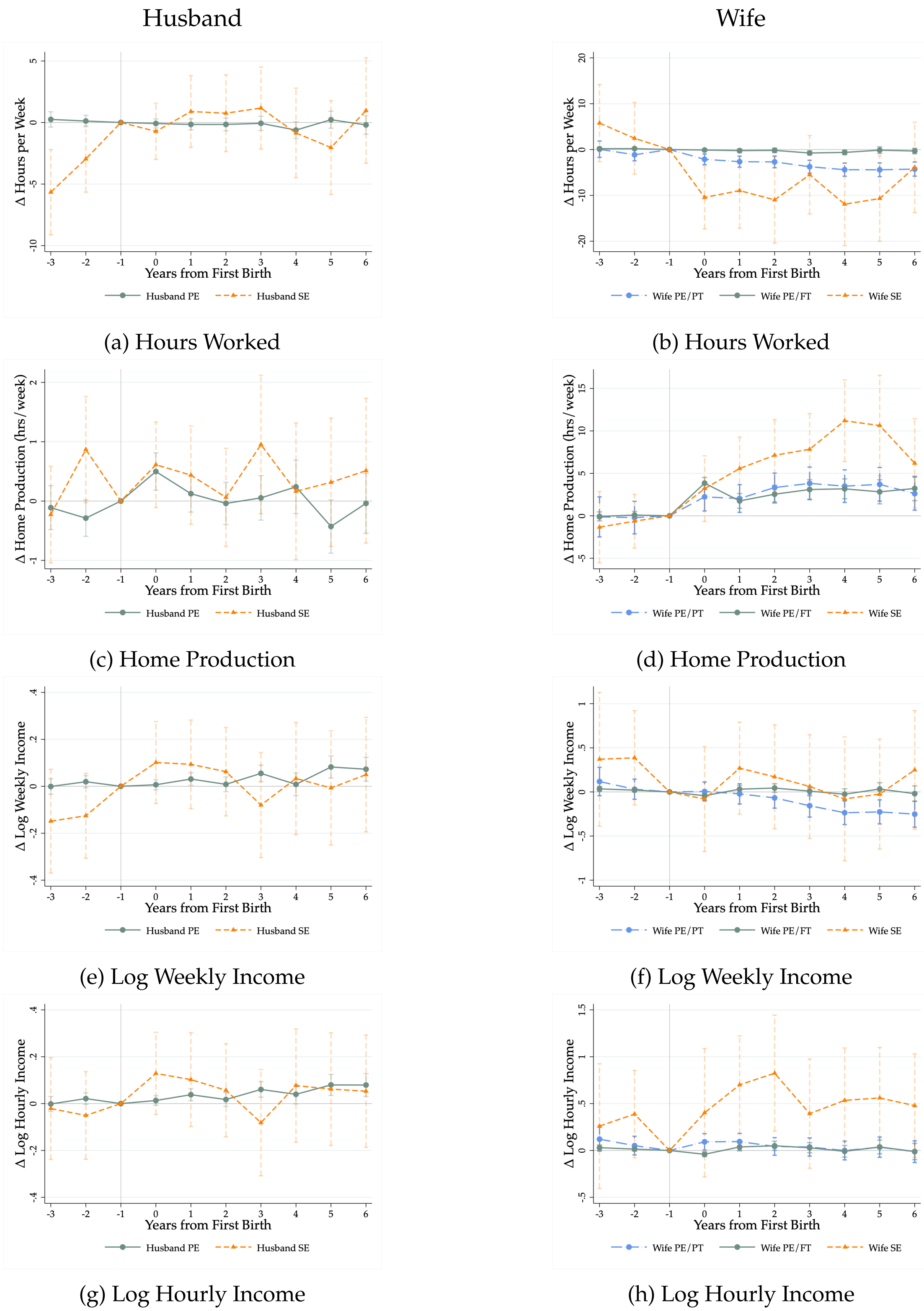
Figure B.2 complements the event-study evidence with the cross-sectional distribution of weekly earnings by work arrangement.

Self-employed workers also adjust hours more within individuals over time. Figure B.3 shows the distribution of within-person standard deviations in weekly hours, computed from individuals observed in at least two waves in the same work arrangement. Self-employed husbands and wives

[54] Authors note that "These conversions were developed for this research project, and do not constitute official guidance on classification conversion. They should be used with caution."

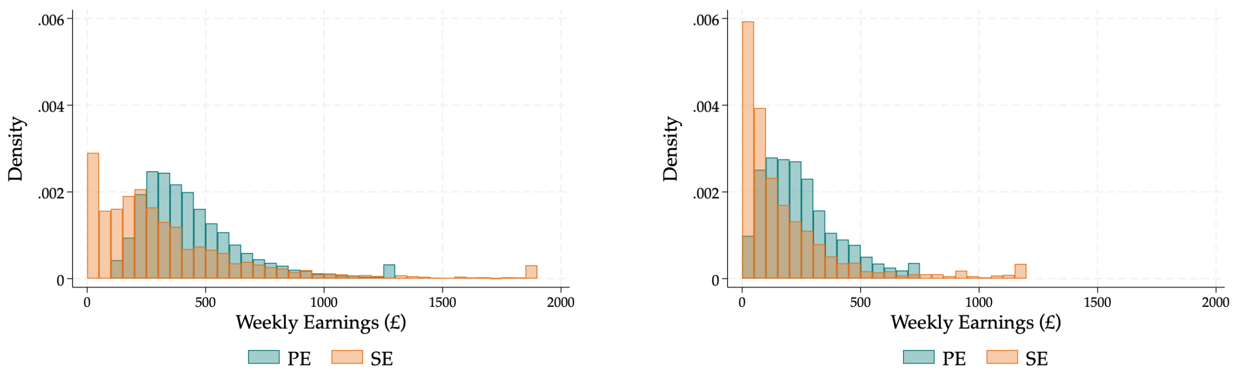
[55] In the report, BLS acknowledges that there are some weak connections listed in Table 1.

[56] Note that some data are not collected. e.g. Occupation 55-1011 "Air Crew Officers" is a military occupation. O*NET does not collect data on military occupations. The military services are the best resource for information on this occupation.



Note: Event-study estimates by work arrangement around first birth. Hours panels show absolute changes in hours per week relative to $t = -1$; income panels show log-point changes relative to $t = -1$. Wife panels: PE/PT, PE/FT, SE; husband panels: PE and SE. Controls include age, year, education, and marital status; standard errors clustered at the individual level. Confidence intervals at 90%.
Source: British Household Panel Study (BHPS).

Figure B.1: Event Study: Labor Market Outcomes by Gender and Work Arrangement

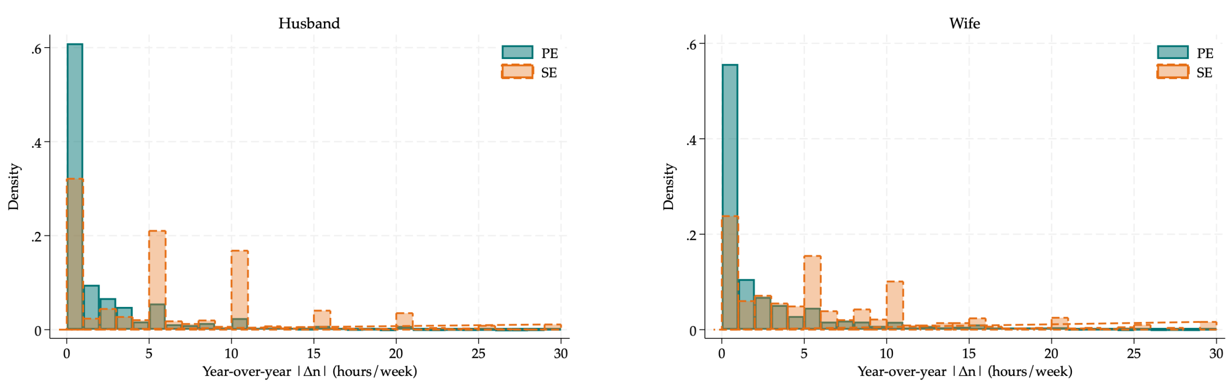


(a) Husband (b) Wife

Note: Bars report the share of person-year observations in £50 bins. Self-employment earnings are bottom-coded at zero in the BHPS, so the self-employment spike at the zero bin collects observations with zero or negative earnings recorded as zero; all other bins report positive earnings.
Source: British Household Panel Study (BHPS).

Figure B.2: Distribution of Weekly Earnings by Work Arrangement

show substantially wider within-person hour variation than their paid-employed counterparts, confirming that the dispersion visible in Figure 2 reflects genuine scheduling flexibility rather than permanent sorting into different jobs. Figure B.4 documents the same pattern using the five-quarter Labour Force Survey.



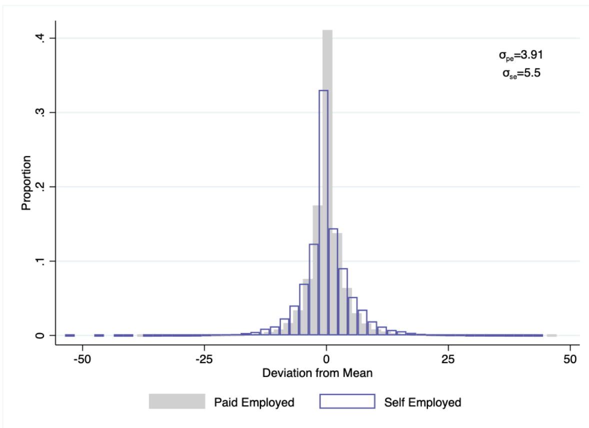
(a) Husband (b) Wife

Note: Distribution of within-person standard deviations in weekly hours worked, for individuals observed in at least two waves in the same work arrangement.
Source: British Household Panel Study (BHPS).

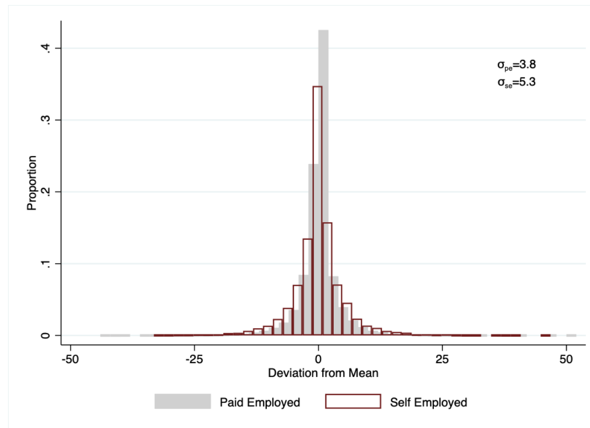
Figure B.3: Within-Person Hours Variation by Work Arrangement

Appendix B.2 Self-Employment: Work Flexibility

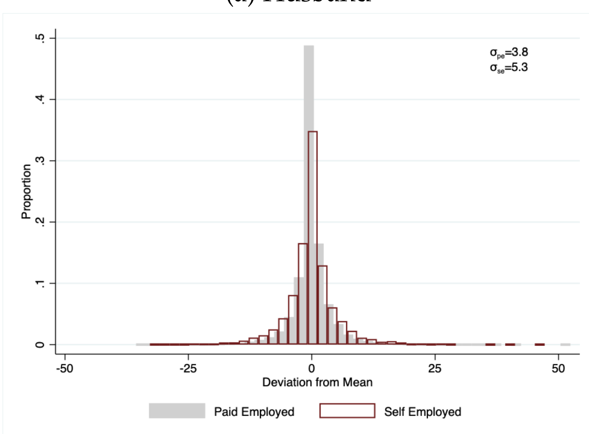
Time-use diaries from the UK Time Use Survey show how self-employed parents use scheduling flexibility. Self-employed mothers keep a morning work rate similar to paid employees but carry more work into the afternoon and evening: 22% work during the 3–4pm childcare window versus



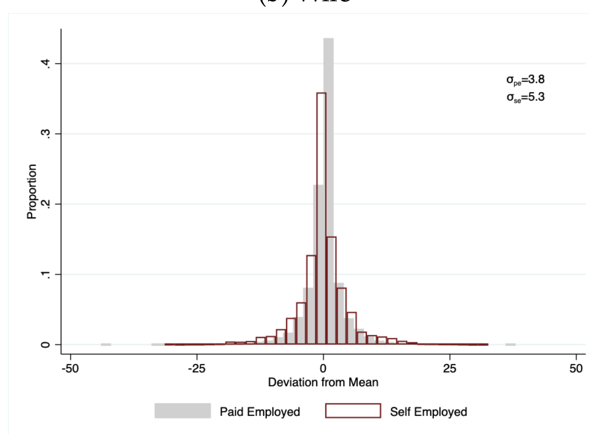
(a) Husband



(b) Wife



(c) Husband in Paid Employment



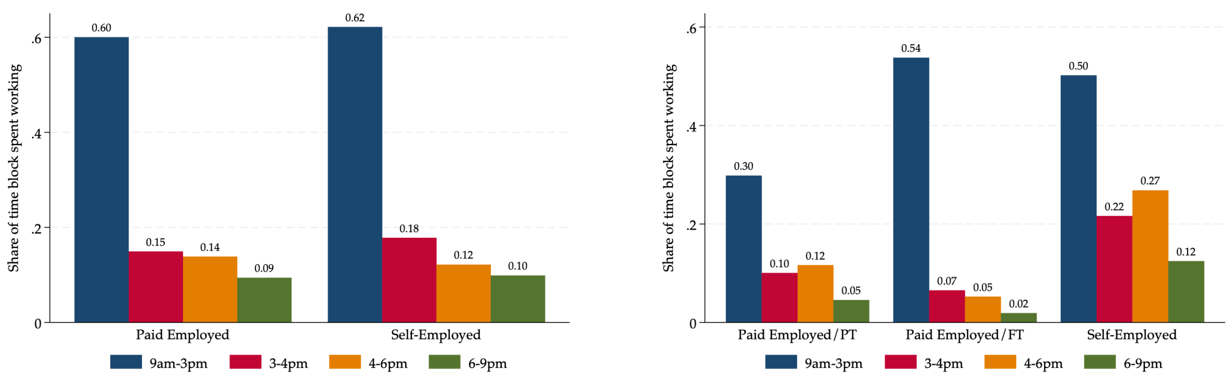
(d) Husband in Self-Employment

Note: Each panel shows deviations from a respondent's mean hours across five consecutive quarters, for married individuals with a youngest child aged 5. Controls include the husband's work arrangement.
Source: Five-Quarter Longitudinal Labour Force Survey.

Figure B.4: Within-Individual Variation in Hours Worked by Work Arrangement

7% of paid-employed mothers, and 27% during 4–6pm versus 5%. This shows up as split shifts, work in two or more non-contiguous blocks: 33% of self-employed mothers split their day versus 7% of paid employees, stepping away during the afternoon childcare window and resuming later. Fathers show the same pattern. These diaries indicate that the flexibility of self-employment is used to interleave market work with childcare, not simply to reduce total hours.

Self-employed parents are also more likely to interleave childcare and housework within their market work spans: among those with a work span of at least three hours, self-employed mothers record childcare as a secondary activity during work at nearly twice the rate of paid-employed mothers, and self-employed fathers show a similar pattern relative to paid-employed fathers. Figure B.7 shows that work-from-home rates are higher for both self-employed wives (44% vs. 12% for paid-employed wives, rising to 56% when a young child is present) and self-employed husbands, providing the physical proximity that makes schedule interruptions feasible. Together, these patterns show that self-employment flexibility operates through the timing and structure of the workday, not only through total hours.



(a) Fathers (b) Mothers

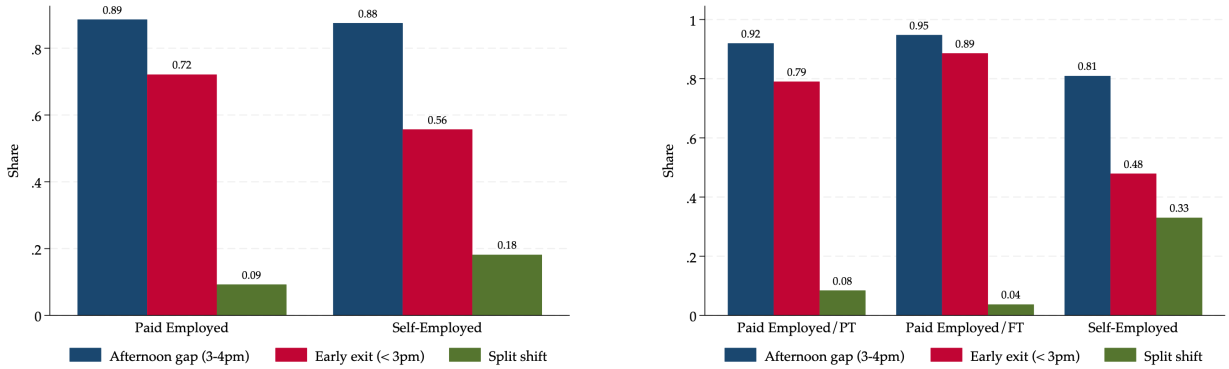
Note: Share of each time block spent working by work arrangement.
Source: UK Time Use Survey (2000–2001).

Figure B.5: Work Rate by Time Block

Appendix B.3 Self-Employment: Incorporation and Sectoral Sorting

The self-employment rise documented in Section 3 is driven almost entirely by unincorporated self-employment among women. Figure B.8 shows event-study estimates for incorporated and unincorporated self-employment separately, conditional on remaining employed. The unincorporated rate rises sharply for wives after childbirth, while the incorporated rate remains flat. For husbands, neither form of self-employment shows a substantial post-birth adjustment. This pattern is consistent with mothers sorting into sole-trader arrangements rather than into formal businesses with employees.

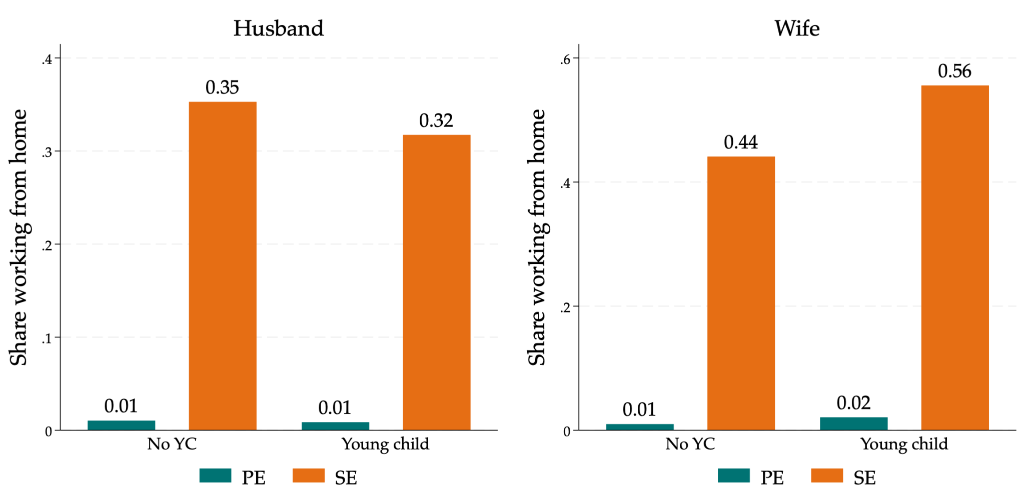
Paid employed and self-employed workers sort into systematically different occupations and industries. Both spouses avoid large-employer, fixed-schedule sectors when entering self-



(a) Fathers (b) Mothers

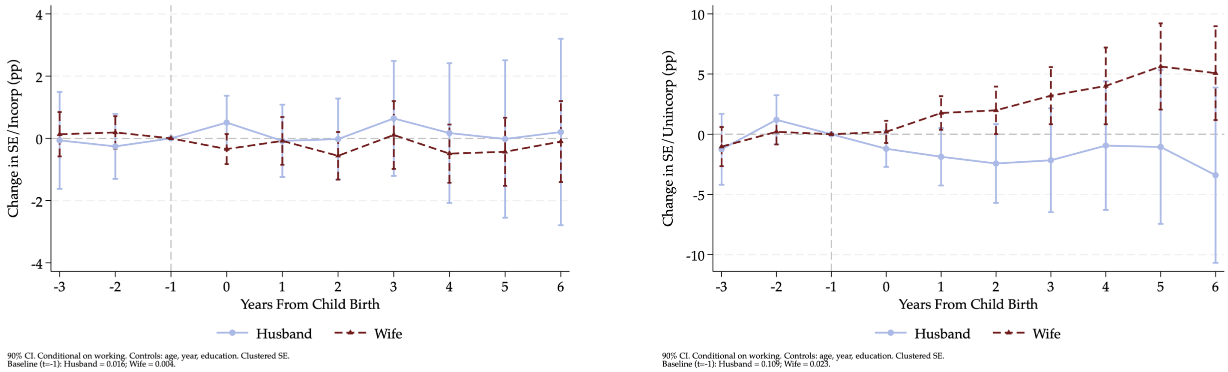
Note: Afternoon gap = share with a work break during the 3–4pm block; early exit = share finishing work before 3pm; split shift = share working in two or more non-contiguous time blocks.
Source: UK Time Use Survey (2000–2001).

Figure B.6: Schedule Interruptions by Work Arrangement



Note: Share of workers usually working from home, by work arrangement, separately for husbands and wives and by the presence of a young child (under 5).
Source: BHPS.

Figure B.7: Working from Home by Work Arrangement



(a) Incorporated Self-Employment

(b) Unincorporated Self-Employment

Note: Both panels are conditional on being employed. The y-axis shows percentage point changes relative to the year before first birth ($t = -1$). Controls include age, year, education, and marital status; standard errors clustered at the individual level. Confidence intervals at 90%.
Source: British Household Panel Study (BHPS).

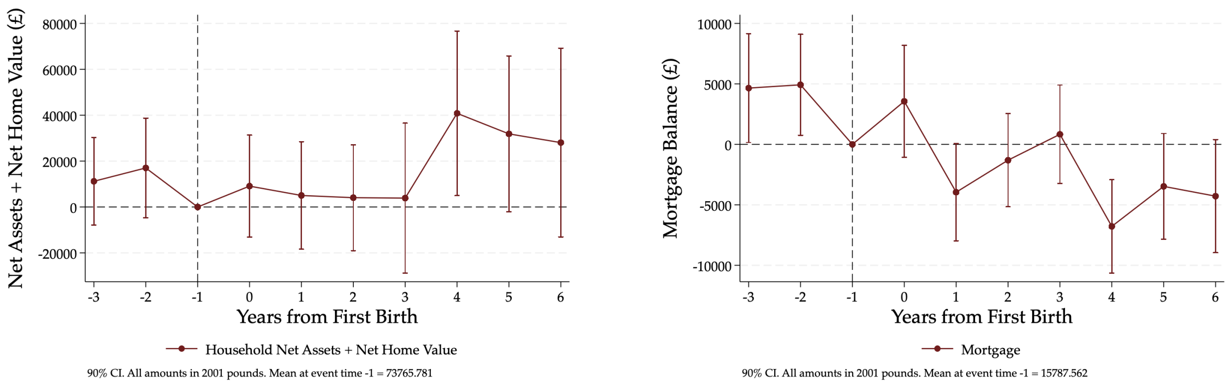
Figure B.8: Event Study: Self-Employment by Incorporation Status (Conditional on Working)

employment: Manufacturing, Public Administration, and Financial Intermediation are heavily PE-concentrated for husbands and wives. Self-employed husbands concentrates in Crafts and Construction, with little response to family stage. Self-employed wives concentrate at the top of the occupational ladder and in personal services, while almost disappearing from clerical and technical roles, which tend to have more rigid schedules. The presence of a young child shifts self-employed wives further toward services and Health and Social Work.

Firm-level data reinforce these patterns. Paid employees work for large organizations, with a plurality at establishments of fifty or more employees, while self-employed workers are overwhelmingly in establishments of fewer than ten people. Within self-employment, roughly one-third of husbands and one-fifth to one-third of wives are employers with paid employees; the remainder are own-account workers. The share of self-employed wives with employees falls sharply when a young child is present.

Appendix B.4 Household Assets

Households accumulate housing wealth in anticipation of childbirth and begin repayment afterward. Figure B.9 shows that net household assets, the sum of net assets and net home value, rise after some years after post birth. The right panel reveals the mechanism: mortgage balances peak pre-birth, indicating that households take on larger mortgages before childbirth, likely to secure larger or more stable housing. Net home value then rises as households begin repayment, making home equity the primary form of wealth accumulation around childbirth. These patterns motivate the inclusion of household savings in the structural model.



(a) Net Assets + Net Home Value (b) Mortgage

Note: Household net assets are defined as savings and investments, net of debt. Net home value is the difference between home value and mortgage balance. All amounts are in 2001 £. The sample includes married and cohabiting couples where the husband is always employed. Controls include year fixed effects, both spouses' age and education, household work arrangement, and the couple's annual income. Standard errors clustered at the household level. Confidence intervals at 90%.
Source: British Household Panel Study (BHPS).

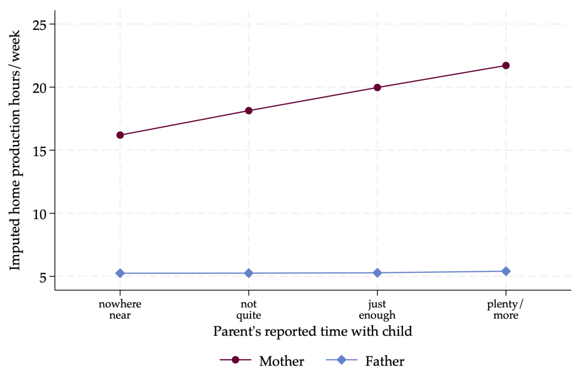
Figure B.9: Event Study: Household Assets

Appendix C Time Use Validation

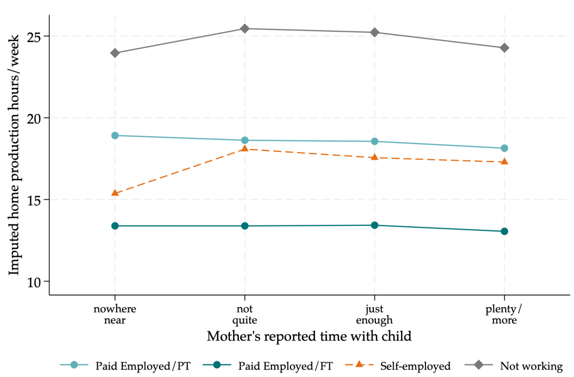
The child production function uses maternal and paternal time inputs imputed from BHPS home-production hours, matched to each MCS household by work arrangement, education, number of children, and the age of the youngest child. The MCS does not record parental time use directly, so I validate these imputed inputs against a question the survey asks at every wave: whether the parent feels they spend enough time with the child. I harmonize the response across sweeps into a common four-point scale, ordered from "nowhere near enough" to "plenty," and ask whether the households the imputation assigns more home-production time are the households that independently report more time with the child.

Figure C.10 Panel (a) plots mean imputed home-production time against the reported-time scale for each parent. Imputed maternal time rises monotonically with reported time, from 16.2 hours per week among mothers who report "nowhere near enough" to 21.7 hours among those who report "plenty". Panel (b) decomposes the maternal gradient by work arrangement. Non-working mothers are assigned the most home-production time, at roughly 25 hours, and report the most time with their children; full-time employed mothers are assigned the least, at roughly 13 hours; part-time and self-employed mothers fall in between. Within a given arrangement the relationship is flat, so the imputation encodes maternal time at the level of the work arrangement, which is the margin along which the model varies parental time. The validation suggests that the two datasets have the same ordering. The BHPS records that non-working women do the most home production, and MCS mothers independently report the most time with their children when not working. Fathers supply a low and arrangement-invariant amount of home production.

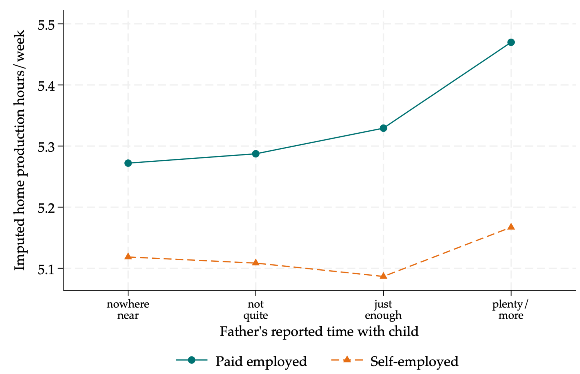
Since, MCS questions are based on subjective adequacy measure, I next validate the imputed



(a) By parent



(b) Mothers, by work arrangement

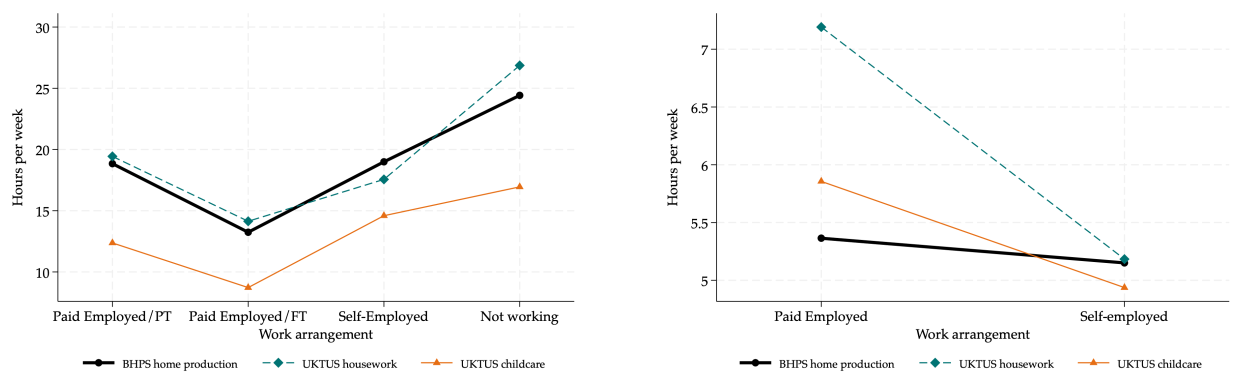


(c) Fathers, by work arrangement

Note: Mean imputed home-production time (BHPS cell means, hours per week) by the parent's reported adequacy of time with the child (MCS, harmonized four-point scale from "nowhere near enough" to "plenty"). Panel (a) plots mothers and fathers; panel (b) plots parents by work arrangement.

Figure C.10: Imputed home-production time and parents' reported time with the child hours against the UK Time Use Survey. I construct weekly housework hours (cooking, cleaning, and home maintenance) and, separately, weekly primary childcare hours (physical care, teaching, playing, and escorting). I average both to the cells used in the imputation and compare them with the BHPS home-production grid. Figure C.11 shows that BHPS home production reproduces diary housework. For mothers the two coincide in level and across arrangements, averaging 18.3 hours per week in the grid against 18.8 in the diaries and tracing the same profile from full-time employment up to non-work. For fathers the ordering is the same, with a small diary-over-survey gap at paid employment that is second-order given how little home production fathers do.

Home production tracks diary housework, with a cell-level correlation of 0.85 for mothers and 0.67 for fathers. It still co-moves with childcare across arrangements (correlation 0.57), so its variation indexes the broader gradient in time spent at home. Hence, it is a good proxy for the parent's time with the child, tracking diary housework and moving with the parents' own reported time-with-child gradient, and enters skill formation as one input.



(a) Mothers (b) Fathers

Note: Mean home-production hours per week by work arrangement. Black line: the BHPS home-production grid (the imputed input). Teal dashed line: UKTUS diary housework. Orange line: UKTUS primary childcare. UKTUS means are diary-weighted and averaged to the imputation cells.

Figure C.11: BHPS home production and UKTUS diary time by work arrangement

Appendix D Childcare Price

I do not distinguish different types of formal childcare and assume homogeneous quality (nursery, childminder, nanny, etc.).[57] I allow childcare costs in the model to vary by the age of the youngest

[57] Three features of the UK regulated sector support this assumption. First, since the Care Standards Act 2000, all formal childcare providers must register with Ofsted and meet minimum National Standards; settings that fail inspection are required to improve or face closure, creating a quality floor that limits the lower tail of quality within the regulated sector. Second, the Working Tax Credit childcare element covered up to 70% of formal childcare costs during the MCS period, compressing price differentials across income groups within the regulated sector and limiting the ability of higher-income families to systematically sort into more expensive, higher-quality settings. Third, Gambaro, Stewart, and Waldfogel (2014) document that in the UK's regulated sector disadvantaged children are more likely than advantaged children to attend settings with qualified staff, so quality does not systematically favor advantaged children within the

child, reflecting the substantial differences in the type and intensity of formal care used across child ages.

Ages 1 and 3: Nursery care. For children under age 2, the average cost of a full-time nursery place in England was £159 per week for 50 hours, implying an hourly rate of £3.18 (Daycare Trust, 2008, Table 1). For children aged 2 and over, the corresponding figure was £149 per week, or £2.98 per hour. In the model, households choose between part-time (15 hours per week) and full-time (30 hours per week) formal childcare. At the full-time intensity, these rates imply annual childcare costs of approximately £4,961 for a child aged 1 and £4,649 for a child aged 3, with part-time costing half.[58]

Age 5: After-school care. Children aged 5 attend primary school and require only wraparound care. The Daycare Trust (Daycare Trust, 2008) reports an average cost of £43 per week for 15 hours of out-of-school club provision in England, implying an hourly rate of £2.87. At 15 hours per week, the annual cost is approximately £2,236, roughly half the cost of full-time nursery care.[59]

Age 7 and above: Childcare as a fraction of income. For families whose youngest child is aged 7 or older, childcare expenditure is modeled as a fixed fraction of household income. School-age children require substantially less formal care: the Childcare and Early Years Survey of Parents 2009 reports a median of 6.5 hours per week in childcare for school-age children, compared with 20 hours for pre-school children, and a median weekly payment of £15 for school-age-only families versus £50 for pre-school families (Smith et al., 2010, pp. 65, 74). Relative to median household income, this implies a childcare expenditure share of approximately 4–5%, consistent with evidence from the Families and Children Study 2005 that the overall average expenditure share was 11%, with lower shares for higher-income households (Hoxhallari, Conolly, and Lyon, 2007). I set the childcare income share for families with older children to 5%. Table D.8 summarizes the calibrated childcare cost parameters.

sector, though settings serving disadvantaged areas do receive somewhat lower inspection ratings. Since the model restricts formal childcare to the regulated sector, quality heterogeneity within my sample is limited.

[58]I use the gross market price rather than adjusting for the free early education entitlement introduced for 3-year-olds in 2004 (12.5 hours per week, later extended to 15 hours), as the entitlement was phased in partway through the sample period and the quality of provision under the free entitlement may differ from that of privately purchased care. These figures are consistent with the average hourly childcare cost of £2.60 (s.e. 0.04) estimated from BHPS expenditure data by Blundell et al. (2016), who pool across child ages.

[59]This is consistent with the median out-of-pocket payment of £2.50 per hour for on-site after-school clubs reported in the Childcare and Early Years Survey of Parents 2009 (Smith et al., 2010).

Table D.8: Calibrated Childcare Cost Parameters

Age of youngest child	Care type	Hourly rate	Hours/week	Annual cost
1 (under 2)	Nursery (full-time)	£3.18	30	£4,961
3 (aged 2–4)	Nursery (full-time)	£2.98	30	£4,649
5	After-school club	£2.87	15	£2,236
7+	5% of net household income			

Note: Source: Daycare Trust (2008) for ages 1, 3, and 5; Smith et al. (2010) and Hoxhallari et al. (2007) for age 7+. Rates are deflated to £2001.

Appendix E Log-Additivity, Exogeneity, and Estimation of the Child Production Function

Appendix E.1 Log-Additivity of Child Skills

I prove that the household conditional value functions are log-additive in child's skills (Mullins, 2022; Garcia-Vazquez, 2023). This reduces the computational burden of solving the model: the optimal policy functions do not depend on child skill levels, which removes child skills from the dynamic state space. However, this does not eliminate forward-looking investment motives. Parents still choose time and formal childcare to shift next-period skills through the CPF, anticipating the continuation value gains from higher future skills.

Proposition 1. For child's age $g_t = g_1, \ldots, g_T$ and $D \in \{cog, ncog\}$, the Stage-2 conditional value function admits the decomposition

$$\widetilde{\mathcal{V}}^{s2}_{q,r}(\Omega^{s2}) = \bar{\mathcal{V}}^{s2}_{q,r}(\Omega^{s2}_{-\theta}) + \sum_{D} \Gamma_{D,g_t} \log \theta_{D,g_t}$$

where the coefficients follow the backward recursion

$$\Gamma_{D,g_t} = \alpha_D + \beta\big(\gamma_{D,1}\,\Gamma_{D,g_t+1} + \gamma_{D',2}\,\Gamma_{D',g_t+1}\big), \qquad \Gamma_{D,g_T+1} = \alpha_D \sum_{s=0}^{J-1} \beta^s,$$

$J = J(a^w_t)$ is the number of remaining model periods from the child's age-7 point, at the wife's age a^w_t, until her terminal age. The coefficient therefore depends on the wife's age, not on the child's age alone: because a given child age occurs at different wife ages under stochastic fertility, Γ_{D,g_t} is more precisely $\Gamma_D(g_t, a^w_t)$ and is solved at each wife age. $\Omega^{s2}_{-\theta}$ is the state space with both child skill levels removed. The terminal condition reflects that skills are fixed after the investment window but continue to enter flow utility in every remaining period. The coefficient values the youngest child's skill to the wife's terminal age.

Proof. The proof proceeds by backward recursion. At the last child investment period g_T (child age 5), parents choose Stage-2 actions $r_{g_T} = (n^w_{g_T}, n^h_{g_T}, \tau^w_{g_T}, \tau^h_{g_T}, cc_{g_T}, a_{g_T+1})$ to solve

$$\begin{aligned}
\widetilde{\mathcal{V}}^{s2}_{q,g_T}(\Omega^{s2}_{g_T}) &= \max_{r_{g_T}\in\mathcal{R}} \Big\{ \mathcal{U}_{g_T}(C_{g_T}, l^w_{g_T}, l^h_{g_T}, cc_{g_T}, q) + \zeta_{q,r_{g_T}} + \beta\mathbb{E}\big[V^{s1}(\Omega^{s1,\prime}_{g_T+1})\big] \Big\} \\
&= \max_{r_{g_T}\in\mathcal{R}} \Big\{ \mathcal{U}_{g_T}(\cdot) + \sum_D \alpha_D \log\theta_{D,g_T} \\
&\quad + \beta\Big(\sum_D \Gamma_{D,g_T+1}\,\mathbb{E}\log\theta_{D,g_T+1} + \mathbb{E}\bar{V}_{g_T+1}(\Omega^{s1,\prime}_{-\theta,g_T+1}) \Big) + \zeta_{q,r_{g_T}} \Big\} \\
\text{s.t.} \quad & (\text{TC}^w), (\text{TC}^h), (\text{Sup}), (\text{BC}), (\text{I}), (\text{IC}^w), (\text{IC}^h), (\text{HC}^w), (\text{CPF}^D)
\end{aligned} \tag{E.1}$$

where the second line uses the fact that $\sum_D \alpha_D \log\theta_{D,g_T}$ enters the current flow utility through the child skills term in $\mathcal{U}_{g_T}$. After the investment window closes, the skill vector is fixed at θ_{D,g_T+1} and adds $\alpha_D \log\theta_{D,g_T+1}$ to flow utility in each of the J remaining periods; collecting this discounted stream gives the coefficient $\Gamma_{D,g_T+1} = \alpha_D \sum_{s=0}^{J-1}\beta^s$, and $\bar{V}_{g_T+1}$ collects the remaining skill-free terms.

Substituting the child production function into $\log\theta_{D,g_T+1}$ yields

$$\begin{aligned}
\widetilde{\mathcal{V}}^{s2}_{q,g_T}(\Omega^{s2}_{g_T}) = \max_{r_{g_T}} \mathbb{E}_\nu \Big[& \mathcal{U}_{g_T}(\cdot) + \sum_D \alpha_D \log\theta_{D,g_T} \\
& + \beta\sum_D \Gamma_{D,g_T+1}\Big(\log A_{D,g_T} + \gamma_{D,1}\log\theta_{D,g_T} + \gamma_{D,2}\log\theta_{D',g_T} \\
& + \gamma_{D,3}\log\tau^w_{g_T} + \gamma_{D,4}\log\tau^w_{g_T}\cdot\theta^w + \gamma_{D,5}\log\tau^h_{g_T} + \gamma_{D,6}\log\tau^h_{g_T}\cdot\theta^h \\
& + \gamma_{D,7}cc^{PT}_{g_T} + \gamma_{D,8}cc^{FT}_{g_T} + \gamma_{D,9}cc^{PT}_{g_T}\cdot\theta^w + \gamma_{D,10}cc^{FT}_{g_T}\cdot\theta^w + \gamma_{D,11}sib_{g_T}\Big) \\
& + \beta\mathbb{E}\bar{V}_{g_T+1}(\cdot) + \zeta_{q,r_{g_T}} \Big] \\
\text{s.t.} \quad & (\text{TC}^w), (\text{TC}^h), (\text{Sup}), (\text{BC}), (\text{I}), (\text{IC}^w), (\text{IC}^h), (\text{HC}^w)
\end{aligned}$$

where the expectation over the production shock ν_{D,g_T+1} drops out because $\mathbb{E}[\nu_{D,g_T+1}] = 0$. Collecting the terms in $\log\theta_{D,g_T}$ from the current flow utility, the own-domain persistence ($\gamma_{D,1}$), and the cross-domain persistence ($\gamma_{D',2}$, after relabeling domains in the cross term), I can rewrite the conditional value function at g_T as

$$\widetilde{\mathcal{V}}^{s2}_{q,g_T}(\Omega^{s2}_{g_T}) = \bar{\mathcal{V}}^{s2}_{q,g_T}(\Omega^{s2}_{-\theta,g_T}) + \sum_D \Gamma_{D,g_T}\log\theta_{D,g_T}$$

with $\Gamma_{D,g_T} = \alpha_D + \beta(\gamma_{D,1}\,\Gamma_{D,g_T+1} + \gamma_{D',2}\,\Gamma_{D',g_T+1})$, the recursion evaluated at the terminal condition, and

$$\begin{aligned}
\bar{\mathcal{V}}^{s2}_{q,g_T}(\Omega^{s2}_{-\theta,g_T}) = \max_{r_{g_T}} \Big[& \mathcal{U}_{g_T}(\cdot) + \beta\sum_D \Gamma_{D,g_T+1}\Big(\log A_{D,g_T} + \gamma_{D,3}\log\tau^w_{g_T} + \gamma_{D,4}\log\tau^w_{g_T}\cdot\theta^w \\
& + \gamma_{D,5}\log\tau^h_{g_T} + \gamma_{D,6}\log\tau^h_{g_T}\cdot\theta^h + \gamma_{D,7}cc^{PT}_{g_T} + \gamma_{D,8}cc^{FT}_{g_T} \\
& + \gamma_{D,9}cc^{PT}_{g_T}\cdot\theta^w + \gamma_{D,10}cc^{FT}_{g_T}\cdot\theta^w + \gamma_{D,11}sib_{g_T}\Big) + \beta\mathbb{E}\bar{V}_{g_T+1}(\cdot) + \zeta_{q,r_{g_T}} \Big] \\
\text{s.t.} \quad & (\text{TC}^w), (\text{TC}^h), (\text{Sup}), (\text{BC}), (\text{I}), (\text{IC}^w), (\text{IC}^h), (\text{HC}^w)
\end{aligned}$$

which proves the proposition for the terminal child investment period g_T.

Now suppose the result holds at $g_t + 1$, so that $\widetilde{\mathcal{V}}^{s2}_{q,g_t+1} = \bar{\mathcal{V}}^{s2}_{q,g_t+1} + \sum_D \Gamma_{D,g_t+1} \log \theta_{D,g_t+1}$. The decomposition extends to the Stage-1 ex-ante value $V^{s1}_{g_t+1}$ because the skill term is common across hours-savings choices r and work arrangements q, so it passes through the maximization and the expectations over $\boldsymbol{\zeta}$ and $\boldsymbol{\xi}$ as an additive constant. For the inductive step, I show the same decomposition holds at g_t. Applying the induction hypothesis to the continuation value, the conditional value function at g_t is

$$\begin{aligned}\widetilde{\mathcal{V}}^{s2}_{q,g_t}(\Omega^{s2}_{g_t}) = \max_{r_{g_t} \in \mathcal{R}} \Big\{ & \mathcal{U}_{g_t}(\cdot) + \sum_D \alpha_D \log \theta_{D,g_t} \\ & + \beta \Big(\sum_D \Gamma_{D,g_t+1} \mathbb{E} \log \theta_{D,g_t+1} + \mathbb{E} \bar{V}_{g_t+1}(\Omega^{s1,\prime}_{-\theta,g_t+1}) \Big) + \zeta_{q,r_{g_t}} \Big\}\end{aligned}$$

where $\bar{V}_{g_t+1}$ collects the remaining skill-free terms. Substituting the CPF for $\log \theta_{D,g_t+1}$ and again collecting the terms in $\log \theta_{D,g_t}$ from the flow utility, the own-domain persistence, and the cross-domain persistence, I obtain

$$\widetilde{\mathcal{V}}^{s2}_{q,g_t}(\Omega^{s2}_{g_t}) = \bar{\mathcal{V}}^{s2}_{q,g_t}(\Omega^{s2}_{-\theta,g_t}) + \sum_D \Gamma_{D,g_t} \log \theta_{D,g_t}$$

with $\Gamma_{D,g_t} = \alpha_D + \beta(\gamma_{D,1} \Gamma_{D,g_t+1} + \gamma_{D',2} \Gamma_{D',g_t+1})$, completing the induction. Note that the CPF input terms enter $\bar{\mathcal{V}}^{s2}_{q,g_t}$ with weight Γ_{D,g_t+1} rather than α_D, since parents internalize that skills produced today persist into future utility. □

Corollary 1 (Skill-Independent Policy Functions). The household's optimal policy functions $\{n^{w*}_{g_t}, n^{h*}_{g_t}, \tau^{w*}_{g_t}, \tau^{h*}_{g_t}, cc^*_{g_t}, a^*_{g_t+1}\}$ are independent of the child skill levels $(\log \theta_{cog,g_t}, \log \theta_{ncog,g_t})$.

Proof. From Proposition 1, the skill term $\sum_D \Gamma_{D,g_t} \log \theta_{D,g_t}$ is additively separable and common across choices, so it cancels in the argmax and the optimal policy depends only on $\bar{\mathcal{V}}^{s2}_{q,r}(\Omega^{s2}_{-\theta})$, not on child skills. The same holds for the Stage-1 work-arrangement choice, since the altruism weights $\alpha_{cog}, \alpha_{ncog}$ are common across arrangements. □

Appendix E.2 Exogeneity of Parental Inputs

The CPF estimation in Section 5.1 addresses two distinct sources of econometric concern: measurement error in observed child skills, and potential endogeneity of parental inputs. The CPF coefficients on parental time and childcare enter the structural model as fixed parameters and carry directly into the counterfactual exercises of Section 7, so it is important that they capture the causal returns. Parental inputs (τ^w, τ^h, cc) are tied to labor supply decisions through the household constraints. I therefore instrument all four inputs with UK tax-and-benefit policy variation to recover the causal effects.

Instrument construction. I construct simulated instruments from external UK tax-and-benefit policy variation, each capturing a distinct policy channel. For each MCS household-wave, I use hourly wages and observed family structure into the FORTAX microsimulator (Shephard, 2009) under counterfactual configurations, so that variation across households comes from the policy regime and household characteristics (children, ages, education). The wage predictions come from four log-wage Mincer regressions estimated on the sample of married and cohabiting couples with children, separately by gender and by paid employment versus self-employment status, with controls for education, age, and age squared.

1. Z_w^{NI}, Z_h^{NI}: the paid employment versus self-employment National Insurance contribution differential for the wife and husband, capturing the Class 1 versus Class 2/4 wedge that shifts sectoral choice.[60]
2. Z^{cc}: the Work Tax Credit (WTC) childcare element value at benchmark hours, shifting the effective price of formal childcare;
3. Z_{2003}^{WTC}: the 2003 Working Tax Credit reform interacted with the number of dependent children, exploiting the response structure of the WTC childcare expansion;
4. Z_{1999}^{WFTC}: the 1999 Working Families Tax Credit expansion interacted with low-income eligibility and dependent children, capturing the extensive-margin labor supply incentive;
5. Z^{χ_3}: the household-level log-log curvature of the FORTAX net-income schedule, capturing rate-schedule progressivity;
6. Z_h^{k30}: the discrete jump in household disposable income at the husband's 29-to-30-hour threshold, where the WTC 30-hour premium activates;
7. Z_w^{IT}, Z_h^{IT}: the wife's and husband's income tax liability at modal PE hours, shifting the after-tax return to work;
8. Z^{CB}: the child benefit plus family transfers at zero earnings, varying with the number and ages of children across policy years;
9. Z_h^{PTR}: the husband's participation tax rate (the change in household disposable income from not working to working at modal PE hours), shifting the husband's extensive margin.

Instrument validity. All instruments feed predicted wages from Mincer regressions and observed family structure into FORTAX policy rules so that variation across households reflects the policy schedule applied to predetermined education, age, and family composition rather than endogenous

[60]Class 1 contributions are paid by employees at 12% on earnings between the Primary and Upper Earnings Limits. Class 2 is a flat weekly contribution paid by the self-employed (£2.05/week in 2003). Class 4 is a profit-based contribution at 8% on profits above the Lower Profit Limit. For most self-employed workers in the sample earnings range, total NI liability is substantially lower than Class 1, creating a net-of-tax incentive toward self-employment.

labor supply choices. The 1999 WFTC and 2003 WTC reform instruments are simulated differentials between the pre- and post-reform statutory schedules, so their variation reflects the legislated policy change interacted with predetermined characteristics, not within-sample exposure to the reform. The 30-hour threshold instrument Z_h^{k30} uses predicted wages at modal paid employment hours rather than actual hours, so the policy discontinuity identifies variation in the WTC premium across household types rather than households' sorting around the kink.

Moment conditions. Let $X = (\tau^w, \tau^h, cc)$ be the vector of parental inputs, $\log\theta = (\log\theta_{cog}, \log\theta_{ncog})$ the lagged latent skills, and X^c the exogenous controls (sibling indicator). The excluded instruments are the cross-domain skill proxies $Z^{skill} = (Z^{ncog}, Z^{cog})$ and the simulated FORTAX policy instruments Z^{FORTAX}.

Estimation combines two blocks of moment conditions. Let $Z^{input} = (Z^{skill}, Z^{FORTAX}, X^c, 1)$ be the instrument vector for the input equations. For each input j,

$$\mathbb{E}\left[Z_i^{input}\left(X_{ji} - \pi_j' Z_i^{input}\right)\right] = 0 \tag{E.2}$$

Let $\hat{v} = X - \hat{\Pi}' Z^{input}$ denote the four input-equation residuals, which enter the outcome equation as control-function corrections. With $Z^{outcome} = (Z^{skill}, X, X^c, \hat{v}, 1)$,

$$\mathbb{E}\left[Z_i^{outcome}\left(y_i - \beta_\theta' \log\theta_i - \beta_X' X_i - \beta_c' X_i^c - \beta_v' \hat{v}_i\right)\right] = 0 \tag{E.3}$$

Consistency. The procedure makes two standard corrections. Cross-domain skill instruments corrects for measurement error in the lagged skills. FORTAX instruments with a control function corrects for input endogeneity, valid under the exclusion restriction $Cov(Z^{FORTAX}, \nu') = 0$. Under these conditions the stacked GMM estimator is consistent (Hansen, 1982; Wooldridge, 2002).

Appendix E.3 Estimation of Child Production Function

Table E.9 reports the OLS and joint GMM estimates. The endogeneity correction raises the parental-input coefficients relative to OLS; what is stable across specifications is the sign pattern and the education gradient. Maternal time is more productive for college children in both domains, and its level is positive and significant for cognitive skills. Part-time childcare raises skills for non-college households with the college interaction offsetting it, consistent with larger returns to formal care for disadvantaged children (Cornelissen et al., 2018; Kottelenberg and Lehrer, 2017), and sibling effects are negative for cognitive and positive for non-cognitive skills.

The diagnostic tests the joint significance of the first-stage residuals. For cognitive skills, the Hausman-type test yields $p = 0.076$, so input exogeneity is not rejected at conventional levels. For non-cognitive skills, $p < 0.001$, so endogeneity is detected and concentrated in part-time formal childcare ($\hat{v}_{cc^{PT}}$ significant at the 1% level; $\hat{v}_{cc^{FT}}$ is not significant at conventional levels). The parental time residuals ($\hat{v}_{\tau^w}$, $\hat{v}_{\tau^h}$) are individually insignificant in both equations, so the correction

Table E.9: Child Production Function: OLS vs Joint GMM

	Cognitive$_{t+1}$				Non-Cognitive$_{t+1}$			
	OLS		Joint GMM		OLS		Joint GMM	
Variable	Coef	SE	Coef	SE	Coef	SE	Coef	SE
Cognitive	0.754	0.089	0.748	0.089	0.148	0.079	0.149	0.082
Non-Cognitive	0.160	0.072	0.154	0.073	0.732	0.065	0.721	0.069
log(Wife's time)	0.056	0.066	0.317	0.127	0.023	0.060	0.209	0.123
log(Wife's time) $\times$ College	0.063	0.012	0.111	0.021	0.052	0.011	0.103	0.021
log(Husband's time)	0.112	0.153	0.059	0.296	-0.024	0.118	-0.028	0.262
log(Husband's time) $\times$ College	0.063	0.017	0.045	0.020	0.071	0.016	0.058	0.019
Childcare (PT)	0.106	0.061	1.117	0.445	0.089	0.050	1.624	0.409
Childcare (FT)	0.110	0.057	0.675	0.286	0.120	0.054	0.496	0.268
Childcare (PT) $\times$ College	-0.154	0.090	-1.120	0.438	-0.158	0.070	-1.657	0.400
Childcare (FT) $\times$ College	-0.017	0.078	-0.517	0.265	0.042	0.074	-0.283	0.251
Sibling	-0.117	0.033	-0.168	0.043	0.117	0.031	0.072	0.040
Endogeneity diagnostics (first-stage residuals)								
$\hat{v}_{\tau^w}$			-0.234	0.166			-0.008	0.164
$\hat{v}_{\tau^h}$			0.039	0.338			-0.038	0.305
$\hat{v}_{cc^{PT}}$			-1.038	0.449			-1.586	0.412
$\hat{v}_{cc^{FT}}$			-0.590	0.293			-0.398	0.276
Joint test (p-value)			0.076				< 0.001	
First-stage instrument strength (joint GMM)								
Sanderson-Windmeijer F: τ^w	51.9							
Sanderson-Windmeijer F: τ^h	69.7							
Sanderson-Windmeijer F: cc^{PT}	19.1							
Sanderson-Windmeijer F: cc^{FT}	36.9							
Kleibergen-Paap rk Wald F	8.7							
Anderson-Rubin test (p)	< 0.001							
Hansen J (all instruments)	117.7 (7 df), $p < 0.001$							
Observations	6,726							

Note: OLS columns instrument only the lagged skill measures with cross-domain lagged scores for measurement-error correction; parental inputs are treated as exogenous. Joint GMM columns stack four first-stage moment conditions for the parental inputs with the second-stage CPF moment condition, correcting simultaneously for measurement error in lagged skills and endogeneity of τ^w, τ^h, cc^{PT}, and cc^{FT} using eleven FORTAX-simulated instruments; see Appendix E. Standard errors are analytical two-step sandwich standard errors clustered on the household, accounting for first-stage estimation uncertainty and the generated-regressor status of $\hat{v}$. The joint test reports the p-value from a Wald test of the null hypothesis that all four first-stage residuals are jointly zero; rejection indicates that the endogeneity correction is active. The Sanderson-Windmeijer F statistics report first-stage instrument strength for each endogenous input and the Kleibergen-Paap rk Wald F the joint statistic; the Anderson-Rubin test is the weak-instrument-robust test that the inputs have no effect, and the first stage is common to both equations. The Hansen J tests the overidentifying restrictions. The joint GMM column corresponds to the point estimates in Table 3.

does not reject the exogeneity of parental time. The supervision constraint is slack in 81% of young-child households ($\bar{T}^{cc} = 20$), so for most, parental time is not mechanically tied to formal care.

The Hansen J rejects the overidentifying restrictions, but this reflects the full instrument set rather than the validity of the estimated technology. Identification rests on the FORTAX exclusion restriction, imposed by construction under Corollary 2, not on the overidentifying restrictions the test evaluates. A parsimonious, theory-tied subset of six, one to two per input, does not reject ($J = 5.0$, 2 df, $p = 0.08$), and with eleven instruments and clustered errors the overidentification test tends to over-reject. The estimates are stable in sign; taken one instrument at a time, the childcare effect is positive on cognitive skills under every instrument and on non-cognitive skills under every well-identified instrument. Because the results are stable across these specifications, I use the full-set coefficients in the structural model, where the additional instruments strengthen identification and efficiency.

The structural estimation in Section 6 uses the joint GMM coefficients.[61] The endogeneity correction ensures that the CPF parameters entering the structural model reflect policy-relevant returns to parental inputs, so that the counterfactual exercises in Section 7 rest on policy-relevant treatment effects rather than on associations that may be confounded by unobserved household heterogeneity.

Appendix F Flexibility - Choice of Hours Grid

I embed flexibility by endowing each work arrangement with a more or less coarse hours menu. Paid employed wives choose from two points corresponding to part-time and full-time hours, while paid employed husbands choose from two points reflecting hours variation within full-time employment. Self-employed workers of both genders choose from a five-point menu spanning a wider range. This difference in menu coarseness directly encodes the scheduling flexibility advantage of self-employment and is a central modeling choice, disciplined by the paid employment rigidity literature, by within-person evidence from the BHPS, and by an objective k-means saturation criterion applied separately to each distribution.

[61]The parental-time inputs are matched from BHPS cell means, so I propagate the matching uncertainty into the standard errors with a two-step bootstrap. Each replication resamples the BHPS by household, recomputes the discretized cell means, re-maps the imputed inputs into the MCS, resamples the MCS by household, and re-estimates the control function. The analytical two-sample correction of Inoue and Solon (2010) provides a smoother alternative. The standard errors widen, modestly in the cognitive equation and more in the imputation-sensitive non-cognitive time terms, but no conclusion changes. Part-time formal childcare, the input the procedure corrects, remains significant (t = 3.2). The education interactions that carry the parental-time channels remain significant, with the father's non-cognitive interaction marginally so (t = 2.2). The education interactions are precisely estimated, so the difference in time productivity by education is identified.

Appendix F.1 Work arrangement rigidity

Wage earners cannot freely adjust weekly hours within a job (Altonji and Paxson, 1988; Aaronson and French, 2004), and the constraint is sharp enough that bunching estimators detect zero paid employment response to tax kinks while recovering substantial bunching among the self-employed (Saez, 2010; Chetty et al., 2011b). Kleven and Waseem (2013); Kleven (2016) review this asymmetry. Self-employment is the canonical residual margin of hours flexibility (Hamilton, 2000; Hurst and Pugsley, 2011; Yurdagul, 2017). I further allow nonlinearity in earnings through the hours-earnings curvature $\eta_{i,j}$ in the household income equation and through the year-over-year hours adjustment evidence below, rather than through grid cardinality alone.

Appendix F.2 Within-person variation in hours worked

From the data sample, paid employed workers exhibit essentially zero year-over-year hours adjustment within the same work arrangement. Among wives who remain paid employed in two consecutive years, 85.4 percent report a year-over-year change of at most 5 hours and the median change is nearly zero. Among husbands who remain paid employed, the corresponding figures are 88.9 percent and zero. Wives and husbands who remain self-employed adjust hours far more freely: only 61.5 and 63.6 percent stay within 5 hours of the previous year, the median change is 5 hours, and the mean is 6.6 and 6.4 hours, respectively. The gap between the two work arrangements is roughly 25 percentage points in the share staying within 5 hours, and 2.5 to 3 times larger in mean hours change, and is nearly identical for wives and husbands, indicating that the asymmetry is a feature of the work arrangement rather than gender. Table F.10 reports the key moments of $|\Delta n|$ and Figure B.3 shows the corresponding density.

Table F.10: Within-person stability of weekly work hours, conditional on staying in the same work arrangement in two consecutive years

			$\|\Delta n\|$ (hours/week)			
	Work arrangement	N	$\% \leq 5$h	Mean	p50	p90
Wife	Paid employed	12,765	85.4	2.6	0.0	8.0
	Self-employed	818	61.5	6.6	5.0	15.0
Husband	Paid employed	14,635	88.9	2.2	0.0	6.0
	Self-employed	2,867	63.6	6.4	5.0	15.0

Note: Sample is BHPS couple-year observations with non-missing work arrangement and weekly work hours in two consecutive years and the same work arrangement in both. $|\Delta n|$ is the absolute year-over-year change in usual weekly work hours. The first column reports the share of person-year pairs with a change of at most 5 hours; the next three report the mean, median, and 90th percentile of $|\Delta n|$. Paid employees exhibit substantially less within-person variation in hours than the self-employed, consistent with the rigid wage-hours bundles documented in Altonji and Paxson (1988) and the bunching evidence in Saez (2010).

Appendix F.3 Grid construction and validation

I locate grid points at weighted k-means cluster centroids of the BHPS distribution within each work arrangement using household weights. For paid employed workers, I use $K = 2$, where the paid employment centroids track the modal part-time and full-time mass points of the paid employed hours distribution, which is unimodal for husbands (spike near 40) and bimodal for wives (part-time near 17 and full-time near 37).[62] For self-employed workers, I use $K = 5$ to capture the broader dispersion of self-employed hours. For home production hours, I do not adjust by work arrangement because this is beyond firms' scope to control workers' hours. Table F.11 reports the resulting centroids alongside empirical quantiles.

Table F.11: Discrete grid points (k-means centroids), by gender

		Quantile					k-means centroid				
Variable	Arrangement	p10	p30	p50	p70	p90	c_1	c_2	c_3	c_4	c_5
Wife											
Work hours	PE ($K = 2$)	15.0	22.0	35.0	37.0	40.0	17.6	36.7	–	–	–
Work hours	SE ($K = 5$)	10.0	20.0	30.0	40.0	54.0	15.6	34.9	47.8	58.9	76.0
Home prod	All ($K = 5$)	5.0	10.0	14.0	20.0	30.0	9.6	20.8	29.3	35.1	43.7
Husband											
Work hours	PE ($K = 2$)	35.0	37.0	39.0	40.0	46.0	32.8	41.0	–	–	–
Work hours	SE ($K = 5$)	35.0	40.0	50.0	55.0	70.0	22.7	39.0	48.3	59.4	75.9
Home prod	All ($K = 3$)	1.0	2.0	4.0	7.0	10.0	2.9	8.3	16.0	–	–

Note: Cells report empirical quantiles (p10–p90) and weighted k-means centroids (c_1–c_5) used in the structural estimation. Work hours grids are arrangement-specific: 2-point under paid employment (rigid schedules) and 5-point under self-employment (flexible hours). Home production grids are common across work arrangements within gender.Grid size follows the pooled k-means fit ladder.

Figures F.12 to F.14 show the weighted k-means centroids used by the structural grid on the marginal distributions, comparing alternative grid sizes. For paid employment, husband hours are close to normal at full-time hours with a thin right tail, hence the $K = 2$ centroids align with the two modes, while the third $K = 3$ centroid lands in the long-hours tail rather than on a modal mass point. For self-employment, the $K = 5$ centroids track the modes of both wife and husband self-employed hours; the $K = 6$ centroids nearly overlap $K = 5$, indicating that the sixth centroid has no distinct mass to summarize. For home production, the $K = 5$ centroids align with the modes of the marginal distribution for the wife and $K = 3$ for the husband.

[62] The discrete K-means grid is an approximation to the continuous hours technology that converges as K increases. the K grid points minimize $\int \min_k (h^* - h_k)^2 f(h^*)\, dh^*$, so adding further grid points leaves estimated moments unchanged once the key modes are captured. Paid employment hours exhibit discrete bunching consistent with firms offering tied wage-hours packages rather than a continuous menu (Rosen, 1976; Altonji and Paxson, 1988). Wives cluster near 17 and 37 hours per week, and $K = 2$ captures these two modes without approximation loss. The self-employment grid uses $K = 5$ because self-employed workers face a more continuous hours technology, and the richer grid approximates the wider distribution.

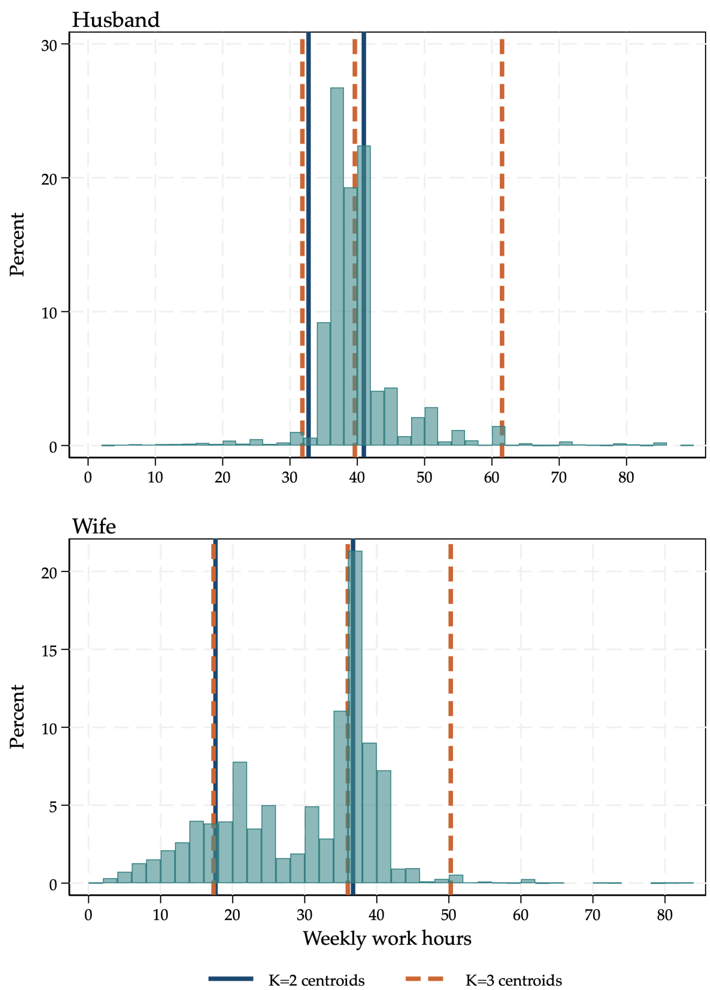


Figure F.12: Distribution of PE weekly work hours with $K = 2$ and $K = 3$ weighted k-means centroids

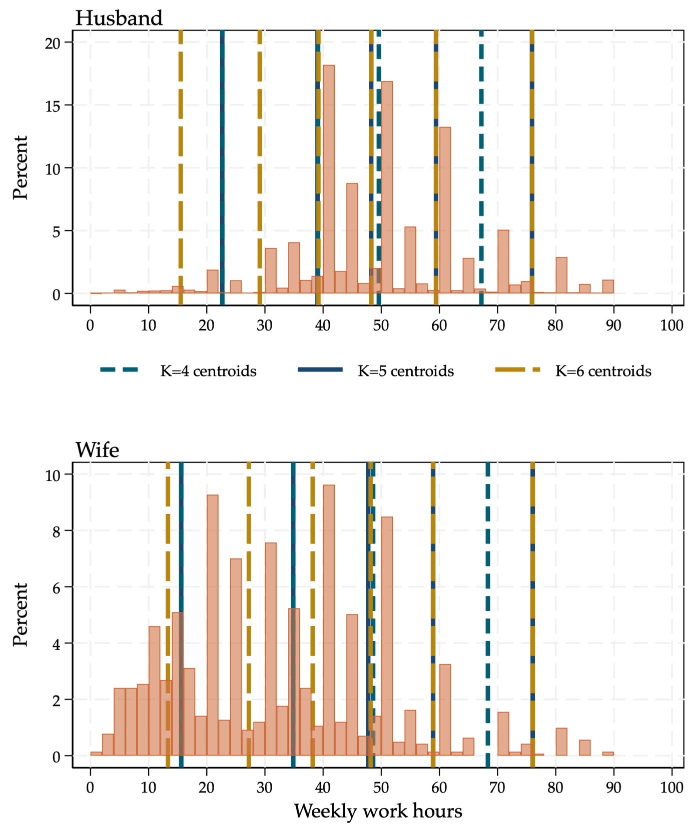


Figure F.13: Distribution of SE weekly work hours with $K = 4$, $K = 5$, and $K = 6$ weighted k-means centroids

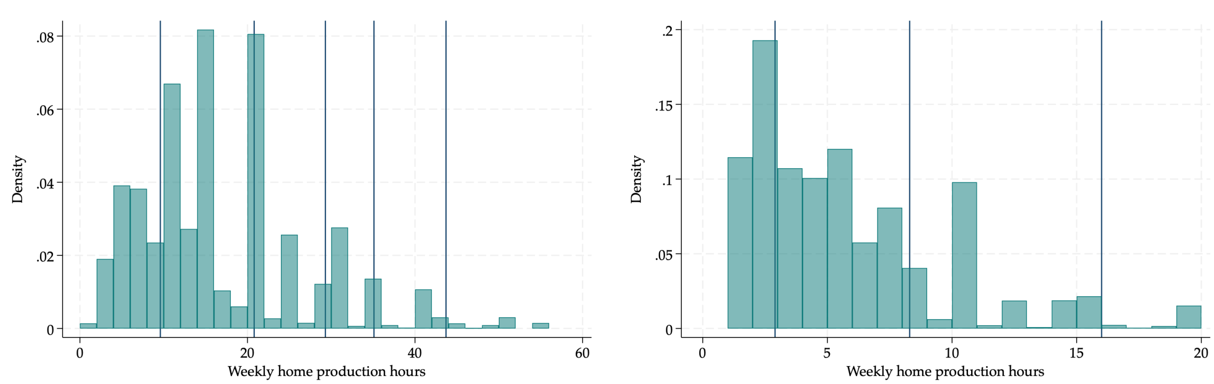


Figure F.14: Distribution of weekly home production hours, pooled across work arrangements, with k-means centroids ($K = 5$ for the wife, $K = 3$ for the husband)

Appendix F.4 Flexibility Policy Expanded Grids

The flexibility policy counterfactual expands only the paid employment work hours menu for the targeted parent, from $K = 2$ to $K = 7$. Home production is common conditional on gender and the policy leaves it unchanged, since firms set work hours but not home production. The expanded menu spans the statutory work band, between the 16-hour threshold that defines work for tax-credit purposes and the Working Time Regulations 48-hour weekly cap; it retains the two baseline paid-employment points and fills the intermediate and longer hours the rigid full-time-or-nothing market does not supply, so its spacing is not uniform but anchors on the baseline points. I construct the menu this way, rather than from the realized self-employed hours distribution, because self-employed hours are skewed toward long hours by selection into self-employment, whereas flexibility grants control over the schedule.[63]

Table F.12: Paid Employment Flexibility Expanded Grids

	Flexible Hours						
	1	2	3	4	5	6	7
Wife							
Work hours n^w (h/wk)	16.0	17.6	24.0	30.0	36.7	41.0	44.0
Husband							
Work hours n^h (h/wk)	24.0	28.0	32.8	37.0	41.0	45.0	48.0

Note: The paid employment work hours menu under the flexibility policy contains the two estimated baseline paid-employment points (wife 17.6/36.7, husband 32.8/41.0) and fills the intermediate and surrounding hours. The wife's floor (16) is the 16-hour tax-credit work threshold and the husband's ceiling (48) is the Working Time Regulations 48-hour cap; each menu spans the part-time to modestly-above-full-time range the respective parent works, within the statutory band. The menu represents the choice set flexibility provides, not the self-employed hours distribution, which is skewed toward long hours by selection into self-employment. Home production is unchanged by the policy and follows the common per-gender menu used throughout the model.

Appendix F.5 Right to Request Flexible Working: Reduced-Form Evidence

The 2003 introduction of the Right to Request Flexible Working offers a natural experiment for whether the reform changed behavior over my sample. From April 2003, employees who are parents of a child under six gained the right to request flexible working, whereas parents of older children did not. I estimate a difference-in-differences on the BHPS estimation sample, comparing mothers whose youngest child is under six (eligible) with mothers whose youngest is aged six to

[63] A grid-convergence check confirms the result is not an artifact of the discretization: the child-skill effects are stable as the hours menu is refined from five to seven to nine points. Coarser menus concentrate choices on the top rung and finer menus spread them across intermediate hours, yet the effect holds, so it reflects genuine hours choices rather than bunching at a single grid point.

eleven (ineligible), before and after the reform,

$$y_{it} = \beta\,(\text{Eligible}_{it} \times \text{Post}_t) + \gamma\,\text{Eligible}_{it} + \mathbf{X}_{it}'\delta + \eta_s + \lambda_t + \varepsilon_{it},$$

where eligibility varies with the youngest child's age, η_s are region fixed effects, λ_t are year fixed effects, and $\mathbf{X}_{it}$ includes age, age squared, college degree, and the number of children. Standard errors are clustered at the individual level. The outcomes are employment, part-time and full-time paid employment, self-employment, weekly hours, and, as the most direct first stage, whether the mother reports flexitime.

Flexitime, a shift in when rather than how many hours, does not map one-to-one to the model's hours menu, and the reform does not move it. Table F.13 shows no significant flexitime response (0.02, s.e. 0.02), matching Xue et al. (2025), who likewise find no flexitime or teleworking response to the 2014 expansion. The margin the model prices, hours, points the other way. The eligible group shifts toward part-time, a 6.6 percentage-point rise, with fewer weekly hours, but this movement fails the pre-trend. Xue et al. (2025) identify the response cleanly on the 2014 expansion, where women's reduced-hours uptake rises from about 3 to 10 percent over six years, with no response among men, an asymmetry consistent with the mother being the margin of adjustment in the model. Fathers in my sample, equally eligible, show no response (Table F.14). The within-sample estimate finds no significant flexitime response, though the confidence interval is too wide to serve as an equivalence bound. I therefore rely on the external evidence of Xue et al. (2025) for the reduced-hours response the structural counterfactual then evaluates at full bite, granting full access to the flexible hours menu.

Table F.13: Right to Request Flexible Working: Difference-in-differences, mothers

	(1) Employed	(2) Part-time	(3) Full-time	(4) Self-employed	(5) Hours	(6) Flexitime
Eligible	-0.148***	-0.105***	-0.057***	0.012	-1.260**	0.002
	(0.018)	(0.021)	(0.018)	(0.009)	(0.514)	(0.010)
Eligible × Post	0.034	0.066**	-0.026	-0.012	-0.821	0.022
	(0.025)	(0.031)	(0.027)	(0.015)	(0.704)	(0.023)
Age	0.074***	0.058***	0.016	0.008	0.133	0.016**
	(0.011)	(0.012)	(0.011)	(0.005)	(0.341)	(0.008)
$Age^2/100$	-0.103***	-0.081***	-0.023	-0.007	-0.219	-0.021**
	(0.015)	(0.017)	(0.015)	(0.008)	(0.480)	(0.011)
College degree	0.118***	0.018	0.106***	0.005	2.490***	-0.019
	(0.018)	(0.021)	(0.020)	(0.010)	(0.521)	(0.013)
Number of children	-0.080***	0.019	-0.110***	0.018***	-3.082***	-0.020***
	(0.011)	(0.013)	(0.011)	(0.006)	(0.322)	(0.007)
Observations	11929	11234	11234	11929	7862	8072

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$
Note: Difference-in-differences estimates for mothers with youngest child aged 0–11 in the BHPS. Eligible indicates a youngest child under six, treated after the April 2003 reform. Controls are age, age squared, college degree, and number of children, with region and year fixed effects; the post main effect is absorbed by the year effects. Standard errors clustered by individual in parentheses. The Eligible × Post coefficient is the reform effect. Flexitime is measured only from 1999 onwards.

Table F.14: Right to Request Flexible Working: Difference-in-differences, fathers

	(1) Part-time	(2) Full-time	(3) Self-employed	(4) Hours	(5) Flexitime
Eligible	0.003	-0.003	0.015	-0.365	0.010
	(0.006)	(0.006)	(0.016)	(0.298)	(0.008)
Eligible $\times$ Post	-0.003	0.003	0.018	0.526	0.016
	(0.013)	(0.013)	(0.024)	(0.466)	(0.019)
Age	-0.007	0.007	0.000	0.171	0.005
	(0.004)	(0.004)	(0.009)	(0.143)	(0.005)
Age^2/100	0.010*	-0.010*	0.008	-0.274	-0.005
	(0.006)	(0.006)	(0.012)	(0.193)	(0.006)
College degree	0.015**	-0.015**	-0.111***	-1.750***	0.061***
	(0.008)	(0.008)	(0.017)	(0.328)	(0.013)
Number of children	0.002	-0.002	0.033***	0.049	-0.004
	(0.004)	(0.004)	(0.011)	(0.187)	(0.006)
Observations	9746	9746	11933	9746	8466

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Difference-in-differences estimates for mothers with youngest child aged 0–11 in the BHPS. Eligible indicates a youngest child under six, treated after the April 2003 reform. Controls are age, age squared, college degree, and number of children, with region and year fixed effects; the post main effect is absorbed by the year effects. Standard errors clustered by individual in parentheses. The Eligible $\times$ Post coefficient is the reform effect. Flexitime is measured only from 1999 onwards. Employment is ommitted because I assume fathers always work.

Appendix G Model Timeline and Computational Details

Appendix G.1 Intratemporal Timeline

The model periods fall into three stages: Stage A (pre-birth), Stage B (young child, ages 1–5), and Stage C (older child, age 7+). Transitions are marked by t^b (the period of birth) and t^B (the period in which the youngest child turns 6).[64]

1. **Stage A**: $t \leq \min\{t^b, T\}$. The household draws work arrangement-specific preference shocks $\boldsymbol{\xi}_t$ and chooses work arrangement q. Income shocks ε_t are drawn (jointly for both spouses if the wife works, husband only otherwise). Given state-specific shocks $\boldsymbol{\zeta}_t$, the household chooses assets, hours, and home production. The wife's human capital evolves, and a child arrives with probability p_t^w estimated from the data as a function of the wife's age, education, and existing number of children (Table I.15); if a child is born, the household transitions to Stage B.

2. **Stage B**: $t^b < t < t^B$. The same sequence as Stage A, with the addition that the household also chooses whether to send the child to formal childcare, determining child-related expenditures. The household remains in Stage B until the youngest child turns 6, then transitions to Stage C.

3. **Stage C**: $t^B \leq t \leq T$. Mirrors Stage A in structure with an older child present; the choice set is unchanged but continuation values reflect the absence of future young-child periods.

Appendix G.2 Model Solution

1. *State space reduction.* Starting from observed individual characteristics from the data, I first enumerate all reachable state space combinations. This reduces the effective state space and substantially eases computation.

2. *Parental education.* Since parental education is a time-invariant binary variable (1 if college-educated, 0 otherwise), I solve the model separately for each of the four possible combinations of parental education levels.

3. *Backwards induction and terminal period setup.* I solve the model by backwards induction, beginning at Stage 2 of the terminal period T. The state space at this stage consists of the mother's human capital (h_T^w), current assets (a_T), number of children (k_T), child's age (g_T), next-period assets (a_{T+1}), income shocks ($\boldsymbol{\varepsilon}_T$), and the choices carried over from the intratemporal Stage 1. Income shocks are discretized via Gaussian quadrature applied to each parent's income process, using the corresponding variance-covariance matrix to obtain nodes and weights. The action space varies with child presence: when no child is present

[64] Households may start in Stage A or B depending on initial conditions. Once the child surpasses age 5, the household enters Stage C. Depending on the arrival of subsequent children, the youngest child may replace the previous one as the focal child.

(either not yet born or after age 5, as described in Section 4), parents allocate time between labor, leisure, and home production; when a child is between ages 1 and 5, they additionally choose whether to use childcare.

4. *Value function computation.* For each point in the state space, I compute the current-period utility and the discounted expected continuation value. I construct the ex-ante value function separately for the cases with and without a child, then combine them using the probability of having a child (p_T^w).[65] The taste shocks over time preferences are then integrated out to obtain the ex-ante value function.

5. *Human capital accumulation.* Depending on the wife's work arrangement choice and hours worked, her human capital evolves according to a stochastic process, which I account for at each node of the state space.

6. *Integration over income and discrete choices.* For each income-shock realization, I compute the Stage-2 ex-ante value function over next-period assets and childcare choices, integrating out the state-specific preference shocks $\boldsymbol{\zeta}_T$. I then integrate out the income shocks using the Gaussian quadrature weights. Households therefore observe the income shock before choosing hours, home production, savings, and childcare.

7. *Work arrangement choice.* At Stage 1, I compute the ex-ante value function over work arrangement choices and integrate out the work-arrangement-specific preference shocks $\boldsymbol{\xi}_T$.

8. *Iteration.* The ex-ante value function obtained at Step 7 serves as the expected continuation value for period $T-1$. Steps 3–7 are then repeated, iterating backwards until the initial period is reached.

Appendix G.3 Model Simulation

Since parental education is time-invariant, there are four possible combinations of spousal education levels: (college, college), (college, no college), (no college, college), and (no college, no college). The simulation then proceeds by initializing each woman at her observed starting age and simulating forward through the life cycle.

1. *Model selection.* At the beginning of each period, I select the model solution corresponding to the couple's education combination.

2. *Stage 1: Work arrangement choice.* I extract the Stage 1 solution, draw work-arrangement-specific preference shocks, and assign the household the work arrangement that yields the highest utility.

[65] At the terminal period, the fertility probability is zero by assumption, since the fertile age range is 21–45.

3. *Stage 2: Hours and savings.* I draw the household's income shock and condition the Stage-2 choice on the realized value, then I extract the Stage 2 solution, draw the relevant state-specific preference shocks, and select the combination of hours worked and next-period assets that maximizes the ex-post value function. If there is a young child, the action space includes childcare choice.The chosen next-period assets a_{t+1} carry forward as the asset state entering the following period.

4. *Human capital update.* Given the chosen work arrangement and hours, I update the wife's human capital according to the accumulation process.

5. *Fertility.* The household draws a fertility shock. If the realization falls below the period-specific fertility probability, a new child is born.

6. *Child skill initialization.* When a new child is born, I draw her initial joint skill vector $(\log \theta_{cog,1}, \log \theta_{ncog,1})$ from $\mathcal{N}(\mathbf{0}, \widehat{\Sigma}_0)$ using the Cholesky decomposition $\widehat{\Sigma}_0 = LL'$, where L is estimated from wave-1 MCS data (Appendix G.6).

7. *Child skill update.* An existing child's cognitive and non-cognitive skills evolve through the production function given the period's parental time and childcare inputs. By log-additivity this does not affect the second-stage behavioral moments, but it generates the age-7 skill paths used in the counterfactuals.

8. *Moment computation.* Steps 1–7 are repeated for all individuals and simulation draws. Household choices and characteristics are stored locally, and the simulated moments are computed as described in Table 2.

Appendix G.4 Model Estimation

The model is estimated by Simulated Method of Moments (SMM). The procedure is as follows.

1. *Model solution.* Given an initial parameter vector, I solve the model following the steps in Appendix G.2.

2. *Simulation.* Using the model solution, I simulate the life cycle forward and compute the simulated moments as described in Appendix G.3.

3. *Minimization.* I compute the weighted distance between the data moments and the simulated moments and evaluate the SMM objective function. I then minimize the objective function over the parameter space to obtain the parameter vector that best aligns the simulated moments with their empirical counterparts.

Appendix G.5 Weighting Matrix and Standard Errors

The weighting matrix is diagonal with elements equal to the inverse bootstrap variance of each data moment, scaled by a dataset-specific constant R_d that adjusts for differences in sample size and measurement precision across BHPS and MCS:[66]

$$W = \text{diag}\left(\frac{1}{\widehat{\text{Var}}(\mathcal{M}_d) \cdot R_d}\right). \tag{G.4}$$

The SMM estimator satisfies (Duffie and Singleton, 1993)

$$\sqrt{N}(\hat{\Theta} - \Theta_0) \xrightarrow{d} \mathcal{N}(0, V), \tag{G.5}$$

where Θ_0 is the true parameter vector and

$$V = \left(1 + \tfrac{1}{S}\right) (G'WG)^{-1} G'W\Omega WG\, (G'WG)^{-1}, \tag{G.6}$$

with $G \equiv \partial\mathcal{M}(\Theta_0)/\partial\Theta' \in \mathbb{R}^{M \times K}$ the Jacobian of the simulated moments, $\Omega \equiv \text{Var}(\sqrt{N}\,\hat{\mathcal{M}})$ the asymptotic variance of the data moments, and $(1 + 1/S)$ accounting for simulation noise. I approximate G by central finite differences with step size $h_k \propto |\hat{\Theta}_k|$, fixing the random seed across evaluations to eliminate simulation noise. I estimate Ω by nonparametric bootstrap over the data moments. Standard errors are $\sqrt{\text{diag}(\hat{V})}$ where

$$\hat{V} = \left(1 + \tfrac{1}{S}\right) (\hat{G}'W\hat{G})^{-1} \hat{G}'W\hat{\Omega}W\hat{G}\, (\hat{G}'W\hat{G})^{-1}. \tag{G.7}$$

Appendix G.6 Initial Distribution of Child's Skills

To assess whether initial skills depend on family characteristics, I regress initial skill measures on family income, siblings, low birth weight, and parental education following Garcia-Vazquez (2023). The income coefficients are not statistically significant ($p = 0.98$ for cognitive, $p = 0.71$ for non-cognitive), and the raw correlations confirm the same pattern (Moschini, 2023). Since the level of initial skills is not separately identified from pre-natal investment, I normalize the initial distribution to be mean-zero and common across households: $(\log\theta_{cog,1}, \log\theta_{ncog,1}) \sim \mathcal{N}(\mathbf{0}, \widehat{\Sigma}_0)$. The estimated variances are $\widehat{\sigma}^2_{cog} = 0.199$ and $\widehat{\sigma}^2_{ncog} = 0.229$, with an implied cross-domain correlation of $\widehat{\rho}_{cog,ncog,1} = -0.12$, consistent with near-orthogonality of cognitive and non-cognitive skills at infancy.

In the first wave ($g_t = 1$), I observe three measures for each skill domain $D \in \{cog, ncog\}$. The

[66] The optimal weighting matrix is the inverse of the full variance–covariance matrix of the moments (McFadden, 1989; Pakes and Pollard, 1989; Duffie and Singleton, 1993). The diagonal inverse-variance approximation is a numerically stable, consistent alternative (Gourieroux and Monfort, 1996; Attanasio, Low, and Sánchez-Marcos, 2008).

measurement system for the latent skill $\theta_{D,1}$ is:

$$\widetilde{\theta}^m_{D,1} = \mu^m_{\theta_D,1} + \alpha^m_{\theta_D,1} \log \theta_{D,1} + \varepsilon^m_{\theta_D,1}, \quad m \in \{1,2,3\}, \tag{G.8}$$

where measurement errors are uncorrelated across measures. Normalizing one reference measure ($\mu^1_{\theta_D,1} = 0$, $\alpha^1_{\theta_D,1} = 1$), the off-diagonal covariances satisfy:

$$s_{12} = \alpha^2_{\theta_D,1}\sigma^2_{\theta_D,1}, \quad s_{13} = \alpha^3_{\theta_D,1}\sigma^2_{\theta_D,1}, \quad s_{23} = \alpha^2_{\theta_D,1}\alpha^3_{\theta_D,1}\sigma^2_{\theta_D,1}, \tag{G.9}$$

yielding the closed-form estimator:

$$\widehat{\sigma}^2_{\theta_D,1} = \frac{s_{12}s_{13}}{s_{23}}. \tag{G.10}$$

For cognitive skills, the three measures are the gross motor, social, and fine motor subscales of the Denver Developmental Screening Test. For non-cognitive skills, they are the Mood and Adaptability subscales of the Carey Infant Temperament Scale and the sleep-time regularity item from the Regularity subscale. The cross-domain covariance $\widehat{\text{cov}}(\log \theta_{cog,1}, \log \theta_{ncog,1})$ is estimated from the covariance between the reference measures under independence of measurement errors across domains.

In the simulation, initial skills are drawn using the Cholesky factor L of $\widehat{\Sigma}_0 = LL'$: for each (household, simulation draw, child) triple, two standard normals (z_1, z_2) are drawn and $\log \theta_{cog,1} = L_{11}z_1$, $\log \theta_{ncog,1} = L_{21}z_1 + L_{22}z_2$. A fixed seed is used so that initial skill draws are identical across counterfactual scenarios, enabling paired comparisons.

Appendix H Identification of Preference Parameter Primitives

This appendix establishes identification of the structural parameters of household preferences, parental income, wife's human capital via SMM. I suppress time subscripts for clarity.

Assets are measured with error: $\tilde{a} = a + \epsilon^a$, where ϵ^a has mean zero and is mean-independent of household choices and states. Child expenditures are also measured with error: $\tilde{M} = M + \epsilon^M$. Consumption is not observed directly; I construct it from the budget constraint (Equation 5) as $\tilde{C} = (1+r)\tilde{a} + \mathcal{T} - \tilde{M} - \tilde{a}'$, using observed assets, next-period assets, child expenditure, and net income, so that $\tilde{C} = C + \epsilon^C$, where

$$\epsilon^C = (1+r)\epsilon^a - \epsilon^{a'} - \epsilon^M.$$

Each component error is mean zero and mean-independent of choices and states, so the same holds for ϵ^C.

Proposition 2. Under the following conditions, the preference parameters and the hours-earnings curvature parameters $\eta_{i,j}$ are identified.

(i) **Tax system regularity.** Conditional on positive hours, $\mathcal{T}(\cdot)$ is differentiable in earnings with $\frac{\partial \mathcal{T}}{\partial Y^i_j} \neq 0$ for $i \in \{w, h\}$, $j \in \{PE, SE\}$.[a]

(ii) **Observability.** Given $n^i > 0$, the joint distribution of $\left(\tilde{C},\, n^i,\, l^i,\, \tau^i,\, \frac{\partial \mathcal{T}}{\partial n^i}\right)$ is observed for $i \in \{w, h\}$. Given $k_t > 0$ and $0 < \tau^i < \bar{T}$, the distribution of (τ^i, l^i) is observed.

(iii) **Measurement error independence.** $(\epsilon^a, \epsilon^{a'}, \epsilon^M)$ are mutually independent and independent of choices, states, and time.

(iv) **Positive consumption.** $\mathbb{E}[\tilde{C} \mid n^i > 0] > 0$ for $i \in \{w, h\}$.

(v) **Work arrangement variation.** Both paid employment and self-employment are observed for each spouse, and the wife is observed in paid employment, self-employment, and non-working. The young-child indicator $\mathbb{1}[ychild = 1]$ and childcare indicators $\mathbb{1}[cc = PT]$, $\mathbb{1}[cc = FT]$ each take both values in the data conditional on each work arrangement.

[a]This follows Gayle and Shephard (2019) and ensures that the tax system does not fully offset marginal changes in earnings.

Appendix H.1 Identification Under Fixed Work Arrangements

Identifying the base leisure preference α^w_l separately from the arrangement-leisure interaction $\alpha^w_{PE,1}$ requires a household configuration that removes the interaction from the first-order conditions. A non-working wife supplies no market labor, so no arrangement-leisure interaction enters her time allocation decision, and conditioning on this configuration in the first step pins down α^w_l directly. The second step then recovers $\alpha^w_{PE,1}$ as the deviation from this baseline, using variation in time allocation across working wives.

I use two household configurations to isolate the intratemporal identification of core preference and technology parameters. The first is $q = 3$ (husband paid employed, wife not working), which anchors the wife's leisure preference α^w_l without work arrangement-leisure interactions. The second is $q = 1$ (both spouses paid employed), which pins down the composite $\alpha^w_l + \alpha^w_{PE,1}$ through the consumption-tax link and thereby identifies the work arrangement-leisure interaction $\alpha^w_{PE,1}$ as a deviation from the non-working anchor.

Anchor: Non-Working Wife ($q \in \{3, 6\}$)

When the wife does not work ($n^w = 0$, $d^w_{NW} = 1$), no work arrangement-leisure interaction enters her utility. The time constraint reduces to $l^w + \tau^w = \bar{T}$, and the relevant FOCs are:

$$\frac{\partial \mathcal{L}}{\partial l^w}: \quad \frac{\alpha^w_l}{l^w} = \lambda^{TC_w} \tag{H.11}$$

$$\frac{\partial \mathcal{L}}{\partial \tau^w}: \quad \frac{\alpha^w_\tau + \alpha^w_{\tau,k}\mathbb{1}[ychild] + \Psi^w \mathbb{1}[ychild]}{\tau^w} = \lambda^{TC_w} \tag{H.12}$$

where $\Psi^w = \beta \sum_D \Gamma_{D,g_t+1}(\gamma_{D,3} + \gamma_{D,4}\,\theta^w)$ collects the child production function returns to maternal time, weighted by the shadow value Γ_{D,g_t+1} of next-period skills from Proposition 1. Equating (H.11) and (H.12) in the no-child subsample ($\mathbb{1}[ychild] = 0$, so Ψ^w drops out):

$$\frac{\alpha^w_l}{\alpha^w_\tau} = \frac{l^w_{NW}}{\tau^w_{NW}} \tag{H.13}$$

This identifies the ratio $\alpha^w_l / \alpha^w_\tau$ where home production is interior. The base level α^w_τ is identified from home production in the no-young-child cells, where, through the discrete choice over the home-production grid, a more negative α^w_τ shifts mass toward lower home production, and it is separated from the young-child interaction $\alpha^w_{\tau,k}$ by the contrast with the young-child allocations.

Level: Working Wife

The Stage 2 Lagrangian, conditional on $q = 1$ (both in paid employment), assets a', and childcare choice cc, is:

$$\begin{aligned}
\mathcal{L} = {} & \log C + (\alpha^w_l + \alpha^w_{PE,1}) \log l^w + \alpha^h_l \log l^h \\
& + [\alpha^w_\tau + \alpha^w_{\tau,k}\mathbb{1}[ychild]] \log \tau^w + [\alpha^h_\tau + \alpha^h_{\tau,k}\mathbb{1}[ychild]] \log \tau^h \\
& + \beta \sum_D \Gamma_{D,g_t+1}\Big[\log A_D + \gamma_{D,1} \log \theta_D + \gamma_{D,2} \log \theta_{D'} + \gamma_{D,3} \log \tau^w + \gamma_{D,4} \log \tau^w \cdot \theta^w \\
& \qquad + \gamma_{D,5} \log \tau^h + \gamma_{D,6} \log \tau^h \cdot \theta^h + \gamma_{D,7} cc^{PT} + \gamma_{D,8} cc^{FT} \\
& \qquad + \gamma_{D,9} cc^{PT} \theta^w + \gamma_{D,10} cc^{FT} \theta^w + \gamma_{D,11} sib \Big] \\
& + \beta \mathbb{E} \bar{V}(a', hc^{w\prime}) \\
& + \lambda^{TC_w}[\bar{T} - l^w - \tau^w - n^w] + \lambda^{TC_h}[\bar{T} - l^h - \tau^h - n^h] \\
& + \lambda^{BC}[(1+r)a + \mathcal{T}(Y^w_{PE}, Y^h_{PE}, \mathbf{x}) - M - C - a']
\end{aligned}$$

where $\alpha^w_{PE,1}$ is the paid employment-leisure interaction and earnings $Y^i_{PE} = y^i_{PE}(n^i)^{\eta_{i,PE}}$ are substituted directly into the net-income function $\mathcal{T}(\cdot)$. Childcare and work arrangement-child interaction terms that do not affect continuous FOCs conditional on (q, cc) are absorbed into additive constants.

Taking first-order conditions:

$$\frac{\partial \mathcal{L}}{\partial C}: \quad \frac{1}{C} = \lambda^{BC} \tag{H.14}$$

$$\frac{\partial \mathcal{L}}{\partial l^w}: \quad \frac{\alpha_l^w + \alpha_{PE,1}^w}{l^w} = \lambda^{TC_w} \tag{H.15}$$

$$\frac{\partial \mathcal{L}}{\partial l^h}: \quad \frac{\alpha_l^h}{l^h} = \lambda^{TC_h} \tag{H.16}$$

$$\frac{\partial \mathcal{L}}{\partial n^w}: \quad \lambda^{TC_w} = \lambda^{BC} \frac{\partial \mathcal{T}}{\partial n^w} \tag{H.17}$$

$$\frac{\partial \mathcal{L}}{\partial n^h}: \quad \lambda^{TC_h} = \lambda^{BC} \frac{\partial \mathcal{T}}{\partial n^h} \tag{H.18}$$

$$\frac{\partial \mathcal{L}}{\partial \tau^w}: \quad \frac{\alpha_\tau^w + \alpha_{\tau,n}^w \, n^w/\bar{n} + \alpha_{\tau,k}^w \mathbb{1}[ychild] + \Psi^w \mathbb{1}[ychild]}{\tau^w} = \lambda^{TC_w} \tag{H.19}$$

$$\frac{\partial \mathcal{L}}{\partial \tau^h}: \quad \frac{\alpha_\tau^h + \alpha_{\tau,n}^h \, n^h/\bar{n} + \alpha_{\tau,k}^h \mathbb{1}[ychild] + \Psi^h \mathbb{1}[ychild]}{\tau^h} = \lambda^{TC_h} \tag{H.20}$$

where Ψ^w is as defined in the non-working anchor block and $\Psi^h = \beta \sum_D \Gamma_{D,g_t+1}(\gamma_{D,5} + \gamma_{D,6}\,\theta^h)$ is the analogous term for paternal time. With the CPF parameters known from external estimation, Γ_{D,g_t+1} is a known linear function of $(\alpha_{cog}, \alpha_{ncog})$.

The marginal net-of-tax return to the wife's hours follows from Equation 6 and the income technology by the chain rule:

$$\frac{\partial \mathcal{T}}{\partial n^w} = \chi_2(\mathbf{x})\,\chi_3(\mathbf{x})\,(Y_{PE}^w + Y_{PE}^h)^{\chi_3(\mathbf{x})-1} \cdot \eta_{w,PE}\, y_{PE}^w\,(n^w)^{\eta_{w,PE}-1} \tag{H.21}$$

The tax parameters (χ_1, χ_2, χ_3) are known. Both y_{PE}^w and n^w are observed. An analogous expression holds for $\partial\mathcal{T}/\partial n^h$. Hours also enter utility through the workload term, contributing $\alpha_{\tau,n}^i(\log \tau^i)/\bar{n}$ to (H.17)–(H.18), and, for the wife, the continuation value through the human-capital transition $\Pi(n^w)$. The sequential argument holds these two channels fixed: the workload interaction $\alpha_{\tau,n}^i$ is identified from the within-person slope of home production in own hours (Step 6), and the human-capital parameters from wage growth and re-entry earnings, so conditional on these the displayed conditions identify the baseline leisure and home-production weights.

1. **Consumption path.** Combining the FOC for a' with (H.14) and the Envelope Theorem yields the Euler equation $1/C = \beta(1+r)\,\mathbb{E}[1/C']$. With β and r preset, the Euler equation governs how the marginal utility of consumption grows across periods. Its level is fixed by the budget constraint together with the borrowing and terminal conditions.

2. **Hours-earnings curvature $\eta_{i,j}$.** The income technology gives $\log Y_j^i = \log y_j^i + \eta_{i,j} \log n^i$. Weekly earnings Y_j^i and hours n^i are both observed. Conditional on the income equation parameters, $\log y_j^i$ is known, and $\eta_{i,j}$ is identified from the work arrangement-specific gradient of $\log Y_j^i$ with respect to $\log n^i$. The model matches the observed nonlinear hours-earnings

profile separately for paid and self-employment, where the curvature reflects the degree of hours flexibility documented in Appendix F.

3. **Husband's leisure preference α_l^h.** Since husband PE is the reference ($\alpha_{PE,1}^h = 0$), the husband's leisure FOC (H.16) contains α_l^h alone. Combining (H.14), (H.16), and (H.18) at an interior solution:

$$\alpha_l^h = \frac{\mathbb{E}\left[l^h \frac{\partial \mathcal{T}}{\partial n^h} \,\middle|\, n^h > 0\right]}{\mathbb{E}[\tilde{C} \mid n^h > 0]}$$

The condition is linear in consumption, $\alpha_l^h C = l^h \partial \mathcal{T} / \partial n^h$, so taking expectations over working households and using $\mathbb{E}[\tilde{C}] = \mathbb{E}[C]$ under the mean-zero measurement error gives the moment above. The household-varying l^h stays inside the expectation, and dividing by $\mathbb{E}[\tilde{C}]$ rather than by $\tilde{C}$ avoids the bias in $\mathbb{E}[1/\tilde{C}]$. Condition (i) ensures the numerator is nonzero; condition (iv) ensures the denominator is positive. Both l^h and the marginal tax rate are computable from data and the known tax system.

4. **Wife's leisure preference α_l^w and paid employment-leisure interaction $\alpha_{PE,1}^w$.** Combining (H.14), (H.15), and (H.17) at an interior solution:

$$\alpha_l^w + \alpha_{PE,1}^w = \frac{\mathbb{E}\left[l^w \frac{\partial \mathcal{T}}{\partial n^w} \,\middle|\, n^w > 0, q = 1\right]}{\mathbb{E}[\tilde{C} \mid n^w > 0, q = 1]} \tag{H.22}$$

This identifies the composite $(\alpha_l^w + \alpha_{PE,1}^w)$ at the level, through the consumption-tax link, and the non-working wife's leisure and participation margin pin α_l^w, leaving $\alpha_{PE,1}^w$ as the residual. For home production, the paid employed wife's τ/l ratio from (H.15) and (H.19),

$$\frac{\tau_{PE}^w}{l_{PE}^w} = \frac{\alpha_\tau^w + \alpha_{\tau,n}^w \, n^w/\bar{n}}{\alpha_l^w + \alpha_{PE,1}^w}$$

is interior only when a young child is present; without one, home production is at its lower bound and the ratio does not hold with equality. The home-production weights are therefore identified from the young-child working allocations (Step 5): that ratio pins the composite $\alpha_\tau^w + \alpha_{\tau,n}^w \, n^w/\bar{n} + \alpha_{\tau,k}^w + \Psi^w$, the within-person slope of home production in own hours (Step 6) separates the workload interaction $\alpha_{\tau,n}^w$, the education variation separates the child-skill return Ψ^w, and the base level α_τ^w is pinned by the no-young-child home production as above.

5. **Home production and child skill preferences $(\alpha_\tau^h, \alpha_{\tau,k}^w, \alpha_{\tau,k}^h, \alpha_{cog}, \alpha_{ncog})$.** The husband's home-production weights are identified symmetrically. The no-young-child home-production level identifies the base weight α_τ^h, his within-person hours slope identifies the workload interaction $\alpha_{\tau,n}^h$, and the young-child interaction $\alpha_{\tau,k}^h$ is identified in the young-child system below, given α_l^h from Step 3.

   The remaining parameters $(\alpha_{\tau,k}^w, \alpha_{\tau,k}^h, \alpha_{cog}, \alpha_{ncog})$ are identified from the young-child subsample ($\mathbb{1}[ychild] = 1$). The τ/l ratio becomes:

$$\frac{\tau^i}{l^i} = \frac{\alpha^i_\tau + \alpha^i_{\tau,k} + \beta \sum_D \Gamma_{D,g_t+1}(\gamma^i_D + \gamma^i_{D,\theta}\,\theta^i)}{\alpha^i_l}$$

where I use shorthand $\gamma^w_D = \gamma_{D,3}$, $\gamma^w_{D,\theta} = \gamma_{D,4}$, $\gamma^h_D = \gamma_{D,5}$, $\gamma^h_{D,\theta} = \gamma_{D,6}$. The shadow value Γ_{D,g_t+1} appears because today's time investment produces next-period skills, which are valued at the Proposition 1 coefficient rather than the flow weight α_D alone. Evaluating at both education levels for both spouses delivers four equations:

$$\begin{aligned}
\text{Wife, } \theta^w = 0: &\quad \alpha^w_{\tau,k} + \beta(\Gamma_{cog,g_t+1}\,\gamma_{cog,3} + \Gamma_{ncog,g_t+1}\,\gamma_{ncog,3}) = \text{known} \\
\text{Wife, } \theta^w = 1: &\quad \alpha^w_{\tau,k} + \beta\big(\Gamma_{cog,g_t+1}(\gamma_{cog,3} + \gamma_{cog,4}) + \Gamma_{ncog,g_t+1}(\gamma_{ncog,3} + \gamma_{ncog,4})\big) = \text{known} \\
\text{Husb., } \theta^h = 0: &\quad \alpha^h_{\tau,k} + \beta(\Gamma_{cog,g_t+1}\,\gamma_{cog,5} + \Gamma_{ncog,g_t+1}\,\gamma_{ncog,5}) = \text{known} \\
\text{Husb., } \theta^h = 1: &\quad \alpha^h_{\tau,k} + \beta\big(\Gamma_{cog,g_t+1}(\gamma_{cog,5} + \gamma_{cog,6}) + \Gamma_{ncog,g_t+1}(\gamma_{ncog,5} + \gamma_{ncog,6})\big) = \text{known}
\end{aligned}$$

where the right-hand side of each equation equals $\alpha^i_l \cdot \mathbb{E}[\tau^i/l^i \mid ychild = 1, \theta^i] - \alpha^i_\tau$, which is computable from data and the parameters identified above. Since the recursion makes each Γ_{D,g_t+1} linear in $(\alpha_{cog}, \alpha_{ncog})$ with coefficients built from the known CPF persistence parameters, this remains a linear system in four unknowns $(\alpha^w_{\tau,k}, \alpha^h_{\tau,k}, \alpha_{cog}, \alpha_{ncog})$. The system has full rank provided the CPF coefficients $(\gamma_{D,3}, \gamma_{D,4})$ and $(\gamma_{D,5}, \gamma_{D,6})$ differ across cognitive and non-cognitive domains, a condition satisfied by the external CPF estimates, which produce domain-specific time-investment elasticities.

6. **Workload interactions** $(\alpha^w_{\tau,n}, \alpha^h_{\tau,n})$**.** With the workload interaction, the weight on the wife's home production is $\alpha^w_\tau + \alpha^w_{\tau,n}\, n^w/\bar{n}$, so the working wife's τ/l ratio in Step 4 identifies the composite weight evaluated at the observed hours point, and one additional moment is required to separate $\alpha^w_{\tau,n}$ from the level. That moment is the within-person slope of home production in own market hours. Any within-person co-movement of home production with hours in excess of this value can only be generated by $\alpha^w_{\tau,n}$, which tilts the ratio as hours rise. The husband's $\alpha^h_{\tau,n}$ is identified symmetrically from his within-person slope.

Appendix H.2 Identification With All Work Arrangements

The baseline identifies $(\alpha^w_l, \alpha^w_{PE,1}, \alpha^h_l, \alpha^w_\tau, \alpha^h_\tau, \alpha^w_{\tau,k}, \alpha^h_{\tau,k}, \alpha_{cog}, \alpha_{ncog}, \eta_{i,j})$ from intratemporal FOCs under two fixed work arrangements ($q \in \{1, 3\}$). The intratemporal arguments carry through conditional on any work arrangement q, since the work arrangement-specific utility terms enter the FOCs only through additive shifts in the leisure condition.

The wife's self-employment-leisure interaction $\alpha^w_{SE,1}$ is identified by evaluating the composite

(H.22) at $d^w_{SE} = 1$ (wife in self-employment):

$$\alpha^w_l + \alpha^w_{SE,1} = \frac{\mathbb{E}\left[l^w \frac{\partial \mathcal{T}}{\partial n^w} \,\middle|\, n^w > 0,\, d^w_{SE} = 1\right]}{\mathbb{E}[\tilde{C} \mid n^w > 0,\, d^w_{SE} = 1]}$$

Since α^w_l is identified from the baseline, $\alpha^w_{SE,1}$ follows by differencing. A parallel argument identifies $\alpha^h_{SE,1}$ from the husband's FOCs at $d^h_{SE} = 1$, with husband PE as reference ($\alpha^h_{PE,1} = 0$).

The remaining parameters, $\alpha^w_{j,2}$, $\alpha^w_{j,3,PT}$, $\alpha^w_{j,3,FT}$, $\alpha^h_{SE,2}$, $\alpha_{cc,PT}$, $\alpha_{cc,FT}$, $\alpha^{ed}_{cc,PT}$, $\alpha^{ed}_{cc,FT}$, are additively separable in the flow utility and do not affect the continuous intratemporal FOCs conditional on (q, cc). They shift the expected value $\mathcal{V}^{s2}_q$ and therefore the Stage 1 work arrangement choice probability:

$$p(d_q = 1 \mid \Omega^{s1}) = \frac{\exp(\mathcal{V}^{s2}_q / \sigma_\xi)}{\sum_{q'} \exp(\mathcal{V}^{s2}_{q'} / \sigma_\xi)}$$

Since all parameters entering $\mathcal{V}^{s2}_q$ except the target are identified from the baseline and the work arrangement-leisure interactions above, the remaining parameters are identified sequentially:

1. **Work arrangement-child interactions $\alpha^w_{j,2}$ for $j \in \{PE, SE\}$.** These shift the value of wife work arrangement j when a young child is present. Variation in wife's work arrangement shares across young-child and non-young-child states, conditional on all other identified parameters, identifies $\alpha^w_{PE,2}$ and $\alpha^w_{SE,2}$. The wife's non-working state ($j = NW$) is the reference ($\alpha^w_{NW,2} = 0$).

2. **Husband's self-employment-child interaction $\alpha^h_{SE,2}$.** Variation in husband work arrangement shares across young-child states identifies $\alpha^h_{SE,2}$, with husband PE as reference ($\alpha^h_{PE,2} = 0$).

3. **Childcare base parameters $\alpha_{cc,PT}, \alpha_{cc,FT}$.** These enter the value function when a young child is present, with $cc = 0$ as reference. Conditional on all other parameters, variation in childcare take-up rates across non-college households with young children identifies both.

4. **Childcare-education interactions $\alpha^{ed}_{cc,PT}, \alpha^{ed}_{cc,FT}$.** These shift the value of childcare for college-educated mothers. The child production function, estimated outside the structural model, implies substantially different skill returns to childcare by maternal education (the college interaction $\gamma_{D,9}$ flips the sign of the part-time childcare effect across education groups). The utility education interactions capture preference heterogeneity in childcare valuation net of the CPF channel. They are identified from the differential childcare take-up rates between college and non-college mothers, conditional on work arrangement. Since the model is solved separately by θ^w, the Bellman equation generates education-specific choice probabilities, and the education-split childcare moments (PT/FT take-up by work arrangement and education) provide the identifying variation.

5. **Childcare-work arrangement interactions $\alpha^w_{j,3,PT}, \alpha^w_{j,3,FT}$ for $j \in \{PE, SE\}$.** These shift the value of childcare conditional on wife's work arrangement. Variation in the joint distribution

of (q, cc) identifies them, with the non-working baseline ($j = NW$) excluded from the sum.

6. **Scale parameters σ_ξ, σ_ζ.** The preference shock scales govern the dispersion of choices around the deterministic optimum. They are identified from the smoothness of observed choice probabilities conditional on state variables: larger σ implies more uniform choice frequencies for a given spread of $\mathcal{V}_q^{s2}$.

7. **Terminal value ψ_{hc}.** The terminal value parameter disciplines the life-cycle savings and human capital investment profiles through Equation 2. It is identified from the wife's labor-force attachment in the later life-cycle periods: with the discount factor preset and the human-capital transition pinned by BHPS wage dynamics, ψ_{hc} alone rationalizes older wives' participation and hours beyond the current wage return to human capital.

Interior solution arguments. The baseline proof assumes interior solutions for hours ($n^i > 0$), home production ($0 < \tau^i < \bar{T}$), and consumption ($C > 0$). Utility and the continuation value enter τ^i only through $\log \tau^i$, so the coefficient on $\log \tau^i$ in the young-child, working-parent problem is the net weight $A_i = \alpha^i_\tau + \alpha^i_{\tau,n}(n^i/\bar{n}) + \alpha^i_{\tau,k} + \Psi^i$, where $\Psi^i > 0$ is the child-skill return to home time. When $A_i > 0$, an Inada condition holds, $A_i \log \tau^i \to -\infty$ as $\tau^i \downarrow 0$, ruling out the lower corner, while $\alpha^i_l \log l^i \to -\infty$ as $l^i \downarrow 0$ rules out the upper corner, so the home-production choice is interior. Interiority is therefore equivalent to the child-skill return dominating the net home-production disutility, $\Psi^i > -\left(\alpha^i_\tau + \alpha^i_{\tau,n}(n^i/\bar{n}) + \alpha^i_{\tau,k}\right)$, which holds at the estimates in the young-child cells. Absent a young child, Ψ^i and $\alpha^i_{\tau,k}$ drop out, the net weight $\alpha^i_\tau + \alpha^i_{\tau,n}(n^i/\bar{n})$ is negative, and home production sits at the grid's positive lower bound. In the data, corner solutions arise through three channels: (1) the wife's non-participation ($n^w = 0$), (2) the borrowing constraint ($a' = \underline{a}$), and (3) the supervision constraint (Equation 4) binding. Each is handled within the model's discrete-choice structure. Non-participation is a separate work arrangement state ($j = NW$) with its own value function, so it does not violate the interior FOCs for working wives. The borrowing constraint, when binding, replaces the Euler equation with $a' = \underline{a}$ and the shadow value of assets becomes $\lambda^{BC} > \beta(1+r)\mathbb{E}[\lambda^{BC\prime}]$; this does not affect the intratemporal leisure-hours-home-production ratios that identify the preference parameters. The supervision constraint, when binding, introduces a Lagrange multiplier λ^{Sup} that shifts the home production FOCs (H.19)–(H.20) by λ^{Sup}/τ^i. In the estimation, the supervision constraint binds only for a subset of childcare-work arrangement cells, and the identification conditions hold in the complement where the constraint is slack. The discrete childcare choice ($cc \in \{0, PT, FT\}$) is itself a corner solution among three alternatives; the childcare parameters are identified through the discrete probabilities as described above. More broadly, the first-order conditions in this appendix are a continuous relaxation, used to show which variation identifies which parameter, while the estimated model chooses over finite grids. Selecting an interior grid point is a value comparison against neighboring points rather than a continuous equality, and the Stage-1 and Stage-2 preference shocks make the resulting choice probabilities smooth functions of the parameters, so the same ratios map into the conditional choice probabilities the estimator matches. The paid-employment hours menu is deliberately coarse. Its two points are a structural representation of the rigidity of paid schedules, not a numerical approximation to a continuous choice, so the hours curvature $\eta_{i,j}$ and the flexibility results are identified from the shares choosing each menu point and from the wider self-employment menu, not from an interior hours condition.

Appendix I Externally Estimated Parameters

Table I.15 shows the fertility probability. Fertility follows a nonlinear life-cycle pattern and college education is associated with a higher likelihood of childbirth. The parity effects are large: a first child raises the probability of a further birth, but the probability declines sharply at higher parities.

Table I.15: Fertility Probability

	Estimate	SE
Probability of Child		
Constant	-15.344	1.025
Age	0.040	0.068
$\text{Age}^2/100$	-1.878	0.109
College	0.241	0.062
1 kid	0.688	0.069
2 kid	-1.099	0.092
3 kid	-4.404	0.711

Note: Probability of having a child follows a logit structure.

Table I.16 reports the estimated tax-and-transfer parameters and Figure I.15 shows the FORTAX fit. The lump-sum term χ_1 is the transfer a household receives at zero earnings and the main channel of redistribution. It is largest for households with a non-working mother, who qualifies for out-of-work support, and reflects Child Benefit and the child element of the tax credits, which scale with family size. Above this floor the schedule is close to proportional, with χ_3 near one, so households retain a roughly constant share of additional gross earnings and redistribution is concentrated at low income. The slope χ_2 and the curvature χ_3 are not separately interpretable, since a lower slope and a higher curvature trade off within the power form and describe the same schedule; I therefore read the fitted net-income function as a whole. Its variation across households is dominated by the mother's employment and by family size.

In fact, paid and self-employed workers face the same income tax schedule but a different National Insurance and social-insurance regime. Employees pay Class 1 contributions on earnings above the primary threshold, at a main rate of about 10 to 11 percent over the sample period, and their employer pays a further secondary contribution. The self-employed instead pay a flat-rate weekly Class 2 contribution together with Class 4 contributions on annual profits, at a lower main rate of about 7 to 8 percent and with no employer contribution (HM Revenue & Customs, 2024).[67] A self-employed worker therefore retains more of a marginal pound of earnings, which features the system where work arrangements are taxed differently (Mirrlees et al., 2011). The self-employed also have more limited access to contributory benefits, including Statutory Sick Pay and contribution-based Jobseeker's Allowance, and self-employment income can be zero or negative when a business makes a loss, which is bottom-coded at zero in the data. The lower National Insurance burden is why the fitted net-income schedule sits slightly above paid employment at low incomes, while the lump-sum term χ_1 captures the difference in contributory-benefit access. The two schedules are otherwise close, so the net tax-and-benefit wedge between the arrangements is modest relative to the pre-tax earnings gap that separates them.

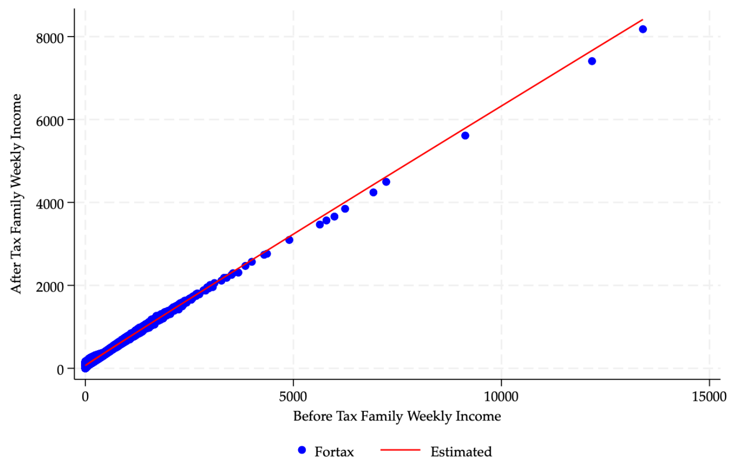


Figure I.15: FORTAX fit

[67] I use this variation for the child production function estimates.

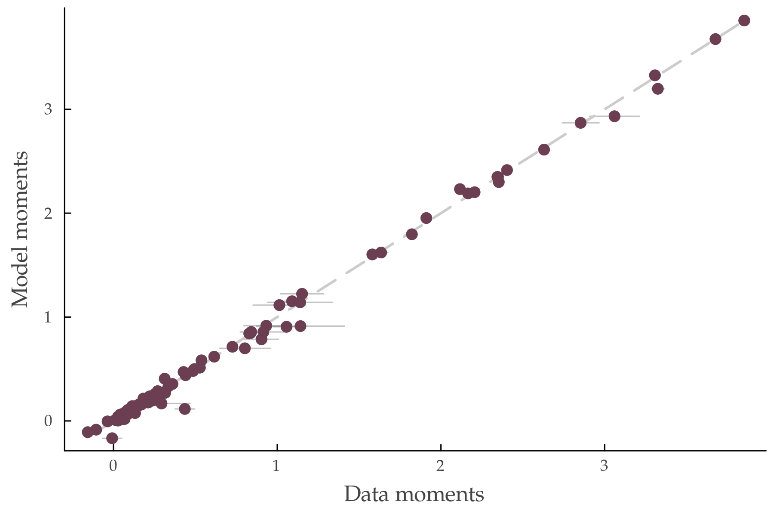
Table I.16: FORTAX Parameters

	Number of Children											
	0			1			2			3		
	χ_1	χ_2	χ_3	χ_1	χ_2	χ_3	χ_1	χ_2	χ_3	χ_1	χ_2	χ_3
Youngest child (age = 0)												
Husband PE												
Wife PE	1.51	1.23	0.92									
Wife SE	-3.32	1.21	0.93									
Wife NW	44.32	0.57	1.01									
Husband SE												
Wife PE	10.06	1.11	0.94									
Wife SE	14.21	1.19	0.92									
Wife NW	32.89	0.84	0.96									
Age = 1												
Husband PE												
Wife PE				69.41	0.75	0.98	84.72	0.54	1.03	74.34	0.74	0.98
Wife SE				79.06	0.61	1.01	77.48	0.67	1.00	156.89	0.08	1.29
Wife NW				132.08	0.11	1.24	126.25	0.16	1.18	128.11	0.15	1.19
Husband SE												
Wife PE				72.76	0.86	0.96	112.52	0.31	1.10	128.97	0.21	1.16
Wife SE				118.26	0.18	1.18	124.18	0.42	1.05	109.68	0.33	1.10
Wife NW				121.24	0.26	1.11	127.30	0.27	1.11	112.42	0.38	1.06
Age = 3												
Husband PE												
Wife PE				69.72	0.68	1.00	70.49	0.70	0.99	81.69	0.54	1.03
Wife SE				122.86	0.12	1.23	72.64	0.77	0.98	32.76	1.86	0.85
Wife NW				122.67	0.18	1.16	128.66	0.15	1.19	137.61	0.13	1.20
Husband SE												
Wife PE				89.49	0.56	1.02	110.15	0.33	1.09	106.46	0.27	1.13
Wife SE				42.88	1.16	0.93	121.00	0.37	1.07	86.72	1.04	0.93
Wife NW				132.62	0.35	1.06	120.15	0.19	1.17	106.41	0.53	1.01
Age = 5												
Husband PE												
Wife PE				40.90	1.11	0.93	50.25	0.99	0.95	35.18	1.15	0.93
Wife SE				82.75	0.31	1.10	45.67	1.12	0.93	73.46	0.64	1.00
Wife NW				116.16	0.20	1.15	133.44	0.15	1.18	126.66	0.18	1.16
Husband SE												
Wife PE				143.49	0.13	1.22	116.79	0.28	1.11	103.98	0.49	1.04
Wife SE				-140.19	6.02	0.73	133.28	0.09	1.28	109.66	0.35	1.09
Wife NW				120.27	0.19	1.16	106.03	0.46	1.03	107.49	0.32	1.09
Age = 7+												
Husband PE												
Wife PE				57.76	0.85	0.97	66.68	0.75	0.98	45.88	0.91	0.96
Wife SE				97.13	0.41	1.06	75.80	0.63	1.01	72.94	0.66	1.00
Wife NW				126.53	0.21	1.14	134.68	0.15	1.19	124.64	0.17	1.16
Husband SE												
Wife PE				76.83	0.70	0.99	81.66	0.59	1.02	113.23	0.29	1.11
Wife SE				75.02	0.90	0.95	113.69	0.51	1.03	94.84	0.87	0.96
Wife NW				124.22	0.31	1.08	115.40	0.35	1.07	108.12	0.48	1.03

Note: PE denotes paid employment; SE denotes self-employment; NW denotes non-working. The tax-and-transfer schedule follows $\mathcal{T} = \chi_1 + \chi_2(Y^w + Y^h)^{\chi_3}$ (Equation 6), where χ_1 is the lump-sum transfer or tax component, χ_2 is the slope of the effective tax rate, and χ_3 captures the degree of progressivity. Age refers to the age of the youngest child; the panel labeled Age = 0 reports parameters for childless households. Parameters are estimated by fitting FORTAX-simulated household disposable income to this functional form separately for each combination of parental work arrangement, number of children, and child age.

Appendix J Model Fit

Figure J.16 summarizes the overall fit. Figures J.17–J.18 and Table J.17 report fit on the three blocks of moments targeted in SMM: work arrangement and childcare shares, hours worked and home production by work arrangement and child presence, and income and human capital dynamics.



Note: Each point is one targeted moment; the horizontal axis is the data value with a one-standard-error band and the vertical axis is the simulated value at the converged estimates. The dashed line is the 45-degree line.

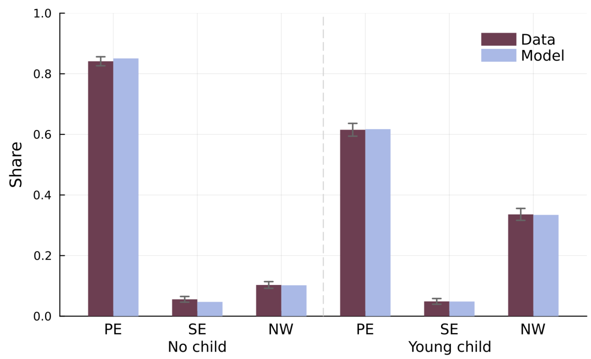
Figure J.16: Overall Model Fit: Simulated vs Data Moments

Table J.19 reports untargeted moments. The model reproduces work arrangement shares split by education within one to six percentage points in the no-child and young-child cells, including the education gradient in maternal non-work with young children and in paternal self-employment. Maternal home production is somewhat under-predicted there: once the investment window closes, the model no longer attaches a child-skill return to her home time, while housework persists in the data for school-age children.

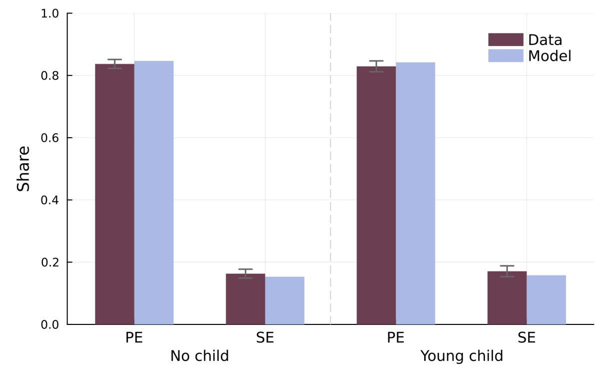
The model's implied labor supply elasticities are untargeted and line up with the quasi-experimental and structural literature. A permanent 5% increase in the wife's hourly wage raises her total hours with an uncompensated elasticity of 0.23, of which roughly one-third operates through participation; with a young child present the elasticity rises to 0.41 and participation accounts for about half.[68] These sit within the 0.1–0.4 range that Bargain, Orsini, and Peichl (2014) estimate for married women across seventeen countries, near the 0.18 median Marshallian hours elasticity in Attanasio et al. (2018), and match the positive, moderate responses, largest for mothers of young children, that Blundell, Duncan, and Meghir (1998) recover from UK tax reforms. The husband's own-wage elasticity is 0.03; because he always works, this is a pure intensive-margin response, which Bargain, Orsini, and Peichl (2014) find to be close to zero for married men. The

[68]Elasticities are computed by re-solving and re-simulating the model after a permanent proportional shift in the wage intercepts of both work arrangements, holding the simulation draws fixed, the standard simulation practice in discrete-choice labor supply models (Bargain, Orsini, and Peichl, 2014).

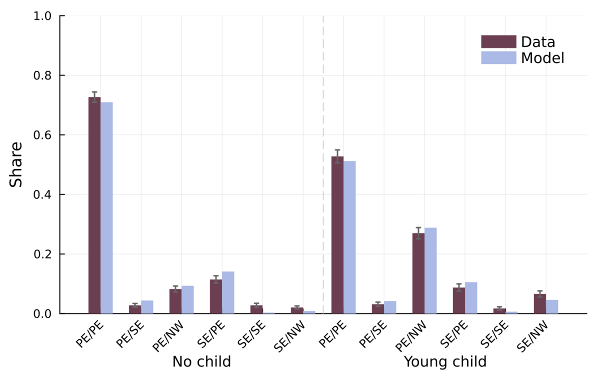
wife's cross-wage elasticity with respect to the husband's wage is -0.31, and -0.62 with a young child, the negative income effect for women with children in Blundell, Duncan, and Meghir (1998). Home production is far less elastic than market hours, much smaller than Gelber and Mitchell (2012) find for an estimate for single women's housework.



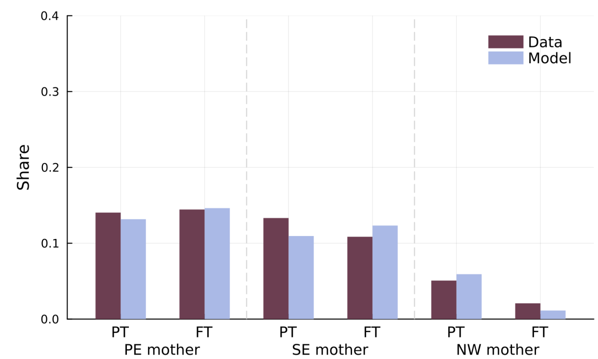
(a) Wife's Work Arrangement



(b) Husband's Work Arrangement



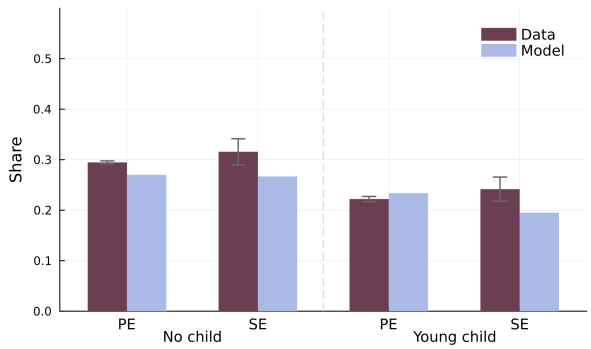
(c) Household Work Arrangement



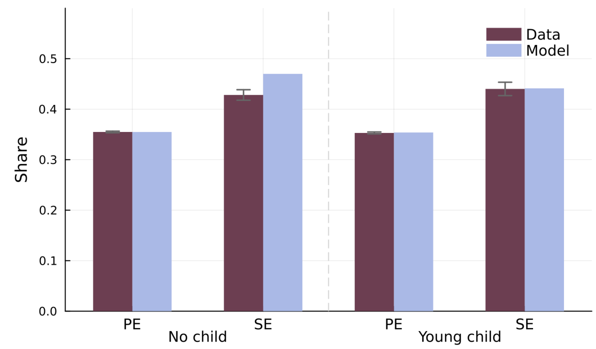
(d) Childcare by Wife's Work Arrangement

Note: The figure compares simulated and empirical shares by work arrangement for wives, husbands, and households.

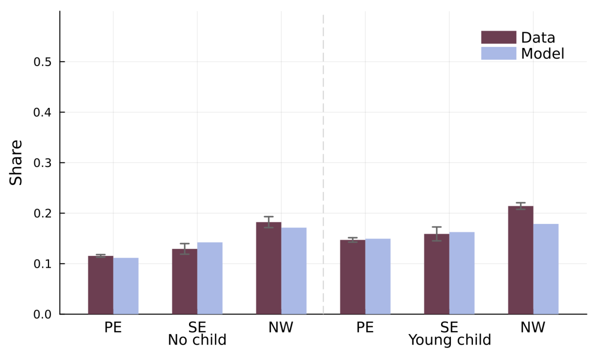
Figure J.17: Work Arrangement and Childcare Fit



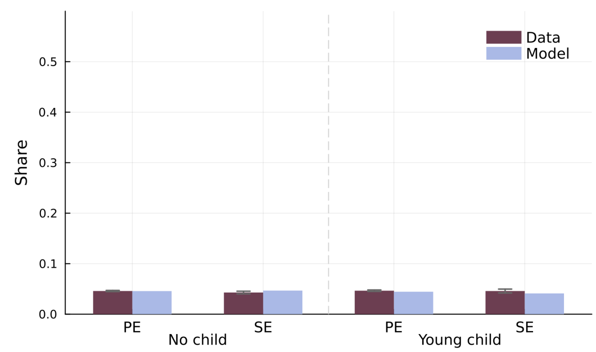
(a) Wife's Hours Worked



(b) Husband's Hours Worked



(c) Wife's Home Production



(d) Husband's Home Production

Note: The figure compares simulated and empirical hours worked and home production by work arrangement type for wives, husbands, and households.

Figure J.18: Hours Fit

Table J.17: Model Fit for Couple's Income

Description	Data	Model
Wife PE		
Log hourly earnings – non-college	1.823	1.796
Log hourly earnings – college	2.353	2.302
Log hourly earnings SD	0.494	0.501
Mean log hours	3.307	3.327
SD log hours	0.360	0.356
Wife SE		
Log hourly earnings – non-college	0.294	0.165
Log hourly earnings – college	1.142	0.915
Log hourly earnings SD	3.060	2.921
Mean log hours	3.325	3.199
SD log hours	0.538	0.585
Husband PE		
Log hourly earnings – non-college	2.206	2.203
Log hourly earnings – college	2.630	2.612
Log hourly earnings SD	0.486	0.480
Log hourly earnings, age 21–29	2.166	2.190
Log hourly earnings, age 30–35	2.349	2.353
Log hourly earnings, age 36–45	2.404	2.416
Log hourly earnings, age 46–51	2.345	2.348
Mean log hours	3.677	3.676
SD log hours	0.083	0.084
Husband SE		
Log hourly earnings – non-college	0.904	0.790
Log hourly earnings – college	1.140	1.143
Log hourly earnings SD	2.853	2.872
Log hourly earnings, age 21–29	1.152	1.244
Log hourly earnings, age 30–35	1.013	1.109
Log hourly earnings, age 36–45	0.933	0.916
Log hourly earnings, age 46–51	0.803	0.703
Mean log hours	3.852	3.856
SD log hours	0.313	0.406
Wage dynamics		
Wife log wage growth	0.033	0.037
Spousal earnings correlation	0.132	0.078
Wife initial wage – non-college	1.581	1.590
Wife initial wage – college	2.116	2.233
Wife wage after non-employment	1.635	1.615
Wife wage after employment	1.911	1.953

Note: PE denotes paid employment; SE denotes self-employment. Wages and hours are in logs. Model columns report simulated moments at the converged estimates.

Table J.18: Model Fit for Wife's Human Capital Dynamics

Description	Data	Model
Home production–hours slope, wife	-0.157	-0.115
Home production–hours slope, husband	-0.036	-0.007
Initial wage SD – non-college	1.057	0.910
Initial wage SD – college	0.916	0.844
Initial wage age-slope – non-college	-0.106	-0.121
Initial wage age-slope – college	-0.008	-0.170
Wage autocovariance (employed)	0.436	0.116
Wage growth, age < 35	0.068	0.012
Wage growth, age ≥ 35	0.061	0.040
Wage growth SD	1.091	1.157
Wage growth, FT vs PT stayers	0.039	0.011

Note: Moments identifying the wife's human capital process (Section 4.3). The autocovariance is computed on education-demeaned log hourly wages for wives employed in consecutive periods; wage growth moments condition on employment at consecutive years. The model value of the autocovariance is bounded by the three-point human capital support.

Table J.19: Untargeted Moments: Model Validation

Description	Data	Model
Wife work arrangement shares by education		
PE (NC, no child)	0.833	0.828
SE (NC, no child)	0.054	0.049
NW (NC, no child)	0.113	0.123
PE (NC, young child)	0.573	0.580
SE (NC, young child)	0.043	0.048
NW (NC, young child)	0.384	0.372
PE (NC, old child)	0.754	0.834
SE (NC, old child)	0.061	0.052
NW (NC, old child)	0.185	0.114
PE (C, no child)	0.863	0.878
SE (C, no child)	0.059	0.045
NW (C, no child)	0.079	0.077
PE (C, young child)	0.709	0.681
SE (C, young child)	0.062	0.050
NW (C, young child)	0.229	0.269
PE (C, old child)	0.759	0.876
SE (C, old child)	0.074	0.051
NW (C, old child)	0.168	0.073
Husband work arrangement shares by education		
SE (NC, no child)	0.184	0.165
SE (NC, young child)	0.203	0.170
SE (NC, old child)	0.210	0.180
SE (C, no child)	0.110	0.137
SE (C, young child)	0.099	0.136
SE (C, old child)	0.137	0.147
Old-child households: hours worked		
Wife, PE	25.240	30.269
Wife, SE	31.343	31.806
Husband, PE	39.802	39.575
Husband, SE	51.241	51.109
Old-child households: home production (share of time endowment)		
Wife, PE	0.159	0.112
Wife, SE	0.167	0.140
Husband, PE	0.049	0.046
Husband, SE	0.044	0.047
Old-child households: log hourly earnings		
Wife, PE	1.893	1.984
Wife, SE	0.309	0.433
Husband, PE	2.389	2.372
Husband, SE	1.015	0.839

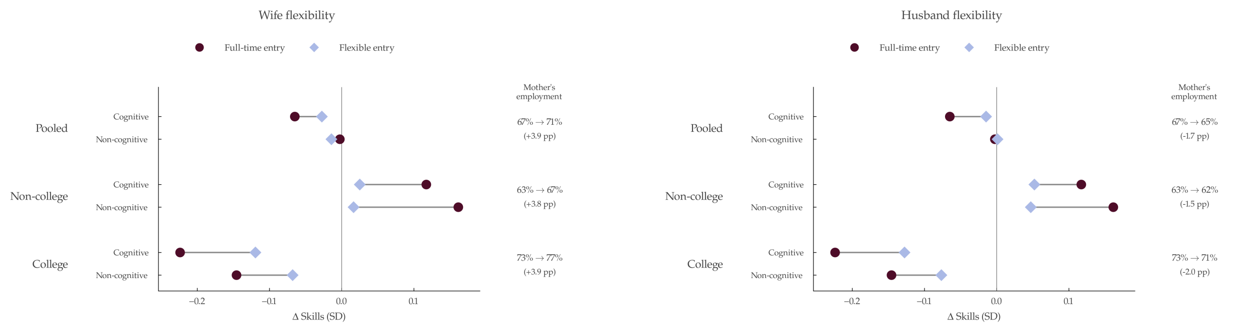
Appendix K Additional Counterfactual Results

Table K.20 lists the parameters governing each counterfactual mechanism and the related trade-offs, and Table K.22 decomposes the policy counterfactuals into the four input channels by maternal education.

Table K.20: Main Mechanisms and Tradeoff

Parameter	Governs	Tradeoff
Utility		
α_{ncog}, α_{cog}	Child skill utility	vs $\alpha^{w}_{\tau,k}$, $\alpha^{h}_{\tau,k}$: parents value child skills but bear private costs of providing the home production time
α^{PT}_{cc}, α^{FT}_{cc}	PT/FT childcare utility	Non-college educated mothers strongly averse, college educated mothers nearly indifferent
CPF technology		
$\gamma_{D,3}$, $\gamma_{D,4}$	Maternal home production	Maternal advantage in the skill technology
$\gamma_{D,5}$, $\gamma_{D,6}$	Paternal home production	Paternal home time does not substitute for released maternal time, so the father's effect on children runs through the budget rather than time
$\gamma_{D,7}$, $\gamma_{D,9}$	PT care	Strong return for non-college educated, near zero for college educated
Flexibility and household constraint		
K_{SE} vs K_{PE}	PE rigidity vs SE flexibility	Self-employment allows intermediate hours that enable home production alongside market work: *flexibility*
σ^2_{SE}	SE earnings variance	Risk channel of SE entry, distinct from the schedule channel in K_{SE}
$K_{SE} + \sigma^2_{SE}$	SE bundles flexibility and risk	Removing K_{SE} vs σ^2_{SE} yields the same ΔSE share but opposite τ^h: flexibility-driven entry selects low-hours SE ($\tau^h \uparrow$); earnings-driven entry selects high-hours SE ($\tau^h \downarrow$)
Supervision constraint	Household dynamics	Spouses cannot optimize independently: market entry by one endogenously shifts the other's home time allocation

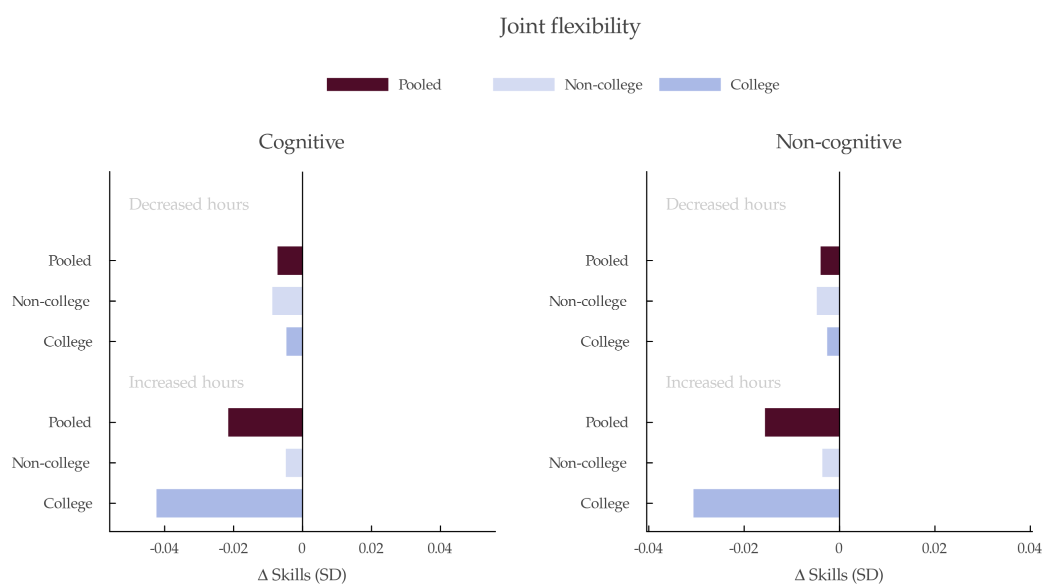
Note: $\alpha^{w}_{\tau,k}$ and $\alpha^{h}_{\tau,k}$ are the young-child interaction terms; with a young child present the flow-utility weight on $\log \tau^i$ is $\alpha^i_\tau + \alpha^i_{\tau,n}(n^i/\bar{n}) + \alpha^i_{\tau,k}$.



(a) Wife-only (b) Husband-only

Note: This figure plots the effect on a child's cognitive and non-cognitive skill at age 7 of the mother entering paid work during the child's ages 1–5, relative to remaining out of work, by maternal education. Circles are full-time entry, the causal treatment-on-treated effect in the baseline. Diamonds are flexible entry: under wife-only flexibility, the intermediate hours she chooses once her own menu widens; under husband-only flexibility, her indirect response through the household's joint allocation, since her own menu is unchanged and only the husband's wider menu shifts home production and care under the supervision constraint. The right column reports the rise in the maternal employment rate.

Figure K.19: Effect of Flexibility on Child Skills



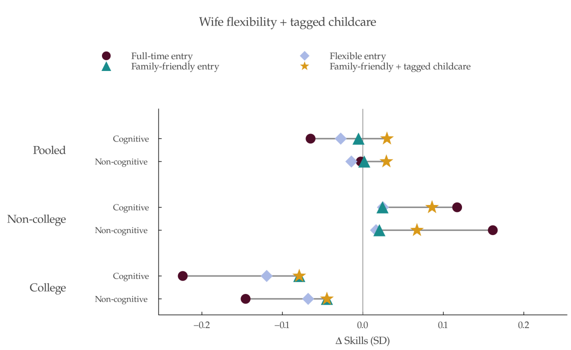
Note: This figure plots the effects of joint flexibility on child's cognitive (left) and non-cognitive (right) skills at age 7, decomposed by the direction of the mother's cumulative hours change over the child's ages 1–5, pooled and by maternal education.

Figure K.20: Redistribution of Child Skills under Flexibility

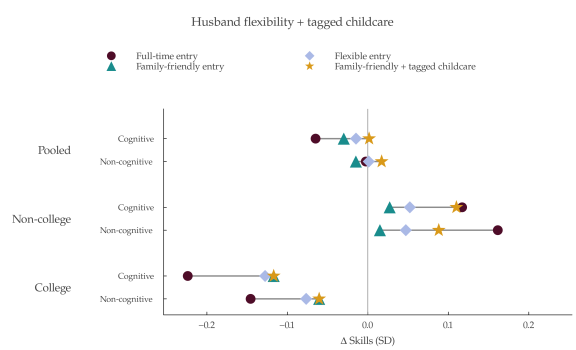
Table K.21: Maternal Employment Entry and Child Skills

Entry margin	$\Delta\tau^w$ (h/wk)	Pooled (SD) ΔCog	Pooled (SD) ΔNcog	Non-college (SD) ΔCog	Non-college (SD) ΔNcog	College (SD) ΔCog	College (SD) ΔNcog
Panel A: entry effect in the rigid baseline (treatment-on-treated)							
NW → PE full-time	−5.25	−0.065	−0.003	+0.117	+0.161	−0.224	−0.146
NW → PE part-time	−2.52	+0.020	+0.051	+0.073	+0.096	−0.130	−0.078
NW → self-employment	−2.37	+0.018	+0.052	+0.064	+0.093	−0.067	−0.025
Panel B: entry effect when full flexibility (RRFW) is offered							
NW → work, full-time	−3.51	−0.077	−0.020	+0.079	+0.061	−0.193	−0.081
NW → work, part-time	−1.25	−0.023	−0.014	+0.022	+0.014	−0.108	−0.066

Note: Effect on a child's skills when the mother enters work during the child's ages 1–5, relative to remaining out of work. Panel A conditions on mothers who make a clean single entry in the baseline simulation; the counterfactual holds the household's last pre-entry time and childcare inputs fixed over the spell, so the effect is causal at fixed selection. Panel B reports the same object for mothers drawn into work by the offer of the full flexibility menu, split by the hours they enter at. $\Delta\tau^w$ is the mean change in the mother's weekly home-production time.



(a) Wife-only



(b) Husband-only

Note: This figure plots the effect on a child's cognitive and non-cognitive skill at age 7 of the mother entering paid work during the child's ages 1–5, relative to remaining out of work, by maternal education. Circles are full-time entry, diamonds full flexible entry, triangles the family-friendly flexibility, and stars the package of family-friendly flexibility with tagged childcare.

Figure K.21: Effects of Family-Friendly Flexibility and Tagged Childcare on Child Skills

Table K.22: CPF Channel Decomposition of Policy Skill Effects

	Non-college Mother					College Mother				
	τ^w	τ^h	cc^{PT}	cc^{FT}	Δ SD	τ^w	τ^h	cc^{PT}	cc^{FT}	Δ SD
Panel A: Work flexibility										
Wife-only										
	Wife: ΔPE +4.2, ΔSE −0.3, ΔNW −3.8; Husb ΔSE +0.3					Wife: ΔPE +4.4, ΔSE −0.5, ΔNW −3.9; Husb ΔSE +0.5				
Δ Input	−0.011	+0.009	−0.001	+0.008		−0.029	−0.000	+0.004	+0.029	
Cog (SD)	**−0.025**	+0.001	+0.001	+0.006	−0.017	**−0.052**	+0.001	+0.000	+0.011	−0.040
Ncog (SD)	**−0.019**	+0.000	+0.001	+0.005	−0.013	**−0.042**	−0.000	−0.000	+0.012	−0.031
Joint										
	Wife: ΔPE +2.6, ΔSE −0.3, ΔNW −2.3; Husb ΔSE −2.6					Wife: ΔPE +2.3, ΔSE −0.3, ΔNW −1.9; Husb ΔSE −2.5				
Δ Input	−0.007	−0.006	−0.001	+0.013		−0.025	−0.001	+0.002	+0.032	
Cog (SD)	**−0.015**	−0.002	−0.000	+0.011	−0.006	**−0.039**	−0.001	+0.000	+0.012	−0.027
Ncog (SD)	**−0.011**	−0.001	−0.000	+0.009	−0.003	**−0.032**	−0.001	+0.000	+0.013	−0.019
Husband-only										
	Wife: ΔPE −1.5, ΔSE +0.0, ΔNW +1.5; Husb ΔSE −2.9					Wife: ΔPE −2.1, ΔSE +0.1, ΔNW +2.0; Husb ΔSE −3.0				
Δ Input	+0.002	−0.012	−0.001	+0.006		−0.002	−0.002	−0.002	+0.008	
Cog (SD)	**+0.008**	−0.003	+0.000	+0.003	+0.008	**+0.006**	−0.001	+0.000	+0.003	+0.009
Ncog (SD)	**+0.006**	−0.001	+0.000	+0.002	+0.007	**+0.005**	−0.001	+0.000	+0.003	+0.008
Panel B: Free formal childcare										
Work-conditional										
	Wife: ΔPE +1.8, ΔSE −0.1, ΔNW −1.6; Husb ΔSE +0.1					Wife: ΔPE +2.8, ΔSE −0.3, ΔNW −2.5; Husb ΔSE +0.3				
Δ Input	−0.063	−0.014	−0.011	+0.098		−0.165	−0.006	−0.026	+0.269	
Cog (SD)	−0.035	−0.002	−0.005	**+0.098**	+0.056	**−0.143**	−0.001	+0.007	+0.092	−0.045
Ncog (SD)	−0.029	−0.001	−0.006	**+0.081**	+0.045	**−0.120**	−0.001	+0.007	+0.096	−0.018
Means-tested										
	Wife: ΔPE −1.7, ΔSE +0.0, ΔNW +1.7; Husb ΔSE +0.5					Wife: ΔPE −3.2, ΔSE +0.1, ΔNW +3.1; Husb ΔSE +0.7				
Δ Input	−0.111	−0.019	−0.015	+0.166		−0.177	−0.008	−0.013	+0.299	
Cog (SD)	−0.059	−0.003	−0.007	**+0.169**	+0.100	**−0.154**	−0.001	+0.003	+0.106	−0.046
Ncog (SD)	−0.048	−0.002	−0.009	**+0.140**	+0.081	**−0.129**	−0.001	+0.004	+0.109	−0.017
Universal										
	Wife: ΔPE −1.2, ΔSE −0.1, ΔNW +1.2; Husb ΔSE +0.1					Wife: ΔPE −2.4, ΔSE −0.1, ΔNW +2.5; Husb ΔSE +0.1				
Δ Input	−0.139	−0.029	−0.018	+0.210		−0.296	−0.012	−0.038	+0.497	
Cog (SD)	−0.075	−0.004	−0.009	**+0.217**	+0.130	**−0.254**	−0.002	+0.009	+0.172	−0.075
Ncog (SD)	−0.061	−0.003	−0.011	**+0.181**	+0.105	**−0.214**	−0.001	+0.010	+0.178	−0.027

Note: Shapley decomposition of the change in mean log skills at age 7 into the four CPF input channels; contributions sum exactly to the total. Panel A grants the RRFW flexible-hours menu to the wife, both spouses, or the husband. Panel B provides free formal childcare that is work-conditional (working families below £100,000), means-tested (bottom third of net income, no work condition), or universal. Input rows report mean changes over young-child periods; the dominant channel in each row is bold.

Table K.23: CPF Channel Decomposition of the Recommended Policies

	Non-college Mother					College Mother				
	τ^w	τ^h	cc^{PT}	cc^{FT}	Δ SD	τ^w	τ^h	cc^{PT}	cc^{FT}	Δ SD
Family-friendly flexibility										
	Wife ΔPE +2.8, ΔSE -0.3, ΔNW -2.5; Husb ΔSE -0.5					Wife ΔPE +1.9, ΔSE -0.3, ΔNW -1.7; Husb ΔSE -0.8				
Δ Input	-0.002	+0.002	+0.000	-0.000		+0.009	+0.001	-0.001	-0.011	
Cog (SD)	-0.000	+0.000	-0.000	-0.001	-0.001	+0.015	+0.000	-0.000	-0.003	+0.011
Ncog (SD)	-0.000	+0.000	-0.000	-0.001	-0.001	+0.012	-0.000	-0.000	-0.004	+0.008
Education-tagged childcare										
	Wife ΔPE -1.2, ΔSE -0.1, ΔNW +1.2; Husb ΔSE +0.1					Wife ΔPE +0.0, ΔSE +0.0, ΔNW +0.0; Husb ΔSE +0.0				
Δ Input	-0.139	-0.029	-0.018	+0.210		+0.000	+0.000	+0.000	+0.000	
Cog (SD)	-0.075	-0.004	-0.009	+0.217	+0.130	+0.000	+0.000	+0.000	+0.000	+0.000
Ncog (SD)	-0.061	-0.003	-0.011	+0.181	+0.105	+0.000	+0.000	+0.000	+0.000	+0.000
Package: flexibility and tagged childcare										
	Wife ΔPE +1.7, ΔSE -0.3, ΔNW -1.3; Husb ΔSE -0.5					Wife ΔPE +1.9, ΔSE -0.3, ΔNW -1.7; Husb ΔSE -0.8				
Δ Input	-0.139	-0.026	-0.018	+0.208		+0.009	+0.001	-0.001	-0.011	
Cog (SD)	-0.074	-0.004	-0.008	+0.214	+0.127	+0.015	+0.000	-0.000	-0.003	+0.011
Ncog (SD)	-0.061	-0.003	-0.010	+0.178	+0.103	+0.012	-0.000	-0.000	-0.004	+0.008

Note: Shapley decomposition of the change in mean log skill at age 7 into the four CPF input channels (mother's time τ^w, father's time τ^h, part-time childcare cc^{PT}, full-time childcare cc^{FT}), whose contributions sum to Δ SD. Arrangement shifts (percentage points) and mean input changes are by maternal education. Family-friendly flexibility offers the reduced-hours menu to both spouses; education-tagged childcare grants free formal childcare to non-college households only (college is stitched to baseline, hence exactly zero); the package combines both. Source: counterfactual_policy_opt.jl.

Appendix L Consumption Equivalent Variation

I measure the policy's effect on household welfare through the consumption equivalent variation (CEV), following De Nardi, Pashchenko, and Porapakkarm (2025). I compare each household with its hypothetical self under the policy counterfactual, defining the CEV as the constant proportional adjustment to its baseline consumption path that leaves it indifferent between the baseline and the policy. Let V_0 denote a household's expected discounted lifetime value,

$$V_0 = \mathbb{E} \sum_{t=1}^{\bar{t}} \beta^{t-1} \mathcal{U}_t,$$

where $\bar{t}$ is the number of model periods over the household's horizon, β the discount factor, and $\mathcal{U}_t$ the flow utility of Equation 1, in which consumption enters as $\log C_t$. Let V_0^{BS} and V_0^{CF} be the value under the baseline and under the policy counterfactual. The CEV λ is the consumption fraction that solves

$$\mathbb{E} \sum_{t=1}^{\bar{t}} \beta^{t-1} \mathcal{U}\Big((1+\lambda) C_t^{BS}, \cdot\Big) = V_0^{CF}.$$

Since $\log C_t$ enters flow utility additively with a unit coefficient in every period, scaling the baseline consumption path by $(1+\lambda)$ raises lifetime value by $\log(1+\lambda) \sum_t \beta^{t-1}$. Solving,

$$\lambda = \exp\left(\frac{V_0^{CF} - V_0^{BS}}{\text{PV}}\right) - 1, \qquad \text{PV} = \sum_{t=1}^{\bar{t}} \beta^{t-1} = \frac{1-\beta^{\bar{t}}}{1-\beta}, \tag{L.23}$$

the exact welfare gain in units of lifetime consumption. I compute λ for each simulated household, average over households with a young-child period by maternal education, and convert to pounds by multiplying by baseline household income. Because I scale λ by baseline income rather than by baseline consumption, these pound figures are income-scaled welfare values, and I apply the same income base to the policy costs.

Flow utility is additively separable in the child's skills and its remaining arguments, so the lifetime value change decomposes exactly into a child-skill and a parental component,

$$\Delta V_0 \equiv V_0^{CF} - V_0^{BS} = \Delta V_0^{\text{skill}} + \Delta V_0^{\text{par}}.$$

The child component is the discounted value of the policy-induced skill path,

$$\Delta V_0^{\text{skill}} = \mathbb{E} \sum_{t=1}^{\bar{t}} \beta^{t-1} \left(\alpha_{cog}\, \Delta \log \theta_{cog,t} + \alpha_{ncog}\, \Delta \log \theta_{ncog,t}\right),$$

where skills are fixed after the investment window (age 7) and ΔV_0^{par} is the value of the changes in consumption, leisure, home production, and work-arrangement amenities. Because this uses the household's own valuation of skills, α_{cog} and α_{ncog}, rather than an external price, I allocate the exact total CEV by the value shares,

$$\lambda^{\text{skill}} = \frac{\Delta V_0^{\text{skill}}}{\Delta V_0}\,\lambda, \qquad \lambda^{\text{par}} = \frac{\Delta V_0^{\text{par}}}{\Delta V_0}\,\lambda, \qquad \lambda^{\text{skill}} + \lambda^{\text{par}} = \lambda, \tag{L.24}$$

so the two components add to the total by construction. The child-skill component λ^{skill} is the part of the total consumption-equivalent attributable to the skill change under this value-share allocation. Note that it is not the consumption-equivalent of the skill change in isolation, $\exp(\Delta V_0^{\text{skill}}/\text{PV}) - 1$, because the CEV is nonlinear in ΔV_0.

Appendix M Discussion on the Unitary Household Framework

I assume that the household model is unitary such that spouses have equal weights ($\mu = 0.5$) and pool resources. It therefore determines aggregate time allocations and child outcomes, but not the intra-household split of welfare. To gauge how much this assumption drives the results, I re-solve the flexibility counterfactual under alternative Pareto weights μ on the wife, holding the remaining parameters fixed. Formally, the estimated flow utility decomposes as $u = u^{\text{sh}} + u^{w} + u^{h}$: the shared component u^{sh} collects consumption, the child-skill returns, and the base childcare cost, while u^{w} and u^{h} collect each spouse's private leisure, home-production, and work-arrangement terms. The weighted objective is $u^{\mu} = u^{\text{sh}} + 2\mu\, u^{w} + 2(1-\mu)\, u^{h}$, so the shared consumption, child-skill, savings, and terminal-value components enter unweighted and μ reweights only the spouses' private time and amenity terms. At $\mu = 0.5$, $2\mu = 2(1-\mu) = 1$ and $u^{\mu} = u$, so the $\mu = 0.5$ row reproduces the estimated baseline exactly.

The equal-weight baseline is consistent with evidence that the wife's weight is approximately 0.5 before childbirth, though it shifts as couples raise young children (Low et al., 2018; Verriest, 2024). Its level also varies with the couple's education. Gayle and Shephard (2019) estimate the weight by education, finding that it rises with the wife's schooling and falls with the husband's, from about 0.32 (college husband, non-college wife) to 0.58 (non-college husband, college wife) and averaging just below equal weighting (Table M.24). In a dynamic setting, Reynoso (2024) finds the same gradient and UK estimates place the wife's share in the same range and moving in the same direction with her relative earnings (Blundell, Chiappori, and Meghir, 2005; Blundell et al., 2007; Chiappori and Meghir, 2015).

I re-solve the flexibility counterfactual under each weight, holding the remaining parameters fixed: assigning each couple its education-specific weight from Table M.24, and uniformly over $\mu \in [0.35, 0.55]$. Table M.25 reports the treatment-on-treated entry effects by the couple's education. The reversal is invariant to the Pareto weights. A mother's full-time entry lowers a college child's skill and raises a non-college child's at every weight in every matching cell. The college loss is essentially independent of the husband's education, -0.21 SD whether her husband is college-educated ($\mu = 0.44$) or not ($\mu = 0.58$); the non-college gain is positive throughout but tracks the wife's weight, $+0.12$ SD for a non-college couple ($\mu = 0.47$) and $+0.03$ SD when she is married to a

Table M.24: Wife's Pareto weight by education

	Wife	
Husband	Non-college	College
Non-college	0.47	0.58
College	0.32	0.44

Note: Wife's Pareto weight μ by the couple's education, from Gayle and Shephard (2019) (Table II), collapsing "some college" into non-college and averaging the underlying cells. The unitary baseline is equal weighting, $\mu = 0.5$.

college husband and carries less weight ($\mu = 0.32$). The bargaining channel moves the magnitude of the gain but not the sign. The full-time entry sign reversal is therefore robust to the Pareto weights, though this check speaks to the entry-effect heterogeneity that drives the results rather than to the flexibility policy's employment and welfare effects.

Table M.25: Flexibility entry effects (ToT) by the couple's education

Wife × husband	Wife's weight μ	Cognitive	Non-cognitive
College × College	0.44	−0.210	−0.157
College × Non-college	0.58	−0.211	−0.094
Non-college × College	0.32	+0.032	+0.068
Non-college × Non-college	0.47	+0.122	+0.173

Note: Treatment-on-treated effect on a child's cognitive and non-cognitive skill at age 7 (SD) of the mother entering paid full-time work during the child's ages 1–5, by the couple's education, re-solving the model with each couple assigned its education-specific Pareto weight μ on the wife from Gayle and Shephard (2019) (Table M.24).